Order, Symmetry and the Swap: A Synthesis of Matrix Transposition on Tapes

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Abstract

Can a deterministic multitape Turing machine transpose a $d \times d$ matrix with $\lceil \log_2 d \rceil$-bit entries in time linear in the input length L? Sixteen earlier papers of this series study the question without deciding it. This paper is a synthesis of them, a diagnosis of where they stop, and two new theorems.

The synthesis states the main results of the series in one place, with their main hypotheses. The diagnosis is this. For oblivious machines the question is now one about a single function, the best rate of a storage code on the bounded-depth cone relation that is left of a linear-time schedule after a genie has revealed εL bits: a transposer needs rate $1 - \varepsilon$ for every ε, pairing gives rate $\frac{1}{2}$, and for the relations of real machines in general no rate above $\frac{1}{2}$ is achieved and no useful bound below one is proved. Conditions on head motion that ignore the data are met by a machine that transposes nothing. Bounds for codes that ignore the machine fail, because abstract relations with the two properties used so far, bounded degree and paired-distance separation, carry codes of rate tending to one. Counting passes or cuts saturates one scale at a time. The common reason is that the $\log d$ of the folklore algorithm is a sum over address scales, and every tool that applies to all machines looks at one scale or at none; the logarithm is proved only where either the coding or the geometry is tied down.

The theorems concern the swap τ on $V \otimes V$, $V = F^d$, and the code $\ker(\tau - K)$ of matrices that an operator $K = \sum_a A_a \otimes B_a$ transposes correctly. *Order*: for every complete flag of V, the code has dimension at most the sum of the dimensions of the images of the flag under the B_a, whatever the A_a are; so two terms, commuting families, and, in characteristic p with $d = p^m$, all affine index maps stay at the dimension $d(d+1)/2$ of the symmetric matrices. *Symmetry*: if each row of a matrix is a fixed linear function of the matrix seen from that row's position, the same function on each of M intervals of rows, then the matrix ranges over a space of dimension at most $(d^2 + Md)/2$, on a cycle and on a line, with equality for equal intervals, where the bound is the dimension of the coarse $M \times M$ block transposition. The proof intersects the code with its own translate. It works in every characteristic and uses a property of machines that cone relations do not record: translated outputs share their decoder. Variants price each change of the decoder of a single output entry, from one row to the next, at half a symbol, where a change of class costs half a row, also in the presence of advice over a finite field; bound decoders with few modulation frequencies; and show that every address digit must be treated asymmetrically, which one unshuffle does for all digits at once. For classes of rows that no single shift respects the bound can fail: at $d = 6$ two such classes carry a code of dimension $25 > 24$, and, by a rule taken from a memo to this paper, two classes carry codes that exceed the bound by $d/4 - 1$ at every $d = 4k \geq 8$, over every field; for every fixed number $M \geq 2$ of classes the gain over the bound is unbounded. A later paper of the series, cited in this revision, bounds two classes of any shape by $\frac{41}{64}d^2 + O(d^{3/2})$ and shows that the second theorem needs the code, but not the decoders, to be linear; for sets of matrices and arbitrary decoders two bounds of this paper remain, the elimination step of the first theorem and the bound for rules that copy one entry, as instances of the acyclic-subgraph bound. The same bound shows that a set of matrices over q symbols

in which every entry is a function of at most two entries, at positions that move with the row, has at most $q^{d(d+1)/2+d-1}$ elements, and that reading any number of entries at boundedly many offsets gives no more than $q^{(1/2+o(1))d^2}$; whether three entries at arbitrary positions of this kind can give more is open. Applied to operators whose rows are arbitrary functions of whole rows of their argument, it shows that a pipeline of stages that each move one address digit or act on rows separately transposes at least $q^{(1/2+\delta)d^2}$ matrices only if it moves at least $\log_2 d - \log_2(1/(2\delta))$ digits, whatever the coding; and on some relations in which every row reads itself and three rows given by increasing maps it gives no better than $q^{(1/2+c)d^2}$, by the monotone expanders of Bourgain and Yehudayoff in the form of Dujmović, Sidiropoulos and Wood.

Both theorems are statements about a linear one-hop model, not about machines, and the question remains open. The coordinate form of the first theorem, its attainment, the pigeonhole lemma, the second theorem on a cycle for arbitrary shift sets and on a line, its attainment for equal intervals, its form for changes at single entries with and without advice, and the twist-class variant are formalized in Lean 4 on the standard axioms. The corollaries that need an algebraically closed field or the structure of p-groups, the copy-rule bound, the modulation and digit corollaries, the passage from flags to bases, one count in a sharper form of the bound for single entries, the gains for a fixed number of classes and the bounds for sets are proved in the text and not formalized. Exact computations, at sizes up to six and for the two-class rule and its blow-ups at larger sizes, are reported as numerical remarks.

Contributions. Hans Brende chose the problem, fixed the verification standard, directed the research, and asked for this synthesis. Claude Fable 5.2, a large language model developed by Anthropic, read the sixteen earlier papers and their memos, wrote the synthesis and the diagnosis, developed the mathematics of Sections 5 and 6 and the exact computations of Section 8, directed the Lean formalization, and wrote this paper. Independent instances of the same model refereed the two theorems; one of them contributed Corollary 5.8, the line form of Theorem 6.4 (Theorem 6.9) and the counterexample of Remark 6.10. The revision that follows the memo [14] was written by Claude Fable 5.2 at Hans Brende's request, with the Lean proofs of Proposition 6.15 and Corollary 6.16; a further instance, which refereed it, pointed out that the charge for revealed bits confused two units, and contributed the bound of Proposition 6.15 with its proof and the combination with advice that became Corollary 6.16. The rule of Proposition 6.17 is the memo's rule with a general coefficient, and the proof of its dimension follows the memo's, where Gemini 4 Argon found the rule and part of the rank count and Claude Opus 5.5 completed the proof; Claude Fable 5.2 observed that the rule works with a general coefficient and over every field, and adapted the proof. A further instance, which refereed the revision that follows the memo's update, observed that a blow-up with a code for each position settles every fixed number of classes, and gave the argument of Lemma 6.18 and Corollary 6.19. The revision that follows the two later papers [25, 19], by Hans Brende and GPT-6 Astra, was written by Claude Fable 5.2, which read both papers, rebuilt their Lean libraries and wrote Section 6.7 and the new form of Problem 9.4. A further instance refereed it against the two papers. It pointed out that Proposition 6.21 is an instance of the acyclic-subgraph bound, gave its form with rows and columns exchanged, and observed that the examples of the second paper lack only a bounded window, so that the question with one function for all rows had to be asked as well. In a second pass it contributed the marker rows of Section 6.7 and their consequence for Problem 9.4(b). The revision after that, on Problem 9.4(b), is by Claude Fable 5.2: it found Propositions 6.22 and 6.23, with the ranking lemma and the two counting arguments, and ran the searches reported in Section 6.7. A further instance refereed the proofs and the scripts and found no error in the proofs. It contributed the examples after Proposition 6.22 (that the ranking lemma is sharp on cycles, the relation at $d = 3$ that attains the bound of window two, and the lower bound 1 for the constant at even d), observed that parts (c) and (d) with a multiplier prime to d follow from Proposition 6.21 after a renumbering, and that one entry of the table at the end of Section 6.7 can be turned into a proof. In a second pass it read the text of this revision against the seventh paper and corrected two sentences of the commentary on Problem 9.4. The revision after that, on whole rows and stages, is again by Claude Fable 5.2: it found Propositions 6.24 to 6.27 and Problem 9.4(c). A further

instance refereed the proofs and the script in two passes. In the drafts it found a strict inequality in Proposition 6.25(b) that fails when there are no scans, and a wrong sign in the count after Proposition 6.25(c); both are corrected. It found that a stage which reads a row together with its shuffle parents reads no coset smaller than the whole space, contrary to an earlier draft, which led to Proposition 6.25(c); it corrected the hypotheses under which the gather passes are linear, and it supplied the extension of Proposition 6.27 to relations in which only the number of rows read is bounded. A third instance read the text of this revision and found no error in the four propositions as they stand; it corrected the status of Proposition 6.26(b), which is cited, and several sentences of the commentary.

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1 Introduction

1.1 The problem

Can a deterministic multitape Turing machine transpose a $d \times d$ matrix with $\lceil \log_2 d \rceil$-bit entries in time linear in the input length $L = d^2 \lceil \log_2 d \rceil$? We call the affirmative statement (T); Section 2.1 states it precisely. The folklore algorithm takes $O(L \log d)$ steps. Transposition is the data-movement step of fast integer multiplication on tapes, and by a reduction of Harvey and van der Hoeven [41] a linear-time multiplier would give (T). So a proof that (T) fails would be the first superlinear lower bound for off-line multiplication on multitape machines, and a proof that it holds would be a new way of moving data.

Sixteen papers and eight memos of this series have studied the question since October 2026. They contain reductions, restricted impossibility theorems, constructions of codes and many formal proofs. They do not contain the answer, and a reader who starts today has to read all of them to learn where the difficulty is. This paper is written for that reader.

1.2 What this paper does

It does three things.

A synthesis (Section 3). We state the main results of the series in one place, by theme: the reductions, what head motion must do, depth reduction, what is proved about storage codes, what is false about them, and adaptive machines. Nothing is reproved. Every statement is cited, with its main hypotheses.

A diagnosis (Section 4). For oblivious machines the series has reduced (T) to one function: the best rate $r(\varepsilon)$ of a storage code on the bounded-depth cone relations that remain of linear-time schedules when a genie reveals εL bits. A transposer needs $r(\varepsilon) \geq 1 - \varepsilon$ for every ε. The matching of each stored symbol with one partner gives $r \geq \frac{1}{2}$, and whether r exceeds $\frac{1}{2}$ at all on the relations of real machines is open in both directions. Three kinds of argument have reached known limits:

- conditions on motion that ignore the data, because a machine that transposes nothing satisfies the cut inequalities behind them; conditions that do see the outputs are single-scale, and one cut serves every pair of row sets;
- bounds for codes that ignore the machine, because abstract relations with the two properties used so far, bounded degree and paired-distance separation, carry codes of rate tending to one;
- counting passes, because it saturates one scale at a time.

They stop for one reason. The logarithm of the folklore bound is a sum over $\log_2 d$ address scales, each costing L, and every tool of the series that applies to all machines looks at one scale or at none; the logarithm is proved only where the coding or the geometry is tied down. The missing statement has to tie what is known to where it can be, for all schedules, and the coding properties proved so far are properties of the cone *relation*, which forgets that neighbouring outputs of a machine are produced by the same rule.

Two theorems (Sections 5 and 6). We prove two rigidity theorems for the swap in a model where the code and its decoder are visible together. Let $V = F^d$ over any field, let τ be the swap on $V \otimes V$, that is, $X \mapsto X^{\mathsf{T}}$ on matrices, and let $K = \sum_a A_a \otimes B_a$ act by $X \mapsto \sum_a A_a X B_a^{\mathsf{T}}$. The code $\ker(\tau - K)$ is the set of matrices that K transposes correctly. The symmetric matrices, for $K = 1$, have dimension $d(d+1)/2$.

- *Order (Theorem FL, Theorem 5.1).* For every complete flag $W_1 \subset \cdots \subset W_d$ of V, with $g(i) = \dim \sum_a B_a(W_i)$,

$$\dim \ker(\tau - K) \le \sum_{i=1}^{d} g(i),$$

  whatever the A_a are. So two Kronecker terms never beat the symmetric matrices, over any field; neither do commuting B_a, for instance any convolutions of any number of rows; and neither does any combination of rotations, reflections, dilations and decimations when $d = p^m$ in characteristic p. The bound is attained for every profile g.

- *Symmetry (Theorem TP, Theorem 6.4).* For any three operators on an N-dimensional space, $\dim \ker(\tau - K) \le \frac{1}{2}(N + \dim \ker[\tau, T] + \operatorname{rank}[K, T])$, an elementary lemma. For the swap, $\dim \ker[\tau, T] \ge d$ always. If row c of X is a fixed linear function of X with its columns rotated by c, the same function for all c in each of M intervals, then

$$\dim\{X\} \le \frac{d^2 + Md}{2},$$

  on a cycle and on a line, with equality for equal intervals: there the largest code has the dimension of the coarse $M \times M$ block transposition. For M equal intervals flags give only rate $1 - 1/(2M)$. Variants bound decoders with s modulation frequencies by rate $s/(s+1)$ and show that every address digit must be treated asymmetrically. The hypothesis on the classes is needed: for $d = 6$, two classes with no almost-period carry a code of dimension $25 > 24$ (Proposition 8.2), found by a SAT solver and checked in Lean, and for every fixed number $M \ge 2$ of classes the gain over the bound is unbounded (Section 6.6, added after a memo to this paper).

In words: to beat the pairing value, the decoder may have neither a hidden order nor a symmetry, on either index, that the swap does not share. The second theorem uses exactly the feature of machines that cone relations forget.

What the theorems are not. They are statements about the linear one-hop model. They are not lower bounds for machines, and (T) remains open. Section 7 lists what they exclude and the three steps that would be needed to carry them to machines; Section 8 gives exact values at small sizes; Section 9 states the problems.

A subsequent memo, and the revisions of this paper. After revision `b2c389a` of this paper in its repository, a memo [14] proved, in Lean, that the two classes of Proposition 8.2 carry a code of dimension 25 over every field in which $3 \neq 0$; that class-uniform codes can be blown up, which gives gains that grow with the number of classes; and bounds for the cost of advice in the one-hop model. Revision `14b157a` of this paper cited the memo where it bears on the text, and corrected three passages:

- the charge for a bit that a genie reveals (Section 7.4 and Problem 9.4): the earlier text charged every revealed bit as a change of regime, at half a row, where advice that enters additively changes no decoder and half a row is the price of a change that affects a whole row;

- Problem 9.1(c), which the memo answered as it was worded and which was then stated for a fixed number of classes;
- the first sentence of Section 7.3, which had dropped a hypothesis of the result it summarizes.

It added Proposition 6.15 and Corollary 6.16, which state the corrected charge, with formal proofs of their bounds in terms of the number of changes, and it updated Remark 8.1(d) and Problem 9.1(b).

An update of the memo then answered that form of Problem 9.1(c) for two classes: for every $d = 4k \geq 8$ it gives an explicit rule, over fields in which $2 \neq 0$, whose code exceeds $(d^2 + 2d)/2$ by $d/4 - 1$, so that the gain of two classes is unbounded. Its proof is written in the memo, which also proves in Lean, for $d = 8$, 12 and 16, that the two classes carry a code of that dimension. It leaves open characteristic 2 and three or more classes. Revision `ac70dc4` of this paper cited that update and added Section 6.6, which settles both.

- Proposition 6.17: the memo's rule works over every field, $\mathbb{F}_2$ included, once its coefficients 2 and $\frac{1}{2}$ are replaced by a nonzero λ and its inverse.
- Lemma 6.18 and Corollary 6.19: a blow-up with a code of its own at each position turns the two-class codes into codes for any fixed number $M \geq 2$ of classes, with a gain of order $Md/8$ over $(d^2 + Md)/2$. So the gain is unbounded for every fixed number $M \geq 2$ of classes, over every field.

These three results are proved in the text and are not formalized. Problem 9.1(c) now asks for the rate of growth.

Two papers of the series followed revision `ac70dc4`. One is about the original runs of adaptive machines [25]. The other is about the class-uniform rules of Section 6, their codes and their composition [19]. Revision `3911d87` cites both where they bear on the text, and changes the following.

- *Bounds for classes of any shape* (Problem 9.1(a), Section 7.3). For two classes the second of these papers proves $\frac{41}{64}d^2 + O(d^{3/2})$, where this paper had $\frac{3}{4}d^2 + O(d)$, and for M classes a deficit of order $M^{-1/2}$, where this paper had $1/(2M)$. For partitions with a period, for which Theorem 6.4 shows that the gain is at most linear in d, it determines the largest code of a lift under hypotheses that are stated with Problem 9.1(c).
- *Which linearity the bounds need* (Section 6.7, new). Theorem TP needs the code to be a subspace, but not the decoders to be linear. For sets of matrices that are not subspaces it fails completely: one decoder for all rows is correct on a fraction $1/(2d)$ of all matrices. Both facts are from that paper. That revision adds Proposition 6.21: the elimination step of Theorem FL and the bound for rules that copy one entry hold for sets and arbitrary functions, as instances of the acyclic-subgraph bound of Bar-Yossef, Birk, Jayram and Kol [2]; and a remark on marker rows, by which a set of matrices with one function for all rows exceeds the value $(d^2 + d)/2$ of subspaces. One sentence of Section 6.5 and one of Section 7.2, which said that Theorem TP needs linear decoders, are corrected, and a sentence is added to Section 6.5.
- *The step to machines* (Section 7.4, Problem 9.4). The earlier text took the form of its question from a count for affine rules. For decoders that are not affine and see the whole matrix there is no such count. What machines add is that the cones a genie leaves are of bounded size, and whether shared decoders of bounded window bound a set of matrices is the new Problem 9.4(b). Section 7.4 also cites the first bound for two hops, and notes that a perfect shuffle of the columns has $d - 1$ classes; Section 7.1 notes what Proposition 6.21 gives for synchronous access.

- *Adaptive machines* (Section 3.6, Section 4.1, Section 4.2). The first of the two papers proves on the machine's own runs a statement of the form that Section 3.6 reported as open, and gives the function of Section 4.1 a form on those runs.

- Problem 9.2 is now said to be about linear decoders.

Proposition 6.21 is proved in the text and is not formalized. The other results of this paper and their formal proofs are unchanged, and so are their numbers.

That revision, `3911d87`, asked Problem 9.4(b) for every window. The revision after it, `a445cd8`, answered part of it, in two propositions added to Section 6.7.

- Proposition 6.22: the acyclic-subgraph bound, applied through an order of the rows or of the columns of the matrix. For window two it gives at most $q^{d(d+1)/2+d-1}$ matrices, for all maps, all functions and all sets, which is attained at $d = 3$ and is within $d/2$ of an example at every even d; and for every window it gives $q^{(1/2+o(1))d^2}$ when the offsets, or the differences of the rows from a permutation, take boundedly many values. Both hold for functions of whole rows, respectively whole columns.

- Proposition 6.23: for every $\varepsilon > 0$, all but εd^2 of the pairs of positions that help each other are related by one of boundedly many translates of the swap, so a code that uses only such pairs stays at the pairing rate; and $o(d^2)$ positions meet all transitive triangles, which makes the theorems of the seventh paper for relations without them available on one-class relations.

- Problem 9.4(b) is now asked for windows of at least three, and Section 6.7 reports computations which indicate that for large windows the acyclic-subgraph bound cannot answer it.

The two propositions are proved in the text and are not formalized. No earlier result changes, and no number of a result or of a table.

A further revision adds four propositions after them, on operators whose rows are arbitrary functions of whole rows of their argument, and Problem 9.4(c).

- Proposition 6.24: such an operator transposes at most q^{σ} matrices, where σ is the least sum, over the orders of the rows, of the place of the last row that each row of the output reads; this is exactly the acyclic-subgraph bound of the relation, it equals $(d^2 + sd)/2$ when each row of the output reads the block of s rows that contains its image under a bijection, and it is at most $wd(d+1)/(w+1)$ when w rows are read.

- Proposition 6.25: in a pipeline of stages that read rows inside cosets of the address space, with arbitrary functions and alphabets between the stages, the dimensions of the cosets add up, and t moved digits give at most $q^{(d^2+2^t d)/2}$ transposed matrices, which partial transposes attain. So a set of rate $\frac{1}{2} + \delta$ needs $\log_2 d - \log_2(1/(2\delta))$ moved digits, for every code. Scans with a buffer of one row enter through advice, and pipelines that mix differently arranged copies of the array through the number of ones of the address, with $\sqrt{\log d}$ in place of $\log d$.

- Proposition 6.26: if the rows read expand, the bound of Proposition 6.24 is at least $(\frac{1}{2} + c)d^2$, and by the monotone expanders of Bourgain and Yehudayoff, in the form with three matchings of Dujmović, Sidiropoulos and Wood, there are such relations made of the identity and three increasing maps, which three forward passes gather when the strings that mark the maps are given. So the acyclic-subgraph bound cannot show that operators of boundedly many such passes transpose only $q^{(1/2+o(1))d^2}$ matrices.

- Proposition 6.27: products of codes on boundedly many positions stay at the pairing rate on every such relation of bounded degree.

These are proved in the text and are not formalized, except that Proposition 6.26(b) applies a cited theorem, of which we use the statement and have not checked the proof. No earlier result changes, and no number of a result or of a table.

1.3 How to read this paper and its formal proofs

The synthesis and the diagnosis contain no numbered results; they cite. The numbered results of Sections 5 and 6 each carry a status line.

- "Lean:" names the formal counterpart in the library `lean/SwapRigidity/`; the Lean statement is authoritative, and the text gives an outline.
- "Lean (partial):" names what is formalized and says what is not.
- "Not formalized" is followed by a complete proof.

Formalized in Lean 4 with Mathlib, on the standard axioms only: the coordinate form of Theorem FL, its attainment and the corollary for a pair of flags; the pigeonhole lemma, its composition property and the lower bound $\dim\ker[\tau, T] \geq d$; Theorem TP on a cycle with an arbitrary shift set, its attainment for equal pieces, and its line form; the bound for decoders that change at single entries (Proposition 6.15), apart from a count, and its form with advice (Corollary 6.16); and the twist-class theorem. Not formalized, and proved in the text: the passage from flags to adapted bases; the corollaries of Theorem FL that need an algebraically closed field, the structure of p-groups or an arithmetic estimate (Corollaries 5.5 and 5.7 to 5.9); the equality case of Proposition 6.2; the copy-rule bound; the count in Proposition 6.15; Corollary 6.12; Corollary 6.13 apart from the abstract lemma it rests on; the gains for a fixed number of classes over every field (Proposition 6.17, Lemma 6.18, and Corollary 6.19); the bounds for sets (Propositions 6.21 to 6.23) and for whole rows and stages (Propositions 6.24, 6.25 and 6.27 and Proposition 6.26(a)); and the translation between class-uniform rules and operators $\sum_a A_a \otimes B_a$. Numerical observations appear only in remarks labelled as numerical and in Section 8, and no proof uses them. Appendix A indexes the declarations and Appendix B explains how to check them.

1.4 Related work

The series. Section 3 is the account of the earlier papers. The closest of them to the new theorems are the eighth, which states the coding problem for linear machines as relative rigidity of the transposition matrix [4]; the second and fourth, which exclude locally affine head motion [24, 6]; and the seventh, whose elimination lemmas and Cayley corollary are the storage-code relatives of Theorems 5.1 and 6.4 [11].

Matrix equations with a transpose. Equations in which X and X^{T} both occur have been studied as $\star$-Sylvester and $\top$-Stein equations. The solution space of $AX + X^{\star}B = 0$ is described through the Kronecker canonical form of a pencil [34]; that of $XA + AX^{\mathsf{T}} = 0$ is found in [33] and applied to congruence orbits; unique solvability of $AXB + CX^{\star}D = E$ is characterized in [35], and of $X = AX^{\mathsf{T}}B + C$ in [31]. Generalized Schur decompositions are used for solvability and numerical solution [32]. Theorem 5.1 uses triangular forms for a different purpose, a dimension bound in the singular case with one side unconstrained. We found no statement of Corollary 5.5, of the bound

for an arbitrary number of terms with one triangularizable side, or of any bound on the rank of the commutation matrix [43] minus a structured sum of Kronecker products. Our search was limited to abstracts and indexes, and we claim no priority for statements that specialists may regard as folklore.

Commutators and network codes. Lemma 6.1 is the fact that a commutator has rank at most twice the rank of either factor; we know no source that states it in the form used here and we give its proof. Transposition as a multi-commodity network-coding instance goes back to [1]. Network codes whose local kernels are sums of cyclic shifts have been studied for multicast [48]; we found nothing on shift-covariant decoders for transposition.

1.5 Organization

Section 2 fixes the problem and the two models. Section 3 is the synthesis and Section 4 the diagnosis. Sections 5 and 6 prove the two theorems. Section 7 is the construction side, Section 8 the exact computations and Section 9 the open problems.

2 The problem and two models

2.1 The question

Throughout, $d \geq 1$ is the side length of a square matrix, $b(d) = \max(1, \lceil \log_2 d \rceil)$ is the entry width, and $L = d^2 b(d)$ is the input length. A machine is a deterministic multitape Turing machine with a read-only input tape, a fixed number k of work tapes, a write-only output tape and a fixed finite alphabet. The input is a $d \times d$ matrix with entries in $\{0,1\}^{b(d)}$, written row by row without delimiters; the output is its transpose in the same encoding.

> **(T)** There are a machine M and a constant c such that, for every d and every input matrix, M halts with the transpose on its output tape within $c(L+1)$ steps.

The authoritative statement is the Lean definition `Transpose.LinearTimeTransposable` of the first paper [9], which the series keeps frozen. A machine is *oblivious* if its head positions at each step, the steps at which it writes output and its halting time depend only on the input length; otherwise it is *adaptive*. A *pass* is a maximal stretch of steps on which every head moves monotonically.

The folklore algorithm transposes in $O(L \log d)$ steps by a recursion on address digits: write the row address and the column address of an entry in binary, and exchange them one digit at a time, each exchange costing $O(L)$. An unrefereed preprint [44] claims $O(L(\log d)^{1-\kappa})$ for a small $\kappa > 0$. Harvey and van der Hoeven [41] reduce transposition to integer multiplication on multitape machines, so a linear-time multiplier would give (T), and a proof that (T) fails would be a superlinear lower bound for off-line multiplication on these machines. No superlinear lower bound is known for transposition, nor for off-line multiplication, sorting or the discrete Fourier transform, on machines with two or more work tapes. The known bounds are for one work tape [37, 36], for machines that only move data [47], and for reversal-bounded conservative machines [42]; the superlinear bounds that follow from the separation of deterministic and nondeterministic linear time [45] concern problems that are hard for nondeterministic linear time [40].

2.2 Storage codes on recovery relations

Most of the series is written in the following language, which we use in Sections 3 and 4.

Definition 2.1 (Storage codes)**.** A *recovery relation* on a finite set V assigns to every $v \in V$ a set $N(v) \subseteq V$; it is *loopless* if $v \notin N(v)$ for all v. A *storage code* on it, over an alphabet α, is a set $C \subseteq \alpha^V$ such that for every v the coordinate w_v of a codeword $w \in C$ is a function of $(w_u)_{u \in N(v)}$. Its *rate* is $\log|C|/(|V|\log|\alpha|)$.

A relation in which $u \in N(v)$ exactly when $v \in N(u)$ is a graph, and the largest $\log_{|\alpha|}|C|$ is then the guessing number of Riis [46]; storage codes are dual to index codes with side information N. On a perfect matching the rate is exactly $\frac{1}{2}$: each vertex copies its partner. We call $\frac{1}{2}$ the *pairing rate*.

The *incidence graph* of a relation has two vertices L_p and R_p for every coordinate p, its port as a helper and its port as a receiver, and an edge L_pR_q whenever $p \in N(q)$. The *paired distance* of p is the distance between L_p and R_p in this graph; for a graph it is the length of the shortest odd closed walk through p. Following the fifth paper [22, Definition 2.3], we call a sequence of relations *separated* if, for every fixed r, all but $o(|V|)$ coordinates have paired distance greater than r. A perfect matching has all paired distances infinite, so separation does not exclude the pairing; it excludes gains that come from short cycles.

For an oblivious machine and a set of steps whose values are revealed (a *genie*), the *cone* of an output bit is the set of input bits on which it can still depend through the time–space dependency graph. The *cone relation* assigns to an input coordinate q the cone of the output bit that must equal it. If the machine transposes, then for some value of the revealed word the inputs that agree with it form a storage code on the cone relation, of high rate when the genie reveals few bits (Section 3.3). The *depth-D ball* of an output bit is the set of input positions read at steps within D dependency steps of the step that writes it, with no genie.

2.3 The linear one-hop model

The two theorems of this paper live in a smaller model, in which the code and the geometry are visible at once. Let F be a field, $V = F^d$, and let τ be the swap $u \otimes w \mapsto w \otimes u$ on $V \otimes V$. Under the identification of $V \otimes V$ with the $d \times d$ matrices $\mathrm{M}_d(F)$, τ is transposition, $X \mapsto X^{\mathsf{T}}$, and an operator

$$K = \sum_a A_a \otimes B_a, \qquad A_a, B_a \in \mathrm{M}_d(F),$$

acts by $X \mapsto \sum_a A_a X B_a^{\mathsf{T}}$. Every operator on $V \otimes V$ has this form for some finite family of terms.

Definition 2.2 (Swap code)**.** Lean: `swapCode`, `mem_swapCode` [`Flag.lean`]
The *swap code* of $K = \sum_a A_a \otimes B_a$ is

$$\ker(\tau - K) = \Big\{ X \in \mathrm{M}_d(F) : X^{\mathsf{T}} = \sum_a A_a X B_a^{\mathsf{T}} \Big\}.$$

Its *rate* is $\dim\ker(\tau - K)/d^2$. (The Lean definition is the set of matrices on the right; its identification with the kernel of an operator is not used in the formal proofs.)

The code is the set of matrices that K transposes correctly. Read row by row, the equation says that every column of X is a fixed linear function of the rows of X: the stored word is X in row-major order, the receivers want it in column-major order, and K is the decoder. For $K = 1 \otimes 1$ the code is the space of symmetric matrices, of dimension $d(d+1)/2$. We call this the *pairing value*; its rate tends to the pairing rate $\frac{1}{2}$. For $K = \tau$ the code is everything.

Entries of b bits are handled by replacing $V \otimes V$ with $V \otimes V \otimes F^b$ and τ with $\tau \otimes 1$; the flag bounds below then hold with an extra factor b (Remark 5.10), and Theorem 6.4 likewise.

Relative rigidity. A bound $\dim\ker(\tau - K) \le (1-\rho)d^2$ for all K in a class says that the transposition matrix is ρd^2-rigid relative to that class: τ is not the sum of a member of the class and a matrix of rank below ρd^2. The eighth paper shows that, for machines with linear data paths, the coding half of the series' programme is exactly such a statement, for the bit-level transposition matrix and the class of matrices whose row for each output bit is supported on its depth-D ball in a schedule [4, Theorem 7.5 and Problem 10.6]; this is a relative form of Valiant's rigidity [49]. The classes of this paper are defined by algebraic structure on one tensor side instead of by a sparsity pattern. The other side is arbitrary.

3 The series in one place

This section states the main results of the sixteen earlier papers and their eight memos, grouped by what they are about and not by when they appeared. Nothing is reproved. Each statement carries its citation and its main hypotheses; the cited theorems have further parameters (the number of tapes, the alphabet, the time constant), which we suppress. Table 1 at the end lists the papers in order. A reader who wants only the conclusion can skip to Section 4.

3.1 The question, passes and the coding formulation

The first paper [9] fixes the formal statement of (T), shows that for linear time it agrees with the simplest case of the formulation of Harvey and van der Hoeven [9, Proposition 11.1, a written proof], and proves the following.

- *Passes.* A transposer with at least one work tape makes $\Omega(\log d)$ passes, whatever it does with its data: for oblivious machines with no premise on the running time, and for all machines that run in time polynomial in L [9, Theorems 5.9 and 5.12]. A memo removes the runtime premise for halting machines that are correct on all inputs [17]. This is the only logarithmic lower bound on a resource of the machine itself that the series proves for all machines. It counts passes, not steps, and it is tight: the folklore algorithm uses $O(\log d)$ passes.

- *Folding.* Along its popular traces, every limit of the rescaled work-head motion of a linear-time transposer satisfies an isoperimetric inequality that forces the heads to fold back (Theorem M*, [9, Theorems 7.8 and 7.9]). The proof is a cut argument: the information that crosses a cut of the time–space diagram is bounded by the states and the work-tape cells that enter across the cut.

- *Coding.* Every linear-time transposer yields a causal network code for complete exchange on the block network of a time–space diagram of linear length, correct on a $2^{-O(d\log d)}$ fraction of the inputs, and on all inputs if the machine is oblivious [9, Theorems 8.10 and 9.9]. The *Tape Coding Conjecture* says that for large d no causal code solves complete exchange on such a fraction of the inputs on the block network of any diagram of linear length; it implies that (T) fails [9, Conjecture 9.1, Theorem 9.2]. Its plain form, for codes correct on all inputs, excludes oblivious linear-time transposers [9, Theorem 9.9]. On these networks every routing has rate $O(1/\log d)$, so a solving code beats routing by a factor $\Omega(\log d)$, and the conjecture is equivalent to saying that coding beats routing by only $o(\log d)$ on tape networks [9, Theorems 8.8 and 9.4].

The pass bound at polynomial running time, Theorem M* and the reduction to codes survive when the machine is correct on only a $2^{-O(d\log d)}$ fraction of the inputs. The same paper studies an abstract model of the one technique claimed to beat $O(L\log d)$, recursion on XOR-mixed address frames: there every gadget costs at least its number of wires, whereas linear time would need less [9,

Theorem 12.4 and Proposition 12.9]; the correspondence between the model and the preprint's recursion is informal. It also records why a proof must use the shape of tapes: a bound $o(\log n)$ on coding gains for bounded-degree graphs in general would give superlinear circuit lower bounds for cyclic shifts and multiplication [9, Proposition 11.2].

The third paper [18] shows that the bit case is at least as hard: linear-time transposition of bit squares of dyadic side implies (T), by one fixed wrapper. The converse is open, and the paper proves barriers for restricted compilers in that direction; one of them is a multiscale obstruction for compilers with boundedly many stages.

3.2 What head motion must do

A second line asks what the heads of a linear-time transposer must do.

- *Not affine.* Along the popular traces, the limits of the rescaled work-head paths of a linear-time transposer fail to be locally affine on sets of times of total measure at least $\frac{1}{2}$, summed over the heads and measured in blocks. Bounded-length words over three explicit palettes of scan operations are excluded, and so are contracting recursions all of whose stages have regular limiting motion [24, Theorems 6.3, 7.1 and 7.5]. No machine whose work heads are home-based with legs made of boundedly many constant-rate runs transposes in linear time, and in every rescaled limit a linear-time transposer has a linear amount of curved travel [6, Theorems 3.10 and 3.12].

- *Macroscopic crossings.* An oblivious linear-time transposer routes a linear amount of information through pairs of interleaved revisits whose excursions are a fixed fraction of L: for every set of at least αL output bits, every set of steps separating them from the earlier ends of such crossings has at least c times as many elements [15, Theorem 5.5]. A fixed fraction of this expansion sits on one coarsely monotone leg of one tape, in a skeleton from a finite family independent of d, and two work heads must each make linearly many long revisits [15, Theorems 6.7 and 4.1]. The crossing and skeleton statements hold on heavy traces of adaptive machines and of a bit simulator [15, Theorem 8.1].

- *Rows to columns.* In the diagram of an oblivious transposer, every set of steps that separates the inputs of a set of rows from the outputs they feed into a set of output rows carries the corresponding information, whatever the coding [15, Lemma 7.1]. When the cuts for different address bits are disjoint this gives a time bound $\Omega(L \log d)$ that no coding can evade [15, Corollary 7.3].

- *Source covers.* For one fixed geometry of returns, a certificate that retains every input on which an output can depend, and charges the information used to simplify the dependence, gives a correctness deficit of at least εL when the return families consist of a two-class core, at most one arbitrary exceptional family and families of at most εL hops, for fixed bounds on the running time, the capacity and the number of span classes [28, Corollary 4.3]. A memo excludes funnels and single-sweep terminal landings and shows where the certificate stops [7]. These hypotheses on the geometry are not derived for arbitrary machines.

  Three results mark the limits of this line.

- *Padding.* For every machine there is a wrapper, with more tapes and states and additive overhead $O(d^2 b)$, whose complete runs satisfy the cut inequality behind Theorem M* with zero slack, and whose normalized head limits satisfy all the conclusions of Theorem M* under a linear time

bound. The wrapper's motion is independent of the computation that follows [24, Theorem 8.1]. It needs no idle tapes, and it also meets the finer entry-cell form of the inequality at every slot length from one cell up to a constant number of rows [12]. So these conditions on motion can be met by a machine that transposes nothing; they do not see which information reaches the outputs.

- *One cut for all pairs.* In the diagram of any run, one set of steps separates the inputs from the outputs for every pair of row sets at once [15, Proposition 10.2]. So counting the cuts of the rows-to-columns inequality cannot by itself go beyond linear time: the disjointness that [15, Corollary 7.3] needs is not automatic.

- *The candidates.* Five explicit linear-time cascade schedules meet the crossing and skeleton conditions at the sizes tested. All five are excluded, for fixed work alphabets, by the regular-recursion obstruction of the second paper [26], and two of them also by explicit cuts of the rows-to-columns inequality [15, Theorem 10.8]. Whether the combined conditions on motion decide the oblivious case is open.

3.3 Depth reduction

The fourth paper [6] splits the oblivious case of (T) into two statements.

- *Depth reduction* (the trajectory cut problem TCP(k)): for every $\varepsilon > 0$, deleting εT edges from the dependency graph of k head walks of length T leaves no path with more than a bounded number of long hops, and then, after periodic steps are also revealed, paths of bounded length.

- *The crux*: an index-coding lower bound on the resulting bounded cones, strong enough to exceed the cost of the revealed steps.

Together they would exclude oblivious linear-time transposers [6, Theorem 4.10 and Corollary 4.11]. Depth reduction is now a theorem. It was proved for home-based walks [6, Theorem 4.20]; then for walks of bounded nesting load, and for one arbitrary head together with heads of bounded nesting load at boundedly many scales [4, Theorems 3.6 and 4.2]; and finally for every number of walks, with a depth $D(k, \varepsilon)$ such that $\log \log D = O(k^2 \log(k/\varepsilon))$ [5, Theorem 7.5]. A memo shows that TCP(k) for all k is equivalent to constant-depth revelation of arbitrarily small entropy per input bit, for all linear-time oblivious computations [30]. The fourteenth paper also refutes Conjectures A′ and A″ of the fourth and eighth papers, which would have made the deep-nesting pairs depth-robust [5, Corollary 9.1]: walks nested at m scales are cut to a depth independent of m. It draws the conclusion that matters here [5, Corollary 9.3]:

> *for oblivious machines, the crux alone would exclude linear-time transposers.*

In the language of Section 2.2: for an oblivious linear-time transposer and every $\varepsilon > 0$, revealing εL bits at suitable steps leaves every output bit a function of the revealed word and of boundedly many input bits, and for some value of the revealed word the inputs compatible with it form a binary storage code of rate at least $1 - \varepsilon$ on the machine's own cone relation. The same genies give depth reduction at the level of codes [5, Corollary 9.6], as conjectured in the eleventh paper. And, given the *native half bound* (every binary storage code on the cone relations of the machine's own traces has rate at most $\frac{1}{2} + o(1)$, which is open), they exclude every linear-time transposer, adaptive or not, whose trace has entropy at most ηL for a fixed $\eta < \frac{1}{2}$ [5, Corollary 9.7].

For machines whose data path is linear over a field, linear depth reduction together with a *linear crux*, a min-rank bound for the ball systems of schedules, excludes oblivious linear-time

transposers [4, Theorem 7.5]. The linear crux is exactly the relative rigidity of the transposition matrix described in Section 2.3 [4, Problem 10.6], and since the fourteenth paper it alone suffices [5, Corollary 9.3(c)].

3.4 Storage codes: what is proved

The crux needs an upper bound on the rate of storage codes on cone relations. The series proved such bounds, at the pairing rate, for many classes of relations.

- *Homomorphisms and small degree.* Rate bounds pull back along homomorphisms, so every storage code on a graph that maps to C_5, to $K_{14/5}$ or to the Petersen graph has rate at most $\frac{1}{2}$ [6, Theorem 6.5 and Corollary 6.9]. On triangle-free graphs of maximum degree three no scalar linear code over any field beats $\frac{1}{2}$ [6, Theorem 6.13]. The Clebsch graph, of degree five, has guessing number 10 on 16 vertices [6, Corollary 6.11], so bounded degree alone is not enough.

- *Bounded complex coefficients.* On locally bipartite graphs of bounded degree, linear codes with bounded complex coefficients have rate at most $\frac{1}{2} + o(1)$ [6, Theorem 6.17]. The same holds for complex linear recovery with bounded coefficients and fixed block width on directed relations of bounded incidence degree that are separated, hence on cone relations after a paid trimming of degrees [22, Theorem 4.5 and Corollary 4.7]. On such relations a gain by linear codes with boundedly many symbols per vertex therefore needs positive characteristic, or complex coefficients that no diagonal rescaling makes bounded.

- *Hyperfinite relations.* Every code over every finite alphabet on relations whose incidence graphs are uniformly hyperfinite and separated has rate at most $\frac{1}{2} + o(1)$; no degree bound is needed [22, Corollary 4.4].

- *Nested elimination.* The exact bound $|C|^2 \le |\alpha|^{|V|}$ holds for all codes, linear or not, on large classes of relations without transitive triangles: triangle-free graphs of maximum degree three; loopless relations whose underlying graph has maximum degree three; relations whose vertices of in-degree at least three induce an acyclic sub-relation; relations partitioned into blocks with private outside parents, each block chordless subcubic, a chorded digon path or internally certified; and Cayley relations $\mathrm{Cay}(G, \{s, s^{-1}, t\})$ of every finite group [11, Theorems 4.3, 5.2, 5.3 and 5.10, Corollary 5.12]. It also holds on the class of the fourth paper's Theorem S, for all codes [16, Theorem 3.6]. The degree bound cannot be raised: a 4-regular triangle-free graph on 13 vertices carries a linear code over $\mathbb{F}_4$ with $|C|^2 = 4^{14}$ [11, Example 7.8].

- *Covers and towers.* For the binary adjacency code of the Clebsch graph, every connected tower of binary lifts has kernel dimension at most $N/2 + 2$ [21, Theorem 9.2]; lifting a binary linear code with a unit-diagonal check along a tower of double covers never raises its rate [4, Theorem 8.1].

- *Memos.* The half bound for circular cliques $K_{p/q}$ extends to every $p/q < 3$ [20]; nested elimination extends to soft recovery with record capacities [27]; and the bounds for supports of maximum degree three extend beyond scalar linear codes, to reversible rules over any alphabet and to linear codes whose rows have at most one port of rank above one [29].

One result of this group is about the relations of machines themselves. For cones of fixed depth in linear-time schedules, after a paid trimming of degrees, only $O(L/\log d)$ matched pairs of an input port and the output that demands it lie within any fixed incidence distance [22, Theorem 3.3, Corollaries 3.7 and 3.8]. So cone relations are separated in the sense of Section 2.2, and almost no output is within bounded distance of its own input bit. No gadget of bounded size can then lift a

code above the pairing rate: a gain above $\frac{1}{2}$ on boundedly many coordinates needs a short path between the two ports of one of them [22, Lemma 4.1 and Theorem 4.2]. The $L/\log d$ comes from a bound in steps for transport at bounded occupancy, which charges every transported label the $\log d$ bits of its row address [22, Lemma 3.4]: for uncoded movement the logarithm is proved, and what the bound does not cover is a cell that carries a combination of many labels.

This led to the *all-code question* [22, Question 9.1]: does every storage code on separated relations of bounded incidence degree have rate at most $\frac{1}{2} + o(1)$?

3.5 Storage codes: what is false

The answer is no.

- *Above the pairing rate.* There are relations of bounded (astronomically large) incidence degree in which every paired distance tends to infinity, with $\mathbb{F}_{127^2}$-linear storage codes of rate above 0.5021, and binary codes of rate above 0.5013; for large primes inert in $\mathbb{Q}(\sqrt{5})$ the rates approach $4 - 2\sqrt{3} \approx 0.536$ [13, Theorems 7.1 and 7.3, Corollary 8.2]. The relations are twisted quotients of a Bruhat–Tits tree, the codes are Tanner codes with Reed–Solomon local codes, and their dimension comes from Riemann–Roch on Hilbert modular surfaces in characteristic p. The undirected form of the question, on graphs of bounded degree with long odd cycles, has a negative answer as well [13, Corollary 9.3]. Tanner operators of the same kind, at $p = 3$, refute the independence of the chain in Jaikin-Zapirain's arbitrary-field form of the Lück approximation conjecture [8, Theorem 5.6].

- *Rates approaching one.* A linear-time transposer, even an adaptive one, would yield binary storage codes of this kind at every large degree cap Δ, with rates tending to one as Δ grows [23, Theorem 3.1]; and such codes exist abstractly, with rate at least $1 - C\Delta^{-\alpha}$ [23, Theorem 5.5]. So, at the scale of that transfer theorem, no rate bound for separated relations of bounded degree in general can exclude a transposer [23, Corollary 6.2]: a lower bound must use a further restriction imposed by the physical computation.

No tape process is known that realizes the recoveries of either family.

3.6 Adaptive machines

Much of the above already covers adaptive machines: the pass bound (at polynomial running time, or by the memo), Theorem M* along popular traces, and the reduction of (T) to the Tape Coding Conjecture. Beyond that the series has the following.

- Every linear-time transposer yields an external-memory transposition with $O(d)$ I/Os on blocks of one row that is oblivious on $2^{L-O(d)}$ inputs, so an external-memory lower bound without the indivisibility assumption would refute (T) [10, Theorem 6.4 and Corollary 6.5].

- The source-cover exclusions extend to machines of sufficiently low adaptivity [10, Theorem 6.1].

- Given the native half bound for the relations of skeleton classes, which is open, a linear-time transposer whose work heads enter every cell boundedly often must carry about $L/2$ bits in its coarse skeleton, hence in the coarse motion of its input head [10, Theorem 4.13 and Corollary 4.16]; the fourteenth paper removes the hypothesis on the work heads from the skeleton statement [5, Corollary 9.7].

	Paper	Main contribution to the question
1	Tape Coding Conjecture [9]	Formal statement; $\Omega(\log d)$ passes; Theorem M*; transposers are network codes
2	Local Coding Barriers [24]	Local coding bounds; affine and regular motion excluded; padding
3	Bit and Word [18]	Transposition of dyadic bit squares implies (T)
4	Curved Motion [6]	Curved motion; depth reduction plus crux; storage codes at rate $\frac{1}{2}$
5	Directed Recovery [22]	Paired-distance separation of cones; half bounds; the all-code question
6	Covers [21]	Half bounds along covers and for locally generated codes
7	Nested Elimination [11]	Exact all-code half bounds by elimination
8	Bounded Nesting [4]	Depth reduction at bounded nesting; the linear crux as rigidity
9	Source Covers [28]	Source-cover certificates and exclusions
10	Directed Reciprocity [16]	All-code bound on the class of Theorem S; deep nesting
11	Merge Depth [10]	Merge depth; adaptive machines; external memory
12	Lück Approximation [8]	A Tanner operator whose kernel survives along a congruence tower and dies along pro-3 towers
13	Above Half Rate [13]	The all-code question has a negative answer
14	Bounded-Stack DAGs [5]	Depth reduction for every number of walks
15	High-Rate Codes [23]	A transposer needs rates tending to one; such codes exist abstractly
16	Macroscopic Crossings [15]	Crossings, skeletons, rows to columns, one cut for all pairs

Table 1: The earlier papers of the series.

- The fifteenth paper extracts its codes from any linear-time transposer through a packet compilation [23, Theorem 3.1] and, at the original bit positions, from the traces of a bit simulator with few traces [23, Theorems 8.1 and 9.2]. The crossing theorems hold on heavy traces of the machine and of that simulator [15, Theorem 8.1]; whether they hold in the machine's own diagram on traces that carry almost all inputs was left open there [15, Problem 11.2]. A paper that followed this one proves a statement of that form at the canonical entry width: on the original run of all but an exponentially small share of the inputs that a linear-time transposer handles, every cut between the outputs of a large set and the macroscopic crossings has size proportional to that set [25, Theorem H.7]. For every entry width, and again outside an exponentially small share of the inputs, it proves that two work heads make linearly many macroscopic returns, and that cuts between a large set of outputs and the reads of their inputs are of linear size [25, Theorem 4.3 and Corollary 5.4].

4 Where the wall is

This section is a diagnosis. It contains no new mathematics. Its claims are consequences of the results cited in Section 3, and where it gives a reading instead of a consequence it says so.

4.1 What is left is one function

For oblivious machines the series has reduced (T) to a question about codes (Section 3.3). Fix the number of tapes, the alphabet and the time constant, and let $\varepsilon > 0$. Depth reduction gives, for every linear-time schedule, a genie that reveals at most εL bits and leaves cones of depth at most D_ε, a bound that depends on ε and on the fixed parameters but not on d. Write $r(\varepsilon)$ for the best rate, in the limit of large d, of a binary storage code on the cone relations that such genies leave on such schedules. Then:

- an oblivious linear-time transposer needs $r(\varepsilon) \geq 1 - \varepsilon$ for every $\varepsilon > 0$;
- a proof that $r(\varepsilon) < 1 - \varepsilon$ for one ε, for each choice of the fixed parameters, would exclude oblivious linear-time transposers [5, Corollary 9.3];
- the pairing gives $r(\varepsilon) \geq \frac{1}{2}$, and whether $r(\varepsilon) > \frac{1}{2}$ for any ε is open in both directions.

The depth bound matters: without it the cone of an output is everything read before it, and the question is empty. The only general upper bound is the elementary one for relations of degree at most Δ, a rate of $1 - 1/(\Delta + 1)$ [6, Proposition 4.14], which is useless at the depths in question. For machines with linear data paths the same reduction ends in the relative rigidity of the transposition matrix on ball patterns (Section 3.3). For adaptive machines the question arises in two forms: the Tape Coding Conjecture of the first paper, which needs no simulator, and the storage-code criteria on the traces of a bit simulator (Section 3.6). A later paper adds a third, on the machine's own runs, at the canonical entry width: for all but an exponentially small share of the inputs, and for every set B of steps of the run, the relation between entries that is left when B is cut out carries storage codes whose deficit is at most a small multiple of the input length plus a constant times $|B|$ [25, Theorem H.5 and Corollary H.6].

For particular classes of schedules the exclusion is proved. The sixteenth paper proves it for schedules whose large output sets can be cut cheaply from the macroscopic crossings, and for schedules with boundedly many macroscopic phases [15, Theorems 5.4, 9.2 and 9.3]; the source covers of the ninth paper exclude restricted geometries for all codes. What is missing is a statement for all schedules.

4.2 Three kinds of argument and their limits

Motion without data. The cut inequality behind Theorem M*, its entry-cell refinement and every conclusion of Theorem M* are satisfied by the padded machine of the second paper, which transposes nothing (Section 3.2). So conditions on head motion that do not refer to the outputs are necessary and cannot be decisive. Conditions that do refer to them go further: the rows-to-columns inequality of the sixteenth paper excludes two of the five candidate schedules. But one cut serves every pair of row sets, so this inequality gives a logarithm only when the cuts for different address bits are disjoint, which is a hypothesis and not a theorem; and what excludes all five candidates is the regular-recursion obstruction, which applies only to schedules all of whose stages have regular limiting motion. Whether conditions of this kind decide the oblivious case is open. A later paper proves conditions of this kind for the original runs of adaptive machines, and shows where they stop: a machine with the same number of tapes and a constant factor more time keeps, on every tape, archives from which the contents of every other tape can be computed, so that requirements on how much information the stores hold are met by redundancy [25, Theorems 4.3, 5.5 and 7.1]; the last of these has a written proof there.

Codes without machines. On abstract separated relations of bounded degree, storage codes exceed the pairing rate, and their rates tend to one polynomially fast as the degree cap grows (Section 3.5). A transposer is only shown to need rates that tend to one very slowly, and the cap grows as ε falls because a genie of cost εL has depth at least of order $1/\varepsilon$ [6, Proposition 4.13]. So no rate bound for separated relations of bounded degree in general can separate a transposer from an abstract code. The two properties of cone relations that the series' coding bounds use, bounded degree after trimming and paired-distance separation, are shared by the counterexamples.

Counting passes. The pass bound of the first paper is the one logarithm proved for a resource of every machine. It saturates one scale at a time: a linear-time run has room for about 2^j legs at scale $2^{-j}L$, while counting at a single scale asks for only $O(j)$ of them, and the threshold of the coarse-pass bound of the sixteenth paper is tight [15, Theorem 10.3 and Corollary 10.4]. That paper states the problem of an inequality in the numbers of legs at all scales at once [15, Problem 11.1]; nothing in the series proves one.

4.3 Why every general tool saturates

The $\log d$ of the folklore algorithm is a sum over $\log_2 d$ address digits. Exchanging one digit of the row address with one digit of the column address costs $O(L)$, and the transposition is the product of these $\log_2 d$ commuting exchanges. Linear time pays for any one of them in full, and for any bounded number.

Every tool of the series *that applies to all machines* is single-scale or scale-free. Cut arguments, macroscopic limits and coarse passes look at one scale at a time, and a linear-time machine has enough steps to satisfy any single-scale requirement. Storage-code bounds are scale-free: they see a relation, not the scale of the addresses that produced it. The one attempt to make a geometric quantity grow with the number of scales failed: walks nested at m scales are cut to a depth independent of m (Section 3.3).

The series does charge scales separately in restricted settings, and they show what the general case lacks.

- For transport at bounded occupancy, each moved label is charged its $\log d$ address bits, in steps [22, Lemma 3.4]. What the bound does not cover is coding.
- For traces whose cuts for different address bits are disjoint, the rows-to-columns inequality gives $\Omega(L \log d)$ against every code [15, Corollary 7.3]. What is not available is the disjointness.
- In the abstract model of XOR-frame recursion every level is charged [9], and compilers with boundedly many stages meet a multiscale obstruction [18]. Both are restricted models.

So the logarithm is proved wherever either the coding or the geometry is tied down, and nowhere else. Corollary 6.13 and Remark 6.14 below make one half of this precise in the linear one-hop model: every address digit must be treated asymmetrically, but one cheap tape operation does that for all digits at once.

4.4 What the missing statement must be

The limits above describe the statement that would decide the oblivious case. It must

- tie what is known to where it can be, for all schedules: the class results of Section 4.1 do this for restricted geometries;
- fail for abstract separated relations, where codes of rate tending to one exist;
- fail for motion that ignores the data, which padding supplies;
- constrain all scales at once;
- and use the shape of tapes, since a bound on coding gains for bounded-degree graphs in general would be a circuit lower bound (Section 3.1).

Since a linear-time multiplier gives a linear-time transposer [41], a proof of such a statement for all machines would also be the first superlinear lower bound for off-line multiplication on multitape machines.

4.5 What a construction cannot avoid

The same results say what a linear-time machine would have to use.

- *No bounded complex coefficients.* Linear recovery with bounded complex coefficients and bounded width stays at the pairing rate on cone relations (Section 3.4). A linear code would need positive characteristic or coefficients that cannot be rescaled to be bounded; nonlinear codes are not covered.
- *Curved motion.* In the limit, head motion must fail to be locally affine on a positive fraction of the time (Section 3.2), and a recursive schedule must have a stage whose limiting motion is not regular.
- *Expanding relations.* The trimmed incidence graphs of the cone relations must not be uniformly hyperfinite (Section 3.4). This is necessary and cheap: a fixed linear-time machine has such cones and still obeys the all-code half bound [22, Corollary 7.9].
- *No gadgets.* After trimming, almost no output is within bounded distance of its own input bit, so no gain above the pairing rate can come from gadgets of bounded size (Section 3.4). The pairing itself is not excluded; it stops at $\frac{1}{2}$.

Above the pairing rate, at a degree cap that is fixed before the length and the paired distances grow, the only codes known come from one source: sections of line bundles on the supersingular loci of Hilbert modular surfaces [13], by a method that originates in [8], together with their amplifications [23]. At degrees that grow with the length other examples exist.

4.6 The rule and the relation

One feature of machines is absent from the coding statements above. A storage code on a relation lets each receiver choose its own decoding function. A Turing machine has one finite transition table, and on a stretch of time during which its heads move in step, the function that produces the output at one position is the function that produces the output at the next position, translated. The cone relation records which inputs each output can depend on. It does not record that neighbouring outputs depend on them *in the same way*.

Sharing decoders is not by itself an obstruction to rate above one half: by their construction, the linear codes of the thirteenth paper are equivariant under large groups with boundedly many orbits of coordinates. But transposition has a feature that an abstract relation lacks: its demand is not invariant under translating one index. Shifting the columns of a matrix shifts the rows of its transpose. The two theorems of this paper are the two cleanest ways in which structure of the decoder on one index side is incompatible with the swap: *order* (Section 5) and *symmetry* (Section 6). They are proved in the linear one-hop model of Section 2.3. They are not theorems about machines, and Section 7 says what would be needed to make them so.

5 Order: the flag bound

Notation is that of Section 2.3: F is a field, $V = F^d$, τ is the swap, $K = \sum_a A_a \otimes B_a$ with finitely many terms, and the swap code is $\ker(\tau - K) = \{X : X^{\mathsf{T}} = \sum_a A_a X B_a^{\mathsf{T}}\}$. A *complete flag* is a chain of subspaces $W_1 \subset W_2 \subset \cdots \subset W_d = V$ with $\dim W_i = i$.

5.1 The theorem

Theorem 5.1 (Theorem FL, the flag bound)**.** Lean (partial): `finrank_swapCode_le_of_flag`, `finrank_swapCode_le_of_staircase`, `finrank_swapCode_le_of_flag_left`, `finrank_swapCode_swap` [`Flag.lean`]

Not formalized: the choice of bases adapted to the two flags (step 1 of the proof). The Lean statement is the resulting coordinate form: if P, Q are invertible and every $P^{-1}B_aQ$ has its column b' supported on the first $g(b'+1)$ rows, the bound holds.

Let $W_1 \subset \cdots \subset W_d = V$ be a complete flag and put $g(i) = \dim \sum_a B_a(W_i)$. Then, whatever the A_a are,

$$\dim \ker(\tau - K) \;\le\; \sum_{i=1}^{d} g(i).$$

The same holds with the roles of the A_a and the B_a exchanged.

Proof. Step 1: bases. The subspaces $U_i = \sum_a B_a(W_i)$ are nested and $\dim U_i = g(i)$. Choose a basis $f_0, \dots, f_{d-1}$ of V with $W_i = \operatorname{span}(f_0, \dots, f_{i-1})$, and a basis $f'_0, \dots, f'_{d-1}$ with $U_i \subseteq \operatorname{span}(f'_0, \dots, f'_{g(i)-1})$ for every i; such a basis exists because the U_i are nested: extend a basis of U_1 through $U_2, \dots, U_d$ and then to V. For a term $A \otimes B$ of K write $[B]_{b,b'}$ for the coefficient of f'_b in $Bf_{b'}$ and $[A]_{a,a'}$ for the coefficient of f_a in $Af'_{a'}$. Since $Bf_{b'} \in U_{b'+1}$, the coefficient $[B]_{b,b'}$ vanishes unless $b < g(b'+1)$.

Step 2: the equations. Write $X = \sum_{a,b} p_{ab}\, f'_a \otimes f_b$. Then $\tau X = \sum p_{ab}\, f_b \otimes f'_a$, whose coefficient at $f_a \otimes f'_b$ is p_{ba}, and the coefficient of KX at $f_a \otimes f'_b$ is $\sum_{\text{terms}} \sum_{a',b'} [A]_{a,a'} [B]_{b,b'}\, p_{a'b'}$. So $\tau X = KX$ says, for all a, b:

$$p_{ba} \;=\; \text{a linear combination of the } p_{a'b'} \text{ with } g(b'+1) > b. \tag{1}$$

Step 3: elimination. Let $I = \{(a', b') : a' < g(b'+1)\}$, so $|I| = \sum_i g(i)$. We claim that a solution p that vanishes on I is zero; then restriction to I is injective on the code and the bound follows. Suppose p vanishes on I, and prove that row b of p vanishes, by downward induction on $b = d-1, \dots, 0$. By (1), p_{ba} is a combination of entries $p_{a'b'}$ with $g(b'+1) > b$. If $a' > b$, row a' vanishes by induction. If $a' \le b$, then $a' \le b < g(b'+1)$, so $(a', b') \in I$ and $p_{a'b'} = 0$. Hence $p_{ba} = 0$.

Step 4: exchange. Since $\tau(\tau - \sum_a A_a \otimes B_a)\tau = \tau - \sum_a B_a \otimes A_a$, the two kernels have the same dimension. □

Outline of the formal proof of the coordinate form. In Lean the columns are numbered from 0 and the function g is shifted accordingly: Lean's $g(b')$ is the $g(b'+1)$ of the statement above. Let Q be the matrix whose columns are the f_j and P the matrix whose columns are the f'_j; then $(P^{-1}B_aQ)_{b,b'}$ is the coefficient $[B_a]_{b,b'}$. The map $X \mapsto P^{-1}X(Q^{-1})^{\mathsf{T}}$ is an isomorphism from the code of (A_a, B_a) onto the code of $(Q^{-1}A_aP,\, P^{-1}B_aQ)$ (`swapCodeEquiv`, `finrank_swapCode_conj`), which reduces the statement to matrices B_a whose column b' vanishes from row $g(b')$ on. For these, a solution that vanishes on $\{(a', b') : a' < g(b')\}$ is zero, by strong induction on $d - 1 - b$ (`eq_zero_of_mem_swapCode_of_staircase`), so restriction to that set is injective (`staircaseRestrict_injective`), and the set has $\sum_{b'} \min(g(b'), d)$ elements (`card_staircase`). The exchange of sides is `finrank_swapCode_swap`. No monotonicity of g is used. □

The bound depends on one tensor side only, and on that side only through how fast the images of a flag grow: we call g the *expansion profile* of the B_a along the flag. The transposed profiles are also available, because $\operatorname{rank}(\tau - K) = \operatorname{rank}(\tau - K^{\mathsf{T}})$ and $K^{\mathsf{T}} = \sum_a A_a^{\mathsf{T}} \otimes B_a^{\mathsf{T}}$.

Proposition 5.2 (The bound is attained for every profile)**.** Lean: `finrank_swapCode_staircase`, `exists_staircase_finrank_eq`, `mem_swapCode_staircase_iff` [`Flag.lean`]

Let $g\colon \{1,\dots,d\} \to \{0,\dots,d\}$ be any function. Take one term for every pair (i,k) with $k < g(i+1)$, namely $A = E_{ik}$ and $B = E_{ki}$, where E_{ik} is the matrix unit. Then column i of every B_a is supported on the first $g(i+1)$ rows, and the swap code is $\{X : X_{ki} = 0$ unless $k < g(i+1)\}$, of dimension exactly $\sum_i g(i)$.

Outline of the formal proof. Here $KX = \sum X_{ki}E_{ik}$ over the chosen pairs, so $X^{\mathsf{T}} = KX$ holds at an entry (i,k) with $k < g(i+1)$ and forces $X_{ki} = 0$ at the others (`mem_swapCode_staircase_iff`). □

Remark 5.3 (Numerical: the bound is not always tight)**.** For a fixed span of second factors the best flag need not give the true maximum over the first factors. In an exhaustive computation over $\mathbb{F}_2$ at $d = 3$, for 4 of the 20 orbits of spans of at most two second factors the maximum is 4 while the best flag, over $\mathbb{F}_2$, $\mathbb{F}_4$ or $\mathbb{F}_8$, gives 5 or 6.

5.2 Corollaries

Write $h = d(d-1)/2$, so that $\dim\ker(\tau - K) \le d(d+1)/2$ is the same as $\operatorname{rank}(\tau - K) \ge h$.

Corollary 5.4 (A pair of flags)**.** Lean (partial): `two_mul_finrank_swapCode_le_of_triangular`, `two_mul_finrank_swapCode_le_of_triangular_left` [`Flag.lean`]

Not formalized: the passage from a pair of flags to bases in which the B_a are upper triangular, as in step 1 of the proof of Theorem 5.1. The Lean statement is the matrix form.

If $P^{-1}B_aQ$ is upper triangular for all a, for some invertible P and Q, then $\dim\ker(\tau - K) \le d(d+1)/2$, for arbitrary A_a. In terms of flags: this holds if there are complete flags W and W' with $B_a(W_i) \subseteq W'_i$ for all a and i.

Outline of the formal proof. This is Theorem 5.1 with $g(i) \le i$, and $\sum_{i \le d} i = d(d+1)/2$ (`two_mul_sum_fin_succ`). Bases adapted to W and W' make the B_a upper triangular, and conversely. □

So the pairing value is the value of a hidden order: a pair of flags is an order in which the t-th generalized column is computed from the first t generalized rows.

Corollary 5.5 (At most two terms, every field)**.** Not formalized; a complete proof is given below.

For all $A_1, A_2, B_1, B_2 \in \mathrm{M}_d(F)$,

$$\dim\{X : X^{\mathsf{T}} = A_1XB_1^{\mathsf{T}} + A_2XB_2^{\mathsf{T}}\} \;\le\; \frac{d(d+1)}{2}.$$

Proof (not formalized). The dimension of a kernel does not change under extension of the field, so let F be algebraically closed. We show that any two linear maps $B_1, B_2\colon V \to V'$ between spaces of the same dimension d admit complete flags with $B_j(W_i) \subseteq W'_i$; Corollary 5.4 then applies. The determinant $\det(\alpha B_1 + \beta B_2)$, in any bases, is a binary form of degree d, so it vanishes at some $(\alpha,\beta) \neq 0$ (at every (α,β) if it is identically zero). Take $v \neq 0$ with $(\alpha B_1 + \beta B_2)v = 0$. Then B_1v and B_2v are proportional; let $W_1 = \langle v\rangle$ and let W'_1 be a line containing both. The maps induced on $V/W_1 \to V'/W'_1$ are again two maps between spaces of equal dimension, and induction on d gives the flags. □

Remark 5.6 (Two is sharp)**.** Three terms can exceed the pairing value. For every d, the operator $\Pi = \sum_{i,j} E_{ij} \otimes E_{ij}$ is $X \mapsto \operatorname{tr}(X) \cdot 1$, of rank one, and

$$\tau - \Pi \;=\; \sum_{i<j} (E_{ij} - E_{ji}) \otimes (E_{ji} - E_{ij})$$

has h terms. So $K = \tau - \Pi$ has h terms and its code $\ker \Pi$, the matrices of trace zero, has dimension $d^2 - 1$, in every characteristic. At $d = 3$ that is three terms and dimension $8 > 6$. At $d = 4$, adding E_{33} to both factors of the third term of the $d = 3$ example, with indices $0, \ldots, 3$, gives three terms and a code of dimension $11 > 10$ (numerical: checked over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$ by `tools/three_term_example.py`). We have no three-term example for $d \geq 5$.

Corollary 5.7 (One triangularizable side)**.** Not formalized; a complete proof is given below.

Suppose that, over the algebraic closure of F, the B_a have a common invariant complete flag. Then $\dim \ker(\tau - K) \leq d(d+1)/2$ *for any number of terms and arbitrary A_a. This holds in particular when*

(a) the B_a commute pairwise, for example when all are circulant;

(b) $\operatorname{char} F = p$ *and all B_a lie in the linear span of one finite p-group of matrices.*

Proof (not formalized). An invariant flag gives Corollary 5.4 with $W' = W$, over the closure, and the dimension does not depend on the field. (a) Commuting matrices over an algebraically closed field have a common eigenvector: an eigenspace of one of them is invariant under the others, and one inducts on the dimension. Passing to the quotient by that eigenvector and inducting again gives a common invariant flag. (b) Let G be a finite p-group of invertible matrices in characteristic p and $z \neq 1$ a central element of order p. Then $(z-1)^p = z^p - 1 = 0$, so $\ker(z-1)$ is a nonzero G-invariant subspace on which $G/\langle z \rangle$ acts; by induction on the order of the group, G fixes a nonzero vector. Passing to the quotient by that vector and inducting on the dimension gives a complete flag invariant under G, hence under its span. □

For circulant B_a the statement reads: *if every column of X is a linear function of the rows of X that commutes with cyclic rotation of the rows' coordinates (any number of rows, each rotated and convolved with kernels that may depend on the column), then X ranges over a space of dimension at most $d(d+1)/2$.* Reading any number of source rows in step with the output cannot beat the symmetric matrices.

In (b) the group must be the same for all B_a. For $p = 2$ and $d = 2^m$ two natural examples are the automorphisms of the hierarchy of dyadic intervals (block swaps, block reversals, XOR translations) and the affine group $\{i \mapsto ui + t : u \text{ odd}\}$ of $\mathbb{Z}/2^m$. They are different subgroups of the symmetric group, and mixing them is not covered: at $d = 4$ a rotation and the transposition (0 1), which swaps the halves of one dyadic block of length two, generate the whole symmetric group. The next corollary covers more of the second example.

Corollary 5.8 (All affine index maps in characteristic p)**.** Not formalized; a complete proof is given below.

Let $\operatorname{char} F = p$ *and $d = p^m$. For $u, t \in \mathbb{Z}/d$ let $\varphi_{u,t}$ be the linear map on $F^{\mathbb{Z}/d}$ with $(\varphi_{u,t} x)_i = x_{ui+t}$; here u need not be a unit. If every B_a lies in the span of the $\varphi_{u,t}$, then* $\dim \ker(\tau - K) \leq d(d+1)/2$.

Proof (not formalized). Let $R = \varphi_{1,1}$ be the rotation. Then $R\varphi_{u,t} = \varphi_{u,t} R^u$, since both send x to $(x_{ui+t+u})_i$. Hence $(R-1)^i \varphi_{u,t} = \varphi_{u,t}(R^u - 1)^i$, and $R^u - 1$ is a multiple of $R - 1$ in the

commutative algebra $F[R]$, so $\varphi_{u,t}$ maps $\ker(R-1)^i$ into itself. As R is a single cycle of length p^m in characteristic p, $R-1$ is nilpotent with a one-dimensional kernel, and the subspaces $\ker(R-1)^i$, $1 \le i \le d$, form a complete flag. Apply Corollary 5.4 with $W' = W$. □

So over $\mathbb{F}_2$, on rows of length 2^m, rotations, reflections, dilations, wrap-around decimations and convolutions all preserve one flag, in any combination and at any depth (Remark 5.11). What leaves that world is a projection: a mask that selects part of a row.

Corollary 5.9 (Blocks and expansion)**.** Not formalized; a complete proof is given below.

(a) If, for suitable invertible P, Q, all $P^{-1}B_aQ$ are block upper triangular with diagonal blocks of sizes $d_1, \dots, d_m$, then $\dim\ker(\tau - K) \le (d^2 + \sum_s d_s^2)/2$.

(b) If some flag has $g(i) \le ci + c_0$ with $c \ge 1$ and $c_0 \ge 0$, then $\dim\ker(\tau - K) \le (1 - \frac{1}{2c})d^2 + (\frac{c_0}{c} + \frac{1}{2})d$.

(c) A sum of k terms has $\dim\ker(\tau - K) \le \sum_{i \le d} \min(d, ki) \le (1 - \frac{1}{2k})d^2 + \frac{d}{2}$.

(d) Let d be prime to $\operatorname{char} F$ *and $s \ge 1$. If every B_a lies in the span of the maps $\varphi_{u,t}$ of Corollary 5.8 with $u \in \{-s, \dots, s\}$, then* $\dim\ker(\tau - K) \le (1 - \frac{1}{2s})d^2 + (\frac{1}{s} + \frac{1}{2})d$.

Proof (not formalized). (a) Take $g(i) = d_1 + \cdots + d_s$ for i in block s; then $\sum_i g(i) = \sum_s d_s(d_1 + \cdots + d_s) = (d^2 + \sum_s d_s^2)/2$. (b) If $d < c_0$ the right side exceeds d^2. Otherwise put $u = (d - c_0)/c$. Then $\sum_i \min(d, ci + c_0) = d^2 - \sum_{i \le u} c(u - i) \le d^2 - \frac{c}{2}(u^2 - u)$, because $\sum_{i \le \lfloor u \rfloor}(u - i) \ge (u^2 - u)/2$, and $d^2 - \frac{(d-c_0)^2}{2c} + \frac{d-c_0}{2}$ is at most the claimed bound. (c) Any flag has $g(i) \le ki$; apply (b). (d) Extend the field so that it contains a primitive d-th root of unity, and let χ_k be the characters of $\mathbb{Z}/d$. Then $\varphi_{u,t}\chi_k$ is a multiple of χ_{uk}. Order the frequencies $0, 1, -1, 2, -2, \dots$ and let W_i be spanned by the first i characters. The frequencies uk with $|u| \le s$ and $\chi_k \in W_i$ lie in an interval of at most $si + 1$ residues, so $g(i) \le si + 1$; apply (b). □

The hypothesis $c \ge 1$ in (b) is needed: for $B = \operatorname{diag}(1_{d/2}, 0)$, $A = 1$ and a flag that alternates kernel and non-kernel vectors, $g(i) = \lfloor i/2 \rfloor$ and the code has dimension $d(d+2)/8$.

Remark 5.10 (Entries of b bits)**.** For $K = \sum_a \tilde{A}_a \otimes B_a$ on $V \otimes V \otimes F^b$ with $\tilde{A}_a \in \operatorname{End}(V \otimes F^b)$ arbitrary and $B_a \in \operatorname{End}(V)$, the proof of Theorem 5.1 gives $\dim\ker(\tau \otimes 1 - K) \le b \sum_i g(i)$, with coefficients $p_{ab} \in F^b$. Operators that mix the column index with the position inside an entry are covered as well: every operator on $V \otimes F^b$ is a sum $\sum_j B_j \otimes C_j$, the C_j can be absorbed into the first factors, and the profile is that of the B_j. For a cyclic shift of the bit string of a row by a number of bits that is not a multiple of b, the B_j are two rotations.

Remark 5.11 (Composition)**.** If the second factors of $K_1, \dots, K_t$ all lie in one algebra $\mathfrak{B} \subseteq \mathrm{M}_d(F)$, so do those of the product $K_t \cdots K_1$, because $(A \otimes B)(A' \otimes B') = AA' \otimes BB'$. So Corollaries 5.7 and 5.8 are statements about linear data paths of any depth: a path that acts on one index through a commutative algebra, or through the affine maps of Corollary 5.8, and on the other index in any way at all, is correct on at most the pairing value.

5.3 What the theorem says

- *Only the weaker side counts.* The bound uses the expansion of the A_a or of the B_a, whichever is smaller. Arbitrary mixing on one index buys nothing while the other index has a flag.

- *A code above the pairing value needs expansion on both sides*: for every flag on either side, the images of the flag must grow faster than the flag. By Corollary 5.9(b), a profile $g(i) \le ci$ with $c \ge 1$ gives rate at most $1 - 1/(2c) + O(1/d)$, so rate $1 - \varepsilon$ needs expansion factor about $1/(2\varepsilon)$ along every flag, on both sides.

- *Expansion is cheap on a tape.* Over the algebraic closure, rotations of a row have an invariant flag, and so do masks by intervals, but the two flags are incompatible: rotations together with the mask of one proper interval generate the full matrix algebra. For rotations with the masks of M intervals J_l, a rotation-invariant flag has $g(i) = \sum_l \min(i, |J_l|) \le Mi$, and no flag does better, because the rotations of W_i span a rotation-invariant space of dimension at least i, and such a space of dimension h restricts to an interval of length m in dimension $\min(h, m)$. For M equal intervals this gives rate $1 - 1/(2M)$ and no more. So the flag bound cannot hold that family to the pairing rate. The next section does.

6 Symmetry: the translate pigeonhole

The flag bound cannot hold rotations with interval masks to the pairing rate (Section 5.3). This section does, by a different argument: a code larger than half the space meets its own translate, and on the intersection a shared decoder forces the data to be translation-covariant.

6.1 The lemma

For operators S, T write $[S, T] = ST - TS$.

Lemma 6.1 (Pigeonhole against a translate)**.** Lean: `two_mul_finrank_ker_sub_le_ker`, `finrank_ker_commutator_sub_le`, `two_mul_finrank_ker_sub_le`, `two_mul_finrank_ker_sub_le_of_commute`, `finrank_range_commutator_le` [`Pigeonhole.lean`]

Let τ, K, T be linear operators on a space of finite dimension N over a field. Then

$$\dim\ker(\tau - K) \;\le\; \tfrac{1}{2}(N + \dim\ker[\tau - K, T]) \;\le\; \tfrac{1}{2}(N + \dim\ker[\tau, T] + \operatorname{rank}[K, T]).$$

Proof. Let $\mathcal{X} = \ker(\tau - K)$, $D = \dim \mathcal{X}$ and $\mathcal{X}_T = \{X \in \mathcal{X} : TX \in \mathcal{X}\}$. The map $\mathcal{X} \to F^N/\mathcal{X}$, $X \mapsto TX + \mathcal{X}$, has kernel $\mathcal{X}_T$ and a target of dimension $N - D$, so $\dim \mathcal{X}_T \ge 2D - N$; T need not be invertible. For $X \in \mathcal{X}_T$, $(\tau - K)TX = 0 = T(\tau - K)X$, so $\mathcal{X}_T \subseteq \ker[\tau - K, T]$. That is the first inequality. For the second, $[\tau - K, T]X = 0$ means $[\tau, T]X = [K, T]X \in \operatorname{im}[K, T]$, and the preimage of a subspace W under a map has dimension at most that of its kernel plus $\dim W$. □

With $S = \tau - K$ the first inequality says $\operatorname{rank}[S, T] \le 2 \operatorname{rank} S$: a commutator has rank at most twice the rank of either factor. The Lean proof takes this route, from $\operatorname{rank}[S, T] \le \operatorname{rank}(ST) + \operatorname{rank}(TS)$ and rank–nullity (`finrank_range_commutator_le`). The lemma is elementary, and we claim no novelty for it. What is used below is the form with τ and K separated: K may nearly commute with a T with which τ does not.

Proposition 6.2 (For the swap, the best case is the pairing value)**.** Lean (partial): `card_le_finrank_ker_swap_commutator`, `two_mul_finrank_ker_swap_sub_one`, `two_mul_finrank_range_swap_sub_one` [`Swap.lean`]

Not formalized: the equality case for $T = 1 \otimes G$.

On $V \otimes V$ with $\dim V = d$, every operator T has $\dim\ker[\tau, T] \ge d$, in every characteristic. Equality holds for $T = 1 \otimes G$, that is $X \mapsto XG^{\mathsf{T}}$, when G is non-derogatory.

Proof. The kernel of $\tau - 1$ is the space of symmetric matrices, of dimension $d(d+1)/2$ in every characteristic, so $\operatorname{rank}(\tau - 1) = d(d-1)/2$, and $[\tau, T] = [\tau - 1, T]$ has rank at most $d(d-1)$ by Lemma 6.1. For $T = 1 \otimes G$, $[\tau, T]X = GX^{\mathsf{T}} - X^{\mathsf{T}}G^{\mathsf{T}}$, so the kernel is the space of $Y = X^{\mathsf{T}}$ with $GY = YG^{\mathsf{T}}$. As G^{T} is similar to G, this space is isomorphic to the centralizer of G, whose dimension is d exactly when G is non-derogatory. □

So Lemma 6.1 never gives less than $(d^2 + d)/2$ for the swap, and it gives exactly the pairing value when K commutes with a suitable T. The rest of the section is about K that commute with a translation up to a boundary.

6.2 Class-uniform rules

Let σ be the cyclic rotation of $F^{\mathbb{Z}/d}$, $(\sigma x)_j = x_{j+1}$. For a matrix X with rows $x_c \in F^{\mathbb{Z}/d}$, $c \in \mathbb{Z}/d$, write $\sigma^\delta X$ for the matrix with rows $\sigma^\delta x_c$.

Definition 6.3 (Class-uniform rules)**.** Lean: `rot`, `IsClassUniformCode` [`ClassUniform.lean`]
Fix a partition of the row index $\mathbb{Z}/d$ into non-empty classes. A *class-uniform rule* is a family of linear maps $\mathcal{A}_J \colon \mathrm{M}_d(F) \to F^{\mathbb{Z}/d}$, one for each class J. A subspace $\mathcal{X} \subseteq \mathrm{M}_d(F)$ is a *code* of the rule if

$$x_c = \mathcal{A}_J(\sigma^c X) \qquad \text{for every } X \in \mathcal{X}, \text{ every class } J \text{ and every } c \in J.$$

In words: row c of X is a fixed linear function of the whole matrix with its columns rotated by c, and the function depends only on the class of c. The receiver of row c looks at the stored word from its own position, through a window of any size, and receivers in one class use the same decoder.

In the notation of Section 2.3 (this translation is not formalized): let $K = \sum_a A_a \otimes B_a$ have second factors in the span of the matrices $P_J\sigma^e$, a rotation followed by the mask P_J of a class, and arbitrary first factors, say $K = \sum_{J,e} A_{J,e} \otimes P_J\sigma^e$. Then $(KX)_{r,c} = \sum_{i,e}(A_{J,e})_{r,i}X_{i,c+e}$ for $c \in J$, so the swap code of K is the largest code of the rule $\mathcal{A}_J(Y)_r = \sum_{i,e}(A_{J,e})_{r,i}Y_{i,e}$, and every rule arises in this way. This is the family of Section 5.3. For a single class the B_a are circulant and Corollary 5.7 applies. If the mask is applied before the rotation, K^{T} has second factors $P_J\sigma^{-e}$, and since $\operatorname{rank}(\tau - K) = \operatorname{rank}(\tau - K^{\mathsf{T}})$ the bounds below hold for that family as well.

Theorem 6.4 (Theorem TP, the translate pigeonhole)**.** Lean: `IsClassUniformCode.finrank_le`, `Link`, `IsClassUniformCode.two_mul_finrank_le`, `IsClassUniformCode.two_mul_finrank_le_of_const` [`ClassUniform.lean`]
Let $\mathcal{X}$ be a code of a class-uniform rule. Let $\Delta_0 \subseteq \mathbb{Z}/d$ be a set of $s + 1 \geq 2$ shifts, and let χ be the number of connected components of the graph on $\mathbb{Z}/d$ in which $c \sim c'$ when c and c' lie in one class and $c' - c \in \Delta_0 - \Delta_0$. Then

$$\dim \mathcal{X} \;\leq\; \frac{s\,d^2 + d\,\chi}{s+1}.$$

In particular, if the classes are M cyclic intervals, then $\dim \mathcal{X} \leq (d^2 + Md)/2$.

Outline of the formal proof. Let $\mathcal{X}' = \{X : \sigma^\delta X \in \mathcal{X} \text{ for all } \delta \in \Delta_0\}$, an intersection of $s+1$ preimages of $\mathcal{X}$; each has codimension at most $d^2 - \dim \mathcal{X}$, so $\dim \mathcal{X}' \geq (s+1)\dim \mathcal{X} - s\,d^2$ (`card_mul_finrank_add_le_iInf_comap`, `mul_finrank_le_add_finrank_iInf_comap`, in `Pigeonhole.lean`). Let $X \in \mathcal{X}'$, $\delta, \delta' \in \Delta_0$ and c, c' in one class J with $c + \delta = c' + \delta'$. The rule applied to $\sigma^\delta X$ at row c gives $\sigma^\delta x_c = \mathcal{A}_J(\sigma^{c+\delta}X)$, and applied to $\sigma^{\delta'}X$ at row c' it gives $\sigma^{\delta'}x_{c'} = \mathcal{A}_J(\sigma^{c'+\delta'}X)$. The right sides agree, so $x_{c'} = \sigma^{c'-c}x_c$ (`IsClassUniformCode.row_eq_of_rot_mem`). Hence each row of X determines the rows in its component, restriction to one row per component is injective on $\mathcal{X}'$, and $\dim \mathcal{X}' \leq d\chi$. The Lean statement takes any set S of rows that meets every component and gives

$(s+1)\dim\mathcal{X} \le s\,d^2 + d|S|$. For M cyclic intervals and $\Delta_0 = \{0,1\}$ the components are the intervals. In Lean this case (`IsClassUniformCode.two_mul_finrank_le`) takes a non-empty set S containing every row whose class differs from that of the row before it, so that $|S| = M$ for $M \ge 2$ intervals and S is any single row for one class (`IsClassUniformCode.two_mul_finrank_le_of_const`). □

The space on which the proof lands is worth naming. For one unit shift it is $\mathcal{H} = \{X : x_{c+1} = \sigma x_c$ whenever $c, c+1$ lie in one class$\}$: inside each interval the matrix is constant along anti-diagonals. It has d free parameters per interval, whatever the length of the interval.

We say that a partition into M classes has an *almost-period* if some single shift δ leaves only M components in the graph of Theorem 6.4 with $\Delta_0 = \{0, \delta\}$, so that the bound is $(d^2 + Md)/2$. Intervals have the almost-period 1, and residue classes modulo M have the almost-period M.

Lemma 6.5 (Extremal codes contain $\mathcal{H}$)**.** Not formalized; a complete proof is given below.

Let the classes be M cyclic intervals and let $\mathcal{X}$ be a code of a class-uniform rule with $2\dim\mathcal{X} = d^2 + Md$. Then $\mathcal{X} \cap \sigma^{-1}\mathcal{X} = \mathcal{H}$; in particular $\mathcal{H} \subseteq \mathcal{X}$.

Proof (not formalized). By the proof of Theorem 6.4, $\mathcal{X}' = \mathcal{X} \cap \sigma^{-1}\mathcal{X}$ lies in $\mathcal{H}$ and has dimension at least $2\dim\mathcal{X} - d^2 = Md = \dim\mathcal{H}$. □

Proposition 6.6 (Equal pieces: the bound is attained)**.** Lean: `blockSymmetric`, `isClassUniformCode_blockSymmetric`, `two_mul_finrank_blockSymmetric`, `blockSymmetric_attains_bound` [`BlockSymmetric.lean`]

Let $d = Mm$ with $M, m \ge 1$ and let the classes be the M intervals of length m. The block-symmetric *matrices, those whose $m \times m$ blocks in the $M \times M$ grid are all symmetric, are a code of a class-uniform rule, of dimension $(d^2 + Md)/2$.*

Outline of the formal proof. Write a row index as $c = sm + k$ and a column index as $r = tm + l$ with $0 \le k, l < m$. Block symmetry says $X_{sm+k,\,tm+l} = X_{sm+l,\,tm+k}$. The right side is the entry of $\sigma^c X$ in row $sm + l$ and column $tm + k - c = (t - s)m$, which depends on c only through its block s. So $x_c = \mathcal{A}_s(\sigma^c X)$ with $\mathcal{A}_s(Y)_{tm+l} = Y_{sm+l,\,(t-s)m}$ (`isClassUniformCode_blockSymmetric`). There are M^2 blocks, each with $m(m+1)/2$ free entries, and $M^2 m(m+1)/2 = (d^2 + Md)/2$ (`two_mul_finrank_blockSymmetric`); the set of the M first rows of the blocks satisfies the hypothesis of Theorem 6.4 for the unit shift (`blockSymmetric_attains_bound`). □

The operator K of this rule is the coarse transposition of the grid: $X \mapsto (\mathcal{A}_{s(c)}(\sigma^c X))_c$, read as a matrix with these columns, has the entry $X_{sm+l,\,tm+k}$ in position $(tm + l,\ sm + k)$, so block (s, t) is moved to position (t, s) with its inner coordinates kept.

So the largest code of a class-uniform rule with M equal interval pieces has dimension exactly $(d^2 + Md)/2$, the value of the $M \times M$ coarse transposition, against the bound $1 - 1/(2M)$ that flags give for the same family. For $M = d$ the pieces are single rows, the rule is unrestricted and the code is everything. In between, each additional piece raises the bound by $d/2$.

Remark 6.7 (Numerical: what the exact value does not say)**.** Exhaustive computations over $\mathbb{F}_2$ show the following. (a) Extremal codes are far from unique: for $d = 4$ and classes $\{0,1\}, \{2,3\}$ there are 59 392 codes of dimension 12, in 327 orbits under the evident symmetries, and the block-symmetric code is one orbit. (b) Coding beats copying inside the bound: for $d = 4$ and pieces $\{0\}, \{1,2,3\}$ the bound 12 is attained, but no rule that copies one stored entry to each output entry exceeds 11. (c) The bound is not attained for every partition into unequal intervals: by Lemma 6.5 and a solver, the pieces $\{0\}, \{1\}, \{2,3,4\}$ at $d = 5$ have largest code $19 < 20$ (Remark 8.1).

Proposition 6.8 (Copy rules, every partition)**.** Not formalized; a complete proof is given below.

Call a class-uniform rule a copy rule *if every output entry is a multiple of one stored entry:* $X_{c,r} = \lambda_{J,r}\, X_{\pi_J(r),\,\rho_J(r)+c}$ *for* $c \in J$*. A code of a copy rule with* M *classes, of any shape, has dimension at most* $(d^2 + Md)/2$*.*

Proof (not formalized). Let $f(c,r) = (\pi_J(r), \rho_J(r) + c)$ for $c \in J$, a map on the d^2 positions. A matrix in the code is determined, on each component of the functional graph of f, by its value at one position on the cycle of that component, so the dimension is at most the number of cycles. A fixed point (c,r) has $\pi_J(r) = c$ with J the class of c, so for each r there is at most one fixed point per class: at most Md in all. Every other cycle has at least two positions. □

So for copy rules the shape of the classes does not matter, while for coded rules Theorem 6.4 needs the classes to have an almost-period. The difference is real: Proposition 8.2 exhibits two classes with no almost-period and a coded rule whose code has dimension greater than $(d^2 + Md)/2$, and Section 6.6 gives, for every fixed number of classes, partitions whose codes exceed it by an unbounded amount. How far coded rules on such classes can go is Problem 9.1.

6.3 On a line

A tape is a line, not a cycle. Let rows and columns be indexed by $\{0, \dots, d-1\}$ and let the classes be intervals. A *two-sided rule on a line* is given by coefficients $\mathcal{A}_{J,i}[r][e]$, $-d < e < d$, with

$$X_{c,r} = \sum_i \sum_{j=0}^{d-1} \mathcal{A}_{J,i}[r][j-c]\, X_{i,j} \qquad (c \in J).$$

The decoder of row c again sees the stored word from position c, through a window that is cut off at the ends instead of wrapping around. Every class-uniform rule is of this form, with the same classes: take $\mathcal{A}_{J,i}[r][e]$ to depend on e modulo d (this remark is not formalized). On a line, an interval is a set of consecutive rows that does not wrap around.

Theorem 6.9 (The line form of Theorem TP)**.** Lean: `IsLineUniformCode`, `IsLineUniformCode.two_mul_finrank_le`, `isLineUniformCode_of_coeff`, `lineView`, `truncShift` [`Line.lean`]

A code of a two-sided rule on a line with M *interval classes has dimension at most* $(d^2 + Md)/2$*.*

Outline of the formal proof. Let $(TX)_{i,j} = X_{i,j+1}$ for $j < d-1$ and $(TX)_{i,d-1} = 0$, and $\mathcal{X}' = \{X \in \mathcal{X} : TX \in \mathcal{X}\}$, so that $\dim \mathcal{X}' \geq 2 \dim \mathcal{X} - d^2$. Let $X \in \mathcal{X}'$ and let $c, c+1$ lie in one class J. The rule for TX at row c reads $X_{c,r+1} = \sum_i \sum_{j \geq 1} \mathcal{A}_{J,i}[r][j-(c+1)] X_{i,j}$, with $X_{c,d}$ read as 0, and the rule for X at row $c+1$ has the same sum with $j \geq 0$. So

$$X_{c+1,r} - X_{c,r+1} = \sum_i \mathcal{A}_{J,i}[r][-(c+1)]\, X_{i,0}.$$

Suppose that column 0 of X vanishes. Then inside each class, with rows $a, \dots, a + m_J - 1$, the matrix is constant along anti-diagonals. An anti-diagonal that meets column 0 inside the class is zero, and so is one whose first position in the class is in the last column below row a, because $X_{c,d} = 0$. The remaining anti-diagonals start in row a at a column $r \geq m_J$: there are $d - m_J$ of them. So $\dim \mathcal{X}' \leq d + \sum_J (d - m_J) = Md$. In Lean the decoder of a class is an arbitrary linear function of the zero-padded window $(X_{i,e+c})_{i,e}$ (`lineView`); the coefficient form above is an instance (`isLineUniformCode_of_coeff`). The classes need not be intervals there: the bound is $(d^2 + d|S|)/2$ for any set S of rows that contains row 0 and every row whose class differs from that of the row before it. The key step is `IsLineUniformCode.row_succ_eq`. □

With one class this gives the pairing value for second factors in the span of the truncated shifts, the Toeplitz matrices, which do not commute and are not covered by Corollary 5.7; the symmetric matrices attain it (`isLineUniformCode_symmetric`). For M equal intervals the block-symmetric matrices attain the bound on a line as well, by the remark above; this is not formalized.

Remark 6.10 (Numerical: on a line, other shifts fail)**.** For $d = 3$ and classes $\{0, 2\}, \{1\}$, Theorem 6.4 with $\Delta_0 = \{0, 2\}$ gives 7 on a cycle. On a line the largest code of a two-sided rule has dimension 8 (exhaustive over $\mathbb{F}_2$); one such code is $\{X : X_{0,0} = X_{2,2}\}$. So Theorem 6.9 does not extend to shift sets other than $\{0, 1\}$.

6.4 Twists, digits and composition

A rule need not commute with a translation to be caught; it is enough that it intertwines it with boundedly many others.

Theorem 6.11 (Twist classes)**.** Lean: `succ_mul_finrank_swapCode_le_of_twist_classes`, `two_mul_finrank_swapCode_le_of_twist`, `twistSpace`, `succ_mul_finrank_swapCode_le`, `finrank_le_sum_finrank_twistSpace` [`Twist.lean`]

Let $G \in \mathrm{M}_d(F)$, let the terms of $K = \sum_a A_a \otimes B_a$ fall into classes $1, \dots, s$, and let $G'_1, \dots, G'_s$ commute pairwise, with $GB_a^{\mathsf{T}} = B_a^{\mathsf{T}} G'_\kappa$ for every a in class κ. Then

$$\dim \ker(\tau - K) \;\le\; \frac{s\,d^2 + \sum_\kappa h_\kappa}{s+1}, \qquad h_\kappa = \dim\{P : G^{\mathsf{T}} P = P G'_\kappa\}.$$

Outline of the formal proof. Let $\mathcal{X}' = \{X : XG^t \in \ker(\tau - K),\ 0 \le t \le s\}$, of dimension at least $(s+1)\dim\ker(\tau - K) - s\,d^2$ (`succ_mul_finrank_swapCode_le`). For $X \in \mathcal{X}'$ and $P = X^{\mathsf{T}}$ the hypothesis gives $(G^{\mathsf{T}})^t P = \sum_\kappa T_\kappa G'^t_\kappa$ for $0 \le t \le s$, with $T_\kappa = \sum_{a \in \kappa} A_a X B_a^{\mathsf{T}}$. Applying $P \mapsto G^{\mathsf{T}} P - P G'_\kappa$ removes class κ from such a system and lowers s by one, and its kernel is the space counted by h_κ; induction on the number of classes gives $\dim \mathcal{X}' \le \sum_\kappa h_\kappa$ (`finrank_le_sum_finrank_twistSpace`). This is the statement that $\prod_\kappa (L - R_\kappa) P = 0$ for the commuting operators L, left multiplication by G^{T}, and R_κ, right multiplication by G'_κ. □

Corollary 6.12 (Modulation frequencies)**.** Not formalized; a complete proof is given below.

Let F contain a primitive d-th root of unity ω and let $Z = \mathrm{diag}(1, \omega, \dots, \omega^{d-1})$. If every B_a^{T} is a product $Z^k C$ with C circulant and k in a fixed set of s frequencies, then $\dim\ker(\tau - K) \le \frac{s}{s+1}(d^2 + d)$.

Proof (not formalized). $\sigma Z = \omega Z \sigma$, so $\sigma Z^k C = Z^k C\,(\omega^k \sigma)$: take $G = \sigma$ and $G'_k = \omega^k \sigma$, which commute. The matrices $\sigma^{\mathsf{T}} = \sigma^{-1}$ and $\omega^k \sigma$ are diagonalizable with the same simple spectrum, the d-th roots of unity, so $h_k = d$. □

One frequency is the pairing rate, and all d frequencies span the full matrix algebra. The mask of an interval contains many frequencies (all of them when its length is prime to d), which is why interval masks needed Theorem 6.4.

Corollary 6.13 (Every address digit must be broken)**.** Lean (partial): `two_mul_finrank_ker_sub_le_of_commute` [`Pigeonhole.lean`]

Not formalized: everything specific to the statement: that K commutes with T, the dimension count $\dim\ker[\tau, T] = d^2/q$, and the sharpness. Lean has only the abstract lemma.

Write the index set as $\mathbb{Z}/m \times \mathbb{Z}/q$, $d = mq$, so that the second coordinate is one digit of the address. Suppose every B_a has the form $B'_a \otimes C_a$ with all C_a in $F[G]$ for one non-derogatory $q \times q$ matrix G, for example with all C_a circulant. Then

$$\dim \ker(\tau - K) \;\le\; \tfrac{1}{2}\Big(d^2 + \frac{d^2}{q}\Big),$$

with equality for the partial transpose that exchanges the first digits of the two indices and leaves the second digits in place.

Proof (not formalized). K commutes with $T = 1 \otimes (1_m \otimes G)$, and as in Proposition 6.2 $\ker[\tau, T]$ is isomorphic to the centralizer of $1_m \otimes G$, which is $\mathrm{M}_m(F) \otimes F[G]$, of dimension $m^2 q = d^2/q$. Apply Lemma 6.1. The partial transpose is $\tau_m \otimes 1$ on $(F^m \otimes F^m) \otimes (F^q \otimes F^q)$; its second factors are $E_{u'u} \otimes 1_q$, and its swap code is $\mathrm{M}_m(F) \otimes \{\text{symmetric } q \times q \text{ matrices}\}$, of dimension $m^2 q(q+1)/2$. □

The transposition is the product of the commuting exchanges of the digits of the two addresses (Section 4.3). Corollary 6.13 says that a data path which treats any one digit of one address uniformly is at rate $\frac{1}{2} + \frac{1}{2q}$, however it treats the others; by Lemma 6.1, rate $1-\varepsilon$ needs $\operatorname{rank}[K, T] \ge (1 - 2\varepsilon - \frac{1}{q})d^2$ for the rotation T of every digit of either index. This is a requirement at every scale.

Remark 6.14 (Composition, and why this gives no logarithm)**.** Theorem 6.4 is a statement about one hop: the class-uniform rules with M classes are not closed under composition, and two of them compose to a rule with up to d classes. Lemma 6.1 does compose. Since $[K_2 K_1, T] = K_2[K_1, T] + [K_2, T]K_1$, $\operatorname{rank}[K_2 K_1, T] \le \operatorname{rank}[K_1, T] + \operatorname{rank}[K_2, T]$ (`finrank_range_commutator_comp_le`), and by induction

$$\operatorname{rank}[K_t \cdots K_1, T] \;\le\; \sum_i \operatorname{rank}[K_i, T].$$

For $T = 1 \otimes \sigma$ and a hop whose second factors lie in the span of the $P_J \sigma^e$ with M interval classes, $[K_i, T]X$ is supported on the M columns at which the class changes, so $\operatorname{rank}[K_i, T] \le Md$ and t such hops give $\dim \ker(\tau - K_t \cdots K_1) \le (d^2 + d + tMd)/2$.

The same subadditivity shows why Corollary 6.13 does not produce a factor $\log d$. An exact transposer needs $\sum_i \operatorname{rank}[K_i, T] \ge d^2 - d^2/q$ for the rotation T of every digit. But let U be the perfect unshuffle in base q on one index, which moves every digit to the next position and is a linear-time operation on tapes, and let G_j be the rotation of digit j of that index. Then $UG_j = G_{j+1}U$, so $[U, G_j] = (G_{j+1} - G_j)U$ has rank $d - d/q$, and $\operatorname{rank}[1 \otimes U, 1 \otimes G_j] = d^2 - d^2/q$ for every j at once: one operation supplies the whole requirement for all digits of that index. The requirement is real at every scale and can be met at all scales together.

In a class-uniform rule a change of class changes the decoders of all d entries of a row at once. The argument of Theorem 6.4 for the unit shift also prices a change that affects a single entry.

Proposition 6.15 (Changes that affect single entries)**.** Lean (partial): `IsEntryRuleCode`, `IsEntryRuleCode.apply_succ_eq`, `IsEntryRuleCode.two_mul_finrank_le`, `IsEntryRuleCode.two_mul_finrank_le_of_changes` [`Entry.lean`]
Not formalized: the count in the first inequality. The Lean statement takes any set P of positions that contains $(c+1, r)$ whenever $a_{c,r} \ne a_{c+1,r}$ and that meets every anti-diagonal, and gives $2 \dim \mathcal{X} \le d^2 + |P|$; the proof below exhibits such a set with $\sum_t \max(1, m_t)$ elements. The bound $d^2 + d + E$ is formalized, for any set of pairs that contains those counted by E.

Let $a_{c,r}$, for $c, r \in \mathbb{Z}/d$, be linear functionals on $\mathrm{M}_d(F)$, and let $\mathcal{X} \subseteq \mathrm{M}_d(F)$ be a subspace with $X_{c,r} = a_{c,r}(\sigma^c X)$ for all $X \in \mathcal{X}$ and all c, r. For $t \in \mathbb{Z}/d$ let m_t be the number of $c \in \mathbb{Z}/d$ with

$a_{c,r} \neq a_{c+1,r}$ *for* $r = t-1-c$*, and let* $E = \sum_t m_t$ *be the number of pairs* (c,r) *with* $a_{c,r} \neq a_{c+1,r}$*. Then*

$$2\dim \mathcal{X} \;\leq\; d^2 + \sum_{t \in \mathbb{Z}/d} \max(1, m_t) \;\leq\; d^2 + d + E.$$

Proof. Let $\mathcal{X}' = \{X \in \mathcal{X} : \sigma X \in \mathcal{X}\}$. As in the proof of Theorem 6.4, $\dim \mathcal{X}' \geq 2\dim \mathcal{X} - d^2$. Let $X \in \mathcal{X}'$. The rule applied to X at the entry $(c+1, r)$ gives $X_{c+1,r} = a_{c+1,r}(\sigma^{c+1}X)$, and applied to σX at the entry (c, r) it gives $X_{c,r+1} = (\sigma X)_{c,r} = a_{c,r}(\sigma^{c+1}X)$. So $X_{c+1,r} = X_{c,r+1}$ whenever $a_{c,r} = a_{c+1,r}$ (`IsEntryRuleCode.apply_succ_eq`). Both entries lie on the anti-diagonal $\{(i,j) : i + j = t\}$ with $t = c + r + 1$. Join the positions $(c, t-c)$ and $(c+1, t-1-c)$ of this anti-diagonal by an edge, for each $c \in \mathbb{Z}/d$: the d positions and d edges form a cycle, X takes equal values at the two ends of every edge with $a_{c,t-1-c} = a_{c+1,t-1-c}$, and m_t of the edges fail this condition. Let P consist of the end $(c+1, t-1-c)$ of every failing edge, and of one position on every anti-diagonal that has no failing edge; it has $\sum_t \max(1, m_t)$ elements, since the ends of distinct failing edges are distinct. A position (i,j) outside P is not the end $(c+1, t-1-c)$ of a failing edge, so $X_{i,j} = X_{i-1,j+1}$. Walking from any position in the direction of decreasing row index one therefore stays on one anti-diagonal, on which X is constant along the walk, and reaches a position of P, because P meets every anti-diagonal. So an element of $\mathcal{X}'$ that vanishes on P is zero, $\dim \mathcal{X}' \leq |P|$ (`IsEntryRuleCode.two_mul_finrank_le`), and $2\dim \mathcal{X} - d^2 \leq \sum_t \max(1, m_t) \leq d + E$ (`IsEntryRuleCode.two_mul_finrank_le_of_changes`). □

A class-uniform rule is the case in which $a_{c,r}$ depends on c only through the class J of c, with $a_{c,r} = \mathcal{A}_J(\,\cdot\,)_r$ (in Lean, `IsClassUniformCode.exists_isEntryRuleCode`). If its decoders change only at the rows of a non-empty set S, then $m_t \leq |S|$ for every t, and Proposition 6.15 gives $2\dim \mathcal{X} \leq d^2 + d|S|$ (`IsEntryRuleCode.two_mul_finrank_le_of_rows`); for M cyclic intervals this is the bound $(d^2 + Md)/2$ of Theorem 6.4 again. The use of Proposition 6.15 is for rules that are not class-uniform. Each place where the decoder of an output entry changes from one row to the next raises the bound by at most half a symbol; a change of class is d such changes and raises it by half a row, and altering the decoder of one entry at one row is two of them. The script `tools/entry_form_check.py` checks the inequality for every rule at $d = 2$ over $\mathbb{F}_2$ and for random and structured rules up to $d = 5$; on the structured rules the bound is often attained, and the same inequality with the changes grouped by diagonals or by rows fails for some of them.

Over a finite field the argument of the memo [14] for advice applies to these rules as well.

Corollary 6.16 (Advice and changes at single entries)**.** Lean: `IsEntryRuleCodeWithAdvice`, `entryRuleSpace`, `IsEntryRuleCodeWithAdvice.natCard_fiber_le`, `IsEntryRuleCodeWithAdvice.natCard_pow_two_mul_finrank_le` [`EntryAdvice.lean`]

Let F *be a finite field,* Ω *a finite set and* $w\colon \mathrm{M}_d(F) \to \Omega$ *any map. For every* $\omega \in \Omega$ *let* $a^\omega_{c,r}$ *be linear functionals on* $\mathrm{M}_d(F)$ *and* $b^\omega_{c,r}$ *elements of* F*, for* $c, r \in \mathbb{Z}/d$*. Let* $\mathcal{X} \subseteq \mathrm{M}_d(F)$ *be a subspace such that* $X_{c,r} = a^\omega_{c,r}(\sigma^c X) + b^\omega_{c,r}$ *for all* c, r *and every* $X \in \mathcal{X}$ *with* $w(X) = \omega$*. Suppose that for every* ω *at most* E *pairs* (c,r) *have* $a^\omega_{c,r} \neq a^\omega_{c+1,r}$*. Then for every* ω *at most* $|F|^{\lfloor (d^2+d+E)/2 \rfloor}$ *matrices* $X \in \mathcal{X}$ *have* $w(X) = \omega$*, and*

$$|F|^{2\dim \mathcal{X}} \;\leq\; |\Omega|^2 \cdot |F|^{d^2+d+E}.$$

In particular $2\dim \mathcal{X} \leq d^2 + d + E + 2q$ *when* $|\Omega| = |F|^q$*.*

Outline of the formal proof. Fix ω and a matrix $X_0 \in \mathcal{X}$ with $w(X_0) = \omega$, if there is one. For every $X \in \mathcal{X}$ with $w(X) = \omega$ the offsets cancel in $X - X_0$, which lies in $\mathcal{X}$ and satisfies the rule a^ω without offsets. So it lies in a code of that rule (`entryRuleSpace`), of dimension at most $(d^2 + d + E)/2$ by Proposition 6.15 (`IsEntryRuleCodeWithAdvice.natCard_fiber_le`). The sets $\{X \in \mathcal{X} : w(X) =$

$\omega\}$ partition $\mathcal{X}$, so $|F|^{\dim \mathcal{X}} \le |\Omega| \cdot |F|^{\lfloor (d^2+d+E)/2 \rfloor}$; square (`IsEntryRuleCodeWithAdvice.natCard_pow_two_mul_finrank_le`). The last statement follows because $|F| \ge 2$. The counting is that of [14, Theorem 5.5], which has class-uniform rules and Theorem 6.4 where this proof has entry rules and Proposition 6.15. □

Here the advice w is any function of the matrix, and its value may select the functionals of all entries and add a constant to each. The corollary separates two prices. The advice itself costs at most its q symbols in the whole code, and nothing among the matrices that share one of its values. Each change of a decoder from one row to the next costs at most half a symbol in both counts.

6.5 What the theorem says

- *For translation-covariant rules, the half in the half bound is a pigeonhole.* A code larger than half the space meets its translate in at least $2D - d^2$ dimensions, and translation-covariant data has d dimensions per uniform piece. The argument does not depend on the characteristic, on the degree, on the window of the decoder, or on the other tensor side. It does depend on exact covariance under a unit shift inside each class, on the classes being the same for every output row, on the code being a subspace, and on there being one hop. It does not depend on the decoders being linear (Section 6.7). Proposition 6.15 relaxes the second of these: every output row, that is, every coordinate r of the rows of X, may have its own change points, at half a symbol for each. The flag bounds of Section 5 are not shown to be instances of it.

- *No symmetry that the swap lacks.* By Lemma 6.1, a code that exceeds the pairing value by εd^2 needs $\operatorname{rank}[K, T] + \dim \ker[\tau, T] \ge d + 2\varepsilon d^2$ for every operator T. Together with Section 5: to beat the pairing rate, the decoder may have neither a hidden order nor a symmetry, on either index, that the swap does not share.

- *It uses the rule, not the relation.* The hypothesis of Theorem 6.4 is that translated receivers share a decoder. That is the feature of machines singled out in Section 4.6, and it is invisible in the cone relation. For sets that are not subspaces the hypothesis loses its force when a decoder sees the whole matrix; what it gives for decoders that read boundedly many entries is open (Section 6.7, Problem 9.4(b)).

- *The folklore step is optimal in dimension.* With M equal pieces, nothing beats the $M \times M$ coarse transposition by even one dimension, although many other codes equal it (Remark 6.7).

6.6 Without an almost-period: an unbounded gain at every fixed number of classes

Theorem 6.4 gives the bound $(d^2 + Md)/2$ only for classes with an almost-period, and Proposition 8.2 shows that without one a code can be larger, by one dimension at $d = 6$. The subsequent memo [14] shows that for two classes the difference is unbounded. For every $d = 4k \ge 8$ it gives an explicit rule, over fields in which $2 \ne 0$, whose code exceeds $(d^2 + 2d)/2$ by $d/4 - 1$ [14, Definition 4.7 and Theorem 4.8]. The proof is written in the memo; in Lean the memo proves, for $d = 8$, 12 and 16, that the two classes carry a code of that dimension [14, Theorem 4.9]. The memo's rule has the coefficients 2 and $\frac{1}{2}$, and the memo leaves characteristic 2 open. Here we observe that these two coefficients can be any nonzero λ and its inverse, and that the proof then holds over every field (Proposition 6.17). With $\lambda = 1$ in characteristic 2, every coefficient of the rule is 0 or 1. A blow-up that treats the positions differently then gives an unbounded gain for every fixed number $M \ge 2$ of classes (Lemma 6.18 and Corollary 6.19), which the memo also leaves open for three or more classes.

Let $d = 4k \geq 8$. The two classes are $J_0 = \{0, 2, \ldots, d-4\}$, the even residues other than $d-2$, and $J_1 = \mathbb{Z}/d \setminus J_0$, the odd residues together with $d-2$. Write $\mathcal{N} = \mathbb{Z}/d \setminus \{0, d-2\}$, and for $j \in \mathbb{Z}/d$ let $e_j \in F^{\mathbb{Z}/d}$ be the indicator vector of j if $j \in \mathcal{N}$, and $e_j = 0$ if $j \in \{0, d-2\}$. Fix a nonzero $\lambda \in F$. For $m \in \{0,1\}$ and $r \in \mathbb{Z}/d$ define $w_r^{(m)} \in F^{\mathbb{Z}/d}$ by

$$w_r^{(0)} = \lambda e_{r+3}, \qquad w_r^{(1)} = e_{r+1} \qquad \text{for even } r,$$

and for odd r by

$$w_r^{(0)} = e_{r+5} \quad (r \neq d-7), \qquad w_{d-7}^{(0)} = \alpha = \sum_{i \in \mathcal{N} \text{ even}} (-1)^{i/2} e_i,$$

$$w_r^{(1)} = \lambda^{-1} e_{r+3} \quad (r \neq d-5), \qquad w_{d-5}^{(1)} = \beta = \sum_{i \text{ odd}} e_i - \lambda^{-1} \sum_{i \in \mathcal{N} \text{ even}} e_i.$$

Since 4 divides d, the sign $(-1)^{i/2}$ of an even residue i is well defined. Let $a_r^{(m)} \in F^{\mathbb{Z}/d}$ agree with $w_r^{(m)}$ on $\mathcal{N}$, and at the two remaining residues let

$$a_r^{(m)}(0) = [m=0] - \sum_{i \in \mathcal{N} \cap J_0} w_r^{(m)}(i), \qquad a_r^{(m)}(d-2) = [m=1] - \sum_{i \in \mathcal{N} \cap J_1} w_r^{(m)}(i),$$

where $[\,\cdot\,]$ is 1 if the condition holds and 0 otherwise. The rule has the decoders

$$\mathcal{A}_{J_m}(Y)_r = \sum_{i \in \mathbb{Z}/d} a_r^{(m)}(i)\, Y_{i,\,r-i} \qquad (m \in \{0,1\},\ r \in \mathbb{Z}/d).$$

For $\lambda = 2$ this is the parity-coset rule of [14, Definition 4.7].

Proposition 6.17 (Two classes: a gain of $d/4 - 1$ over every field)**.** Not formalized; a complete proof is given below.

Let F be a field, $\lambda \in F$ nonzero, and $d = 4k \geq 8$. The largest code of the class-uniform rule above, for the classes J_0 and J_1 of $\mathbb{Z}/d$, has dimension

$$\frac{d^2 + 2d}{2} + \frac{d}{4} - 1.$$

So over every field, including $\mathbb{F}_2$, and for every $d = 4k \geq 8$, the classes J_0 and J_1 of $\mathbb{Z}/d$ carry a code of a class-uniform rule that exceeds $(d^2 + 2d)/2$ by $d/4 - 1$.

Proof (not formalized). The proof is that of [14, Theorem 4.8], which is the case $\lambda = 2$ over a field in which $2 \neq 0$. We give it in full, because the statement is more general and one step is different: for $s \equiv 3 \pmod 4$ the memo uses that a certain coordinate equals $-2 \neq 0$, and we use a fixed point instead.

The code splits into anti-diagonals. Write $J_{m(c)}$ for the class of c. Since $(\sigma^c X)_{i,r-i} = X_{i,c+r-i}$, the equation of the rule at row c and coordinate r is $X_{c,r} = \sum_i a_r^{(m(c))}(i)\, X_{i,c+r-i}$, and all its entries lie on the anti-diagonal $\{(i,j) : i + j = c + r\}$. So the largest code $\mathcal{X}$ is the direct sum, over $s \in \mathbb{Z}/d$, of the spaces $\mathcal{X}_s$ of the vectors $z \in F^{\mathbb{Z}/d}$, with $z_i = X_{i,s-i}$, that satisfy the d equations

$$z_c = \sum_{i \in \mathbb{Z}/d} a_{s-c}^{(m(c))}(i)\, z_i \qquad (c \in \mathbb{Z}/d).$$

Reduction to a rank. By the choice of $a_r^{(m)}$ at 0 and $d-2$, $\sum_{i \in J_0} a_r^{(m)}(i) = [m=0]$ and $\sum_{i \in J_1} a_r^{(m)}(i) = [m=1]$. So the indicator vectors of J_0 and of J_1 lie in $\mathcal{X}_s$. They take the values

$(1,0)$ and $(0,1)$ at the residues $(0, d-2)$, so $\mathcal{X}_s$ is the direct sum of their span and of the space of solutions with $z_0 = z_{d-2} = 0$. For such z the equation of row c says that z is orthogonal to

$$R_c = w_{s-c}^{(m(c))} - e_c,$$

a vector supported on $\mathcal{N}$ (recall $e_0 = e_{d-2} = 0$). Hence $\dim \mathcal{X}_s = 2 + (d-2-r_s) = d - r_s$, where r_s is the rank of the vectors R_c, $c \in \mathbb{Z}/d$. We show that

$$r_s = 2k-2 \ \text{ if } s \equiv 1 \pmod 4 \text{ and } s \neq d-7, \qquad r_s = 2k-1 \ \text{ otherwise.}$$

Since $d - 7 \equiv 1 \pmod 4$, the first case holds for $k-1$ residues s, and $\dim \mathcal{X} = \sum_s (d - r_s) = d(d-2k+1) + (k-1) = (d^2+2d)/2 + k - 1$. In what follows the rows $c \in J_0$ use $w_{s-c}^{(0)}$ and the rows $c \in J_1$ use $w_{s-c}^{(1)}$; all odd residues lie in $\mathcal{N}$.

Even s. For even $c \in \mathcal{N}$ let $\tau(c) = s + 3 - c$, which is odd. Row c lies in J_0 and $s - c$ is even, so $R_c = \lambda e_{\tau(c)} - e_c$. Row $\tau(c)$ lies in J_1, and $s - \tau(c) = c - 3$ is odd and different from $d-5$, so $R_{\tau(c)} = \lambda^{-1} e_c - e_{\tau(c)} = -\lambda^{-1} R_c$. As c runs over the even residues of $\mathcal{N}$, $\tau(c)$ runs over the odd residues other than $s+3$ and $s+5$. Three of the four remaining rows are

$$R_0 = \lambda e_{s+3}, \qquad R_{d-2} = e_{s+3}, \qquad R_{s+3} = \lambda^{-1} e_0 - e_{s+3} = -e_{s+3},$$

the last because $s - (s+3) = d - 3 \neq d - 5$. The fourth is row $s+5$, with $s - (s+5) = d - 5$: $R_{s+5} = \beta - e_{s+5}$. Now

$$\beta - e_{s+5} = e_{s+3} + \sum_c (e_{\tau(c)} - \lambda^{-1} e_c) = e_{s+3} + \lambda^{-1} \sum_c R_c,$$

with the sums over the even $c \in \mathcal{N}$. So the rows span the same space as e_{s+3} and the $2k-2$ vectors R_c, $c \in \mathcal{N}$ even. These $2k-1$ vectors are independent: the supports $\{c, \tau(c)\}$ are pairwise disjoint and do not contain $s+3$. So $r_s = 2k-1$.

Odd s: the odd rows. An odd row c lies in J_1 and $s - c$ is even, so $R_c = e_{T(c)} - e_c$ with $T(c) = s + 1 - c$, an involution of the odd residues. Then $R_{T(c)} = -R_c$, and $R_c = 0$ at a fixed point. So the odd rows span a space O, supported on the odd residues, whose dimension is the number of 2-cycles of T and all of whose vectors vanish at the fixed points of T. The fixed points are the solutions of $2c \equiv s + 1 \pmod{4k}$, that is, $c \equiv (s+1)/2 \pmod{2k}$: two residues, which are odd if and only if $s \equiv 1 \pmod 4$. So $\dim O = k - 1$ if $s \equiv 1 \pmod 4$, and $\dim O = k$ if $s \equiv 3 \pmod 4$.

Odd $s \neq d-7$: the even rows. Let $U(c) = s + 5 - c$, an involution of the even residues, with $U(0) = s+5$ and $U(d-2) = s+7$. The even rows are supported on the even residues, so r_s is $\dim O$ plus the rank of the even rows. They are the following.

- For $c \in J_0$ with $c \neq s+7$, $s - c$ is odd and different from $d-7$, so $R_c = e_{U(c)} - e_c$. In particular $R_0 = e_{s+5}$.
- Row $d-2$ lies in J_1, and $s - (d-2) = s+2$ is odd and different from $d-5$, so $R_{d-2} = \lambda^{-1} e_{s+5} = \lambda^{-1} R_0$.
- Row $s+7$, if it lies in J_0, that is, if $s \neq d-9$, has $s - (s+7) = d - 7$, so $R_{s+7} = \alpha - e_{s+7}$.

Let W be the span of the rows of the first kind. If c is a fixed point of U, then $R_c = 0$. If $c \in \{0, s+5\}$ and $s + 5 \neq 0$, the two rows are $R_0 = e_{s+5}$ and $R_{s+5} = e_0 - e_{s+5} = -e_{s+5}$; row $s+5$ is of the first kind because $s + 5 \neq d - 2$. For every other $c \in J_0 \setminus \{s+7\}$, both c and $U(c)$ lie in $\mathcal{N} \setminus \{s+7\}$ and $R_{U(c)} = -R_c$: indeed $U(c)$ equals 0, $d-2$ or $s+7$ only for $c = s+5$, $c = s+7$

or $c = d-2$, and $d-2 \notin J_0$. So W has a basis consisting of e_{s+5}, if $s \neq d-5$, and of one vector $e_{U(c)} - e_c$ for each 2-cycle of U inside $\mathcal{N} \setminus \{s+7\}$; and every vector of W vanishes at $s+7$ and at the fixed points of U. The fixed points of U are the solutions of $2c \equiv s+5 \pmod{4k}$: two even residues if $s \equiv 3 \pmod 4$, and none if $s \equiv 1 \pmod 4$.

- $s \equiv 1 \pmod 4$. The $2k$ even residues form k two-cycles of U. Two of them are $\{0, s+5\}$ and $\{d-2, s+7\}$, which are distinct because $s \neq d-7$, so $k-2$ lie inside $\mathcal{N} \setminus \{s+7\}$ and $\dim W = k-1$. The row $\alpha - e_{s+7}$ lies in W: its coordinate at $s+7$ is $(-1)^{(s+7)/2} - 1 = 0$, because $s+7 \equiv 0 \pmod 4$; on a 2-cycle $\{c, U(c)\}$ inside $\mathcal{N}$ we have $c + U(c) = s+5 \equiv 2 \pmod 4$, so $c/2$ and $U(c)/2$ have different parities and the two coordinates of α are opposite; and its coordinate at $s+5$ is matched by a multiple of e_{s+5}. So the even rows have rank $k-1$, and $r_s = (k-1) + (k-1) = 2k-2$.
- $s \equiv 3 \pmod 4$. Now U has two fixed points and $k-1$ two-cycles. If $s = d-9$, then $s+7 = d-2$ is a fixed point, there is no row of the third kind, and the two-cycles are $\{0, d-4\}$ and $k-2$ others inside $\mathcal{N}$; so the even rows span W, of dimension $k-1$. If $s = d-5$, then 0 and $2k$ are the fixed points, $R_0 = R_{d-2} = 0$, and the two-cycles are $\{d-2, 2\}$ and $k-2$ others inside $\mathcal{N} \setminus \{2\}$, so $\dim W = k-2$; here $s+7 = 2$, and the fixed point $2k$ lies in $\mathcal{N} \setminus \{2\}$ because $k \geq 2$. For every other s, which occurs only when $k \geq 3$, the two fixed points lie in $\mathcal{N} \setminus \{s+7\}$, and the two-cycles are $\{0, s+5\}$, $\{d-2, s+7\}$ and $k-3$ others, so again $\dim W = 1 + (k-3) = k-2$. In the last two cases the row $\alpha - e_{s+7}$ is not in W: its coordinate at a fixed point of U in $\mathcal{N}$, namely at $2k$ or at either of the two, is ± 1, and every vector of W vanishes there. So in all three cases the even rows have rank $k-1$, and $r_s = k + (k-1) = 2k-1$.

$s = d-7$. Here $s \equiv 1 \pmod 4$, so $\dim O = k-1$, and the vectors of O vanish at the two fixed points of T. Now $U(c) = d-2-c$, which exchanges 0 and $d-2$ and has no even fixed point. Row 0 has $s - 0 = d-7$, so $R_0 = \alpha$. For $c \in J_0 \setminus \{0\}$, $R_c = e_{d-2-c} - e_c$ with c and $d-2-c$ in $\mathcal{N}$; these rows span the space E of the vectors on the even residues of $\mathcal{N}$ that sum to zero on each of the $k-1$ pairs $\{c, d-2-c\}$, of dimension $k-1$. The row α lies in E, because $(d-2-c)/2 = 2k-1-c/2$ and $c/2$ have different parities. The remaining row is $d-2$, with $s - (d-2) = d-5$: $R_{d-2} = \beta$. Its part on the odd residues is $\sum_{i \text{ odd}} e_i$, which does not vanish at the fixed points of T, while the part on the odd residues of every vector of $O \oplus E$ lies in O and vanishes there. So $\beta \notin O \oplus E$, and $r_{d-7} = (k-1) + (k-1) + 1 = 2k-1$. □

The two classes have no almost-period [14, Section 4.2], and the gain is at most $d/2$. For the shifts $\{0, 2\}$ the graph of Theorem 6.4 has three components: the odd residues, the set J_0, and the single residue $d-2$, whose neighbours $d-4$ and 0 lie in the other class. So every code for these classes has dimension at most $(d^2+3d)/2$. The largest code of a rule for them therefore exceeds $(d^2+2d)/2$ by at least $d/4-1$ and at most $d/2$, over every field.

An unbounded gain also exists for every fixed number $M \geq 2$ of classes. The memo blows a code up by a factor k, which multiplies the number of classes by k [14, Theorem 4.2]. Its proof decodes a row from the blocks at the position of that row alone, so each position may carry its own code. From here on k denotes the factor of a blow-up, and no longer $d/4$. Fix $k \geq 1$. Every $c \in \mathbb{Z}/dk$ is $c = ka + u$ for a unique $a \in \mathbb{Z}/d$ and a unique *position* $u \in \{0, \ldots, k-1\}$, where c is read as an integer in $[0, dk)$. For $X \in \mathrm{M}_{dk}(F)$ and positions u, v, the *block* $X^{(u,v)} \in \mathrm{M}_d(F)$ is $X^{(u,v)}_{a,t} = X_{ka+u,\, kt+v}$.

Lemma 6.18 (Blow-ups with a code for each position)**.** Not formalized; a complete proof is given below.

For each position u let $\mathcal{X}_u \subseteq \mathrm{M}_d(F)$ be a code of a class-uniform rule for a partition $\mathcal{P}_u$ of $\mathbb{Z}/d$. Then

$$\mathcal{Y} = \{X \in \mathrm{M}_{dk}(F) : X^{(u,v)} \in \mathcal{X}_u \text{ for all positions } u, v\}$$

is a code of a class-uniform rule for the partition of $\mathbb{Z}/dk$ into the classes $\{ka + u : a \in J\}$, for $J \in \mathcal{P}_u$ and $0 \le u < k$, and $\dim \mathcal{Y} = k \sum_u \dim \mathcal{X}_u$.

Proof (not formalized). For equal codes $\mathcal{X}_u$ and equal partitions this is [14, Theorem 4.2], which is proved in Lean there, and the argument is the same. Let $\mathcal{A}^u_J$ be decoders of $\mathcal{X}_u$. For positions u, v let $B_{u,v} \colon \mathrm{M}_{dk}(F) \to \mathrm{M}_d(F)$ be the linear map $B_{u,v}(Y)_{a,t} = Y_{ka+u,\, kt+v-u}$. Let $X \in \mathcal{Y}$ and $c = ka' + u$ with $a' \in J \in \mathcal{P}_u$. For all $a, t \in \mathbb{Z}/d$,

$$(\sigma^{a'} X^{(u,v)})_{a,t} = X_{ka+u,\, k(t+a')+v} = (\sigma^c X)_{ka+u,\, kt+v-u} = B_{u,v}(\sigma^c X)_{a,t},$$

where $k(t + a')$ is well defined in $\mathbb{Z}/dk$ for $t + a' \in \mathbb{Z}/d$: the first equality is the definition of the block, and the second holds because $(kt + v - u) + c = k(t + a') + v$ in $\mathbb{Z}/dk$. Since $X^{(u,v)} \in \mathcal{X}_u$, its row a' is $\mathcal{A}^u_J(\sigma^{a'} X^{(u,v)})$. Hence $X_{c,\, kt+v} = \mathcal{A}^u_J\big(B_{u,v}(\sigma^c X)\big)_t$ for all t and v. The right side is a linear function of $\sigma^c X$ that depends on c only through J and u, that is, through the class of c. So $\mathcal{Y}$ is a code of the rule with the decoders $\mathcal{B}_{J,u}(Y)_{kt+v} = \mathcal{A}^u_J(B_{u,v}(Y))_t$. The map $X \mapsto (X^{(u,v)})_{u,v}$ is a linear isomorphism of $\mathrm{M}_{dk}(F)$ onto $\mathrm{M}_d(F)^{k \times k}$, and it maps $\mathcal{Y}$ onto $\prod_{u,v} \mathcal{X}_u$, which gives the dimension. □

Corollary 6.19 (Every fixed number of classes)**.** Not formalized; a complete proof is given below.
Let $M \ge 2$, $k = \lceil M/2 \rceil$ and $p = \lfloor M/2 \rfloor$, let $d \ge 8$ be divisible by 4, and $n = kd$. Over every field there is a partition of $\mathbb{Z}/n$ into M classes that carries a code of a class-uniform rule of dimension

$$\frac{n^2 + Mn}{2} + kp\Big(\frac{d}{4} - 1\Big) = \frac{n^2 + Mn}{2} + p\Big(\frac{n}{4} - k\Big).$$

So for every fixed $M \ge 2$, over every field including $\mathbb{F}_2$, the amount by which a code of a class-uniform rule for M classes of $\mathbb{Z}/n$ can exceed $(n^2 + Mn)/2$ is unbounded as n grows.

Proof (not formalized). Apply Lemma 6.18 with the code of Proposition 6.17, for the two classes J_0, J_1, at the positions $u < p$, and, if M is odd, with the symmetric matrices at the last position $u = p = k - 1$. The symmetric matrices are a code of a class-uniform rule with one class: $X_{a,t} = X_{t,a} = (\sigma^a X)_{t,0}$, so the decoder $\mathcal{A}(Y)_t = Y_{t,0}$ serves every row; their dimension is $(d^2 + d)/2$. The number of classes is $2p + (k - p) = M$, and

$$\dim \mathcal{Y} = k\Big(p\Big(\frac{d^2 + 2d}{2} + \frac{d}{4} - 1\Big) + (k - p)\frac{d^2 + d}{2}\Big) = \frac{k^2 d^2}{2} + \frac{(2p + k - p)\, kd}{2} + kp\Big(\frac{d}{4} - 1\Big),$$

which is the stated value because $n = kd$ and $2p + k - p = M$. □

For even M all positions carry the code of Proposition 6.17, and the corollary is the memo's blow-up theorem applied to it. The gain $p(n/4 - k)$ is of order $Mn/8$, which is the order of nM again (Section 7.3). So a bounded number of classes with no almost-period gains over the interval value by an amount that is unbounded but, in these examples, linear in the side length. Whether it can grow faster is Problem 9.1(c).

Remark 6.20 (Numerical: the rules over small fields)**.** The script `tools/two_class_family.py` builds the rule of Proposition 6.17 from its formulas and checks, by exact elimination, the dimension of its code and every rank stated in the proof: for $d = 8, 12, \ldots, 48$ over $\mathbb{F}_2$ with $\lambda = 1$, over $\mathbb{F}_3$ and

$\mathbb{F}_5$ with several λ, and over $\mathbb{F}_7$ and $\mathbb{F}_{11}$; and the dimension alone for every $d = 4k \le 100$ over $\mathbb{F}_2$. For $d = 8$ and 12 it also solves the whole system of the rule, without the splitting into anti-diagonals. The codes over $\mathbb{F}_2$ at $d = 8$, 12 and 16, of dimension $41 > 40$, $86 > 84$ and $147 > 144$, are saved in `tools/data/` and checked against the definition by `tools/verify_class_uniform_code.py`. For Corollary 6.19 the script builds the rule of the blow-up for $M = 3, \dots, 7$ at sizes up to $n = 32$ over $\mathbb{F}_2$, and for $M = 3, 4$ at $n = 16$ over $\mathbb{F}_3$, and solves its whole system: in every case the largest code of that rule has exactly the dimension of the corollary. The memo reports exact checks of its rule over $\mathbb{Q}$ for every $d = 4k \le 100$, and further two-class codes, among them one with a gain of 2 for two classes of six elements at $d = 12$ [14, Remark 4.11].

6.7 Decoders that are not linear, and sets that are not subspaces

The decoders of a machine are compositions of its transition function, and they are not linear. The inputs that share a value of a genie's advice form a set, not a subspace. A paper that followed revision `ac70dc4` of this one [19] separates the two kinds of linearity in the bounds of this section. This subsection records what it shows, and states two bounds that need neither kind.

The decoders may be arbitrary maps. The proof of Theorem 6.4 uses the rule in one way: two decoders of one class, applied to the same matrix, return the same row. It uses no other property of the maps $\mathcal{A}_J$. So Theorem 6.4 holds, with the same proof, for arbitrary maps $\mathcal{A}_J \colon \mathrm{M}_d(F) \to F^{\mathbb{Z}/d}$ and a subspace $\mathcal{X}$ that satisfies the equations of Definition 6.3 [19, Lemma 3.1]. Theorem 6.9 holds for arbitrary decoders of the zero-padded window as well. The outline given for it above first derives a linear identity and then sets column 0 to zero; for arbitrary maps one restricts to the matrices with zero column 0 first, and then the two windows that are compared are equal [19, Theorem A.4]. Both forms are proved in Lean there. The proof of Proposition 6.15 is of the same kind: it compares the values of $a_{c,r}$ and of $a_{c+1,r}$ at one matrix. So that proposition holds for arbitrary maps $a_{c,r} \colon \mathrm{M}_d(F) \to F$ as well; this form of it is not formalized. With the pigeonhole in this form, and with each further intersection taken inside the space that the earlier ones leave, the same paper improves the bounds for classes of arbitrary shape (Problem 9.1).

The code may not be an arbitrary set. The count of dimensions in these proofs needs $\mathcal{X}$ to be a subspace, and without that hypothesis nothing remains of them. Over an alphabet of $q \ge 2$ symbols, and for every $d \ge 2$, there are a single map $\mathcal{A}$ and a set of at least $q^{d^2}/(2d)$ matrices X with $x_c = \mathcal{A}(\sigma^c X)$ for every c [19, Theorem A.1 and Corollary A.2]: in the main case the decoder, shown $\sigma^c X$, recognises which matrix of the set this is a rotation of, and with it c. The same happens on an affine subspace: one map $\mathcal{A}$ serves all matrices whose row 0 is a given unit vector, an affine space of dimension $d^2 - d$ [19, Proposition 4.5]. So in Theorem 6.4 the subspace cannot be replaced by a set, or by an affine subspace. For the same reason Corollary 6.16 has no counterpart for decoders that are not affine: its proof subtracts two matrices that share a value of the advice, and a set of matrices that satisfies one rule with one class, with no change of decoder at all, can have $q^{d^2}/(2d)$ elements.

These decoders use the whole of the matrix that they are shown: a marker row, or enough of the matrix to tell its rotations apart. A decoder that reads w entries over an alphabet of q symbols can tell apart at most q^w rows in this way, a bounded number when the alphabet and the window are fixed. (The alphabet has to be fixed: over d symbols, on the matrices with $X_{0,j} = j$ for all j, the single entry $(\sigma^c X)_{0,0}$ is c.) The two bounds that follow are for sets and arbitrary functions, and they are stated for the positions that the decoders read. In the language of Definition 2.1, a recovery relation on the d^2 positions of a matrix assigns to every position (c, r) a set $N(c, r)$ of positions, its *helpers*, and a storage code on it is a set S of $d \times d$ matrices over an alphabet such that the

entry $X_{c,r}$ of a matrix $X \in S$ is a function of the entries of X at the helpers of (c,r); the function may depend on (c,r). If the decoder $\mathcal{A}_J$ of a class-uniform rule computes its coordinate r from the entries of its argument at the positions of a set $W_{J,r}$, then every code of the rule is a storage code on the relation with $N(c,r) = \{(i, e+c) : (i,e) \in W_{J,r}\}$ for $c \in J$. The relation keeps the positions and forgets the decoders, and with them the fact that the decoders of one class are equal.

Both bounds are instances of one elementary fact: a storage code over q symbols has at most $q^{|I|}$ elements whenever every directed cycle of the relation, loops included, passes through the set I of coordinates. This is the form for storage codes of the acyclic-subgraph bound for index codes [2, Theorem 3], which the series has used before [6, Proposition 4.14(a)]. What the proposition adds is the choice of I: the two sets are those of the proofs of Theorem 5.1 and of Proposition 6.8, so that these two bounds of this paper hold for sets and for functions that are not linear. We give the short proofs.

Proposition 6.21 (Two bounds for sets)**.** Not formalized; a complete proof is given below.

Let S be a storage code, over an alphabet of q symbols, on a recovery relation N on the positions of a $d \times d$ matrix.

(a) Order. *Number the rows and the columns $0, \dots, d-1$, and let $g\colon \{0,\dots,d-1\} \to \{0,\dots,d\}$ be any function. If, for every c, every helper of a position in row c lies in a column j with $g(j) > c$, then $|S| \le q^{\sum_j g(j)}$. The same holds with rows and columns exchanged: if, for every r, every helper of a position in column r lies in a row i with $g(i) > r$, then $|S| \le q^{\sum_i g(i)}$.*

(b) One helper. *If every position has exactly one helper, $N(c,r) = \{f(c,r)\}$, then $|S| \le q^z$, where z is the number of cycles of the map f on the positions. If moreover there are a partition of $\mathbb{Z}/d$ into M classes and maps $\pi_J, \rho_J\colon \mathbb{Z}/d \to \mathbb{Z}/d$ with $f(c,r) = (\pi_J(r), \rho_J(r)+c)$ for c in the class J, then $|S| \le q^{(d^2+Md)/2}$.*

Proof (not formalized). (a) Let $I = \{(i,j) : i < g(j)\}$, a set of $\sum_j g(j)$ positions. We show that two matrices $X, X' \in S$ that agree on I are equal; then restriction to I is injective on S. Suppose they agree on I, and prove that they agree on row c by downward induction on $c = d-1, \dots, 0$. Let (i,j) be a helper of a position (c,r), so that $g(j) > c$. If $i > c$, the two matrices agree on row i by induction. If $i \le c$, then $i < g(j)$, so $(i,j) \in I$. Hence X and X' agree at all helpers of (c,r), and so at (c,r). For the second statement apply the first to the set of the transposes of the matrices in S, which is a storage code on the relation with all positions transposed.

(b) Every position leads into a cycle of f under iteration of f. Choose one position on every cycle, and let $X, X' \in S$ agree at the chosen positions. If they agree at $f(p)$, they agree at p. Going around a cycle against the direction of f, from its chosen position, they agree on the whole cycle; and by induction on the number of steps that a position needs to reach a cycle, they agree everywhere. So $|S| \le q^z$. For the second statement, a fixed point (c,r) of f has $\pi_J(r) = c$ with J the class of c, so for every r there is at most one fixed point in each class: at most Md in all. Every other cycle has at least two positions, so $z \le Md + (d^2 - Md)/2$. □

Part (a) is the elimination step of Theorem 5.1: read in coordinates, (1) says that an entry of row b is a combination of entries in the columns b' with $g(b'+1) > b$. Step 3 of that proof shows that a solution which vanishes on I is zero; the same induction, applied to two solutions that agree on I, uses neither that the entries are linear combinations nor that the solutions form a subspace. What does not carry over to sets is step 1, the choice of bases: part (a) is about an order of the coordinates as they are given. Part (b) is Proposition 6.8 for sets, with functions of one stored entry in place of its multiples.

Two instances of part (a). With $g(j) = j + 1$ the helpers of row c lie in the columns $c, \dots, d-1$ and the bound is $q^{d(d+1)/2}$, the number of symmetric matrices: if row c is computed, by any functions, from any number of entries in the columns $c, \dots, d-1$, for every c, the set stays at the pairing value. The bound of [19, Proposition K.5], for a map of the rows followed by a map of the columns, is proved there by a recovery of this kind. On a cycle, let every helper of a position in row c lie in a column $c + e$ with $-B \le e \le B$, read modulo d, where $2B \le d$. Take $g(j) = d$ for $j < B$ and $g(j) = \min(d, j + B + 1)$ for $j \ge B$; then

$$|S| \;\le\; q^{\,d(d+1)/2+2Bd-B(2B+1)}.$$

The rows of the helpers and their number are arbitrary here. So what Section 7.1 calls synchronous access, the reading of source rows in step with the output, gives no more than the pairing value up to a term of order Bd when the offsets are at most B, for sets and for functions that are not linear. For offsets of any size we have a bound only for subspaces and one decoder for all rows (Corollary 5.7 and Theorem 6.4).

Call a recovery relation on the positions a *one-class relation of window* w if there are maps $i_m, j_m \colon \mathbb{Z}/d \to \mathbb{Z}/d$, $1 \le m \le w$, with

$$N(c, r) = \{(i_m(r),\, j_m(r) + c) : 1 \le m \le w\} \qquad \text{for all } c, r.$$

These are the relations of rules with one class whose decoders read at most w entries for each coordinate of the output row. The relation of a rule with M classes of any shape whose decoders read at most w entries for each coordinate is contained in a one-class relation of window Mw, with the helpers of all classes together, and a storage code on a relation is a storage code on every relation with more helpers. For storage codes on one-class relations, the results above give the following.

- For $w = 1$, at most $q^{(d^2+d)/2}$ elements, by part (b).
- If all offsets $j_m(r)$ lie in $\{-B, \dots, B\}$ modulo d, with $2B \le d$: at most $q^{d(d+1)/2+2Bd}$ elements, for every w, by part (a).
- If all differences $i_m(r) - r$ lie in $\{-B, \dots, B\}$ modulo d, with $2B \le d$: the same bound, for every w and all offsets, by part (a) with rows and columns exchanged.
- If the code is a subspace and the function for the position (c, r) does not depend on c, dimension at most $(d^2 + d)/2$, for every w, by Theorem 6.4 in the form for arbitrary decoders.
- For $w \le 2$, at most $q^{d(d+1)/2+d-1}$ elements, by Proposition 6.22(b) below.
- If the offsets $j_m(r)$ take at most $s + 1$ values, or the differences $i_m(r) - \pi(r)$ do for some permutation π: at most $q^{d^2/2+6d^{2-1/(s+1)}}$ elements, for every w, by Proposition 6.22(c) and (d).

The fourth case differs from the general one in two ways: the code is a subspace, and one function serves all rows. For $w \ge 3$ we do not know whether storage codes on one-class relations of window w, over a fixed alphabet, stay at the pairing rate when either of these hypotheses is dropped, or both: this is Problem 9.4(b). A subspace with a function for each position, which may then be taken linear, is the case of rules with boundedly many classes drawn entry by entry. A set with one function for all rows can already exceed the value $(d^2 + d)/2$ of the fourth item, as follows.

Marker rows. A set of matrices can carry markers that tell a decoder the class of its row. Let $\mathcal{X}$ be a code of a class-uniform rule with M classes over a field of q elements, let the decoders read at most w entries for each coordinate, and let $k = \lceil \log_q M \rceil$. Choose k rows $t_1, \dots, t_k$ and pairwise distinct words of k symbols, one for each class, and let S be the set of the matrices $X \in \mathcal{X}$ in which, for every c, the entries $X_{t_1,c}, \dots, X_{t_k,c}$ spell the word of the class of c. On S one function serves all rows: it reads the k entries of $\sigma^c X$ at the positions $(t_a, 0)$, which are the entries $X_{t_a,c}$, learns the class of c, and applies the decoder of that class. So S is a storage code, with one function for all rows, on a one-class relation of window $Mw + k$. If $\mathcal{X}$ is a subspace in which the rows $t_1, \dots, t_k$ are free, S has $q^{\dim \mathcal{X} - kd}$ elements. For the block-symmetric matrices of Proposition 6.6 over $\mathbb{F}_2$, with $M \geq 2$ blocks of side m and the marker rows $0, m, \dots, (k-1)m$, these rows are free, because block symmetry ties an entry of row am only to entries of the other rows of its block. Then S has $2^{(d^2+Md)/2-kd}$ elements and the window is $M + k$. For $M = 8$ that is $2^{d^2/2+d}$ matrices with window 11, where a subspace with one function for all rows has at most $2^{(d^2+d)/2}$. So for sets the bound of the fourth item fails by a multiple of d that grows with the window, and the two cases do not differ as one class and many: one function for all rows can imitate boundedly many classes of rows, where the marker rows are free, and a function for each position gives boundedly many classes drawn entry by entry. The same example over a field of q elements, with $\lceil \log_q M \rceil$ marker rows, has $q^{(d^2+Md)/2-kd}$ matrices.

Order bounds for one-class relations. The acyclic-subgraph bound can be applied to a one-class relation through an order of the rows alone, or of the columns alone. The first settles window two, and the second every relation whose offsets take boundedly many values; with a count of the pairs of positions that help each other, it also shows that a code which uses only such pairs stays at the pairing rate. Call a set of positions *acyclic* if the relation has no directed cycle inside it, a position that is its own helper being a cycle; a storage code over q symbols has at most $q^{d^2-|A|}$ elements for every acyclic set A. We use this in the following form: if H is a function from the positions to a totally ordered set, then the set A of the positions p with $H(p') < H(p)$ for every helper p' of p is acyclic. (Two elements of the code that agree outside A agree at the position of A with the least value of H at which they could differ, since all its helpers lie outside A or have a smaller value.)

Proposition 6.22 (Order bounds for one-class relations)**.** Not formalized; a complete proof is given below.

Let S be a storage code, over an alphabet of q symbols, on a one-class relation with the maps $i_m, j_m \colon \mathbb{Z}/d \to \mathbb{Z}/d$, $1 \leq m \leq w$.

(a) Rows. *For every bijection $\varphi \colon \mathbb{Z}/d \to \{1, \dots, d\}$,*

$$|S| \;\leq\; q^{\sum_r \max_m \varphi(i_m(r))}.$$

(b) Window two. *If $w \leq 2$, then $|S| \leq q^{d(d+1)/2+d-1}$.*

(c) Offsets. *Let $1 \leq u \leq d-1$, $g = \gcd(u, d)$, $j_0 \in \mathbb{Z}/d$ and $B \geq 0$ be such that, for all m and r, the residue $u\,(j_m(r) - j_0)$ is that of an integer of absolute value at most B. Then $|S| \leq q^{d^2/2+2(B+g)d}$. In particular, if the offsets $j_m(r)$, over all m and r, take at most $s+1$ values, where $s \geq 1$, then $|S| \leq q^{d^2/2+6d^{2-1/(s+1)}}$.*

(d) Rows, in the form of (c). *Let π be a permutation of $\mathbb{Z}/d$, and let u, g, B be as in (c) and $b_0 \in \mathbb{Z}/d$ be such that, for all m and r, the residue $u\,(i_m(r) - \pi(r) - b_0)$ is that of an integer of*

absolute value at most B. Then $|S| \le q^{d^2/2+2(B+g)d}$; and if the differences $i_m(r) - \pi(r)$ take at most $s+1$ values, then $|S| \le q^{d^2/2+6d^{2-1/(s+1)}}$.

Proof (not formalized). (a) Take $H(i,e) = \varphi(i)$ and $\Phi(r) = \max_m \varphi(i_m(r))$. A position (c,r) with $\varphi(c) > \Phi(r)$ has all its helpers in rows $i_m(r)$ with $\varphi(i_m(r)) \le \Phi(r) < \varphi(c)$, so it lies in A. The other positions number $\sum_r |\{c : \varphi(c) \le \Phi(r)\}| = \sum_r \Phi(r)$.

(b) We use a lemma on rankings. *Let G be a multigraph with n vertices and at most n edges, in which the two ends of an edge may coincide. Then the vertices can be numbered $1, \dots, n$ so that the sum, over the edges, of the larger of the numbers of the two ends is at most $n(n+1)/2 + n - 1$.* Number from the top: for $k = n, n-1, \dots, 1$, among the k vertices that have no number yet, give the number k to a vertex v_k that lies on the fewest edges of the graph G_k induced on them. Let m_k be the number of edges of G_k, let δ_k be the number of those at v_k, and put $\varepsilon_k = m_k - k$, so that $\varepsilon_{k-1} = \varepsilon_k + 1 - \delta_k$ and $\varepsilon_0 = 0$. An edge has larger number k exactly if it is an edge of G_k at v_k, so the sum of the lemma is

$$\sum_{k=1}^{n} k\,\delta_k = \sum_{k=1}^{n} k\,(1 + \varepsilon_k - \varepsilon_{k-1}) = \frac{n(n+1)}{2} + n\,\varepsilon_n - \sum_{k=1}^{n-1} \varepsilon_k.$$

The numbers of edges at the vertices of G_k sum to at most $2m_k$, so $\delta_k \le \lfloor 2m_k/k \rfloor$. Hence $\varepsilon_k \ge \min(\varepsilon_n, -1)$ for all $k \ge 1$, by induction downwards: if $\varepsilon_k \le -1$ then $2m_k/k < 2$, so $\delta_k \le 1$ and $\varepsilon_{k-1} \ge \varepsilon_k$; and if $\varepsilon_k \ge 0$ then $\delta_k \le 2 + 2\varepsilon_k/k$, so $\varepsilon_{k-1} \ge \varepsilon_k(1 - 2/k) - 1 \ge -1$ for $k \ge 2$. With $\varepsilon_n = 0$ the sum is at most $n(n+1)/2 + n - 1$, and with $\varepsilon_n = -s < 0$ it is at most $n(n+1)/2 - ns + (n-1)s$, which is smaller. This proves the lemma. Now let G be the multigraph on $\mathbb{Z}/d$ with the d edges $\{i_1(r), i_2(r)\}$, $r \in \mathbb{Z}/d$, where the two ends coincide if the two rows do or if $w = 1$. The lemma gives a bijection φ with $\sum_r \max_m \varphi(i_m(r)) \le d(d+1)/2 + d - 1$, and (a) applies.

(c) Put $\psi(e) = u\,(e - j_0) \bmod d$, a multiple of g in $\{0, \dots, d-g\}$, and $H(i,e) = \psi(e)$; every such multiple is the value of ψ at exactly g columns. Write $a(c) = uc \bmod d$. A helper of (c,r) lies in a column $c + j$ with $\psi(c+j) = (a(c) + b) \bmod d$, where b is the integer with $|b| \le B$ in the residue class of $u\,(j - j_0)$. If $B \le a(c) < d - B$ this value is $a(c) + b \le a(c) + B$. So A contains every position (c,r) with $B \le a(c) < d - B - g$ and $\psi(r) > a(c) + B$. For such a c, the number of r with $\psi(r) > a(c) + B$ is $g\,(d/g - 1 - \lfloor (a(c) + B)/g \rfloor) \ge d - g - B - a(c) > 0$. The function $f(x) = d - g - B - x$ is decreasing, and a takes every multiple of g exactly g times, so

$$|A| \;\ge\; g \sum_{t:\ B \le gt < d-B-g} f(gt) \;\ge\; \int_{B+g}^{d-B-g} f(x)\,dx - \frac{g^2}{2} = \frac{(d - 2B - 2g)^2 - g^2}{2}$$

when $d \ge 2B + 2g$: each term $g\,f(gt)$ is at least the integral of f over $[gt, gt+g]$, these intervals cover $[B+g, d-B-g]$, on what they cover before $B+g$ the function is nonnegative, and what they cover after $d - B - g$ has length less than g, where $f > -g$. So $|A| \ge d^2/2 - 2(B+g)d$, which is trivial when $d < 2B + 2g$.

For the second statement let j_0 be one of the values and $t_1, \dots, t_s$ the differences of the others from it, and let $N = \lfloor d^{1/(s+1)} \rfloor$. By Dirichlet's theorem on simultaneous approximation there are integers $1 \le u \le N^s$ and $p_1, \dots, p_s$ with $|ut_k/d - p_k| \le 1/N$ for all k. Then $u \le N^s < d$ (we may take $d \ge 2$), every ut_k is congruent to an integer of absolute value at most $B = \lfloor d/N \rfloor \le 2d^{s/(s+1)}$, and $g \le u \le d^{s/(s+1)}$. The first statement gives the exponent $d^2/2 + 6d^{s/(s+1)} \cdot d$.

(d) Take $H(i,e) = \varphi(i)$ with $\varphi(x) = u\,(x - b_0) \bmod d$, and $a'(r) = u\pi(r) \bmod d$. Every helper of (c,r) lies in a row i with $\varphi(i) = (a'(r) + b) \bmod d$ and $|b| \le B$, so A contains every position with $B \le a'(r) < d - B - g$ and $\varphi(c) > a'(r) + B$. Since π is a bijection, a' takes every multiple of g exactly g times, and the count of (c) applies with the roles of c and r exchanged. The second statement follows as in (c). □

Parts (a) and (b) use only that every helper of a position in column r lies in one of the rows $i_m(r)$, and part (c) only that every helper of a position in row c lies in one of the columns $c+j_m(r)$. So they hold for functions of whole rows and of whole columns. *If every column of a matrix of S is a function of at most two rows of it, the two rows depending on the column, then $|S| \le q^{d(d+1)/2+d-1}$. If every row c is a function of the columns $c+j$, for j in a fixed set of $s+1$ offsets, then $|S| \le q^{d^2/2+6d^{2-1/(s+1)}}$.* In the language of Definition 6.3, the second statement is about one decoder for all rows that reads $s+1$ whole columns of its argument, by any map; with a view of the whole matrix the rate tends to one, as recalled above. With $\varphi(i) = i+1$ for $0 \le i < d$, part (a) contains the third item of the list above. When u is prime to d, the first statements of (c) and (d) follow from the cyclic instance of Proposition 6.21(a) after the rows and the columns have been renumbered (by $uc \bmod d$ and ψ in (c)), with a slightly better exponent; what the two parts add is the case $g > 1$, which the statements about $s+1$ values need. Those two statements say something only when $d > 12^{s+1}$. In terms of the sets $R_k = \varphi^{-1}\{1,\dots,k\}$ and the numbers m_k of columns r with $\{i_m(r)\}_m \subseteq R_k$, the exponent of (a) is $d(d+1)/2 + \sum_{k=1}^{d-1}(k-m_k)$: the pairing value plus the sum of the amounts, of either sign, by which the sets of k rows of the chain $R_1 \subset R_2 \subset \cdots$ fall short of containing the rows of k columns. We call this sum the *deficiency* of the chain.

Part (b) is close to the truth. The ranking lemma is sharp for every n: in a cycle on n vertices any $k < n$ vertices span at most $k-1$ edges, so $\varepsilon_k \le -1$ for $1 \le k < n$ in every numbering, and the identity displayed in the proof gives a sum of at least $n(n+1)/2 + n - 1$. The bound of (b) itself is attained at $d = 3$: with $(i_1, j_1) = (1,2), (0,1), (0,2)$ and $(i_2, j_2) = (2,1), (1,0), (2,0)$ at $r = 0, 1, 2$, six positions are their own helpers and the other three help one another, so the matrices whose entries at those three positions sum to zero form a storage code with q^8 elements over $\mathbb{Z}/q$. For even d, the matrices with $X_{c,r} = X_{r,c}$ whenever $c-r$ is even and $X_{c,r} = X_{r-1,c+1}$ whenever $c-r$ is odd form a storage code on the relation with the helpers (r,c) and $(r-1, c+1)$, with $q^{d^2/2+d}$ elements. Both examples have a function for each position. So for window two and a function for each position the constant C of Problem 9.4(b) lies between 1 and $\frac{3}{2}$; with one function for all rows we only know that it lies between $\frac{1}{2}$ and $\frac{3}{2}$. For functions of whole rows the acyclic-subgraph bound cannot always give a smaller exponent than (b): when every column r is a function of the rows r and $r+1$, at $d = 4$ the least number of positions that meets every cycle is $13 = d(d+1)/2 + d - 1$ (Appendix B).

Call p' a *mutual helper* of p if each of the two positions is a helper of the other.

Proposition 6.23 (Mutual helpers and transitive triangles)**.** Not formalized; a complete proof is given below.

Fix w, and let N be a one-class relation of window w on the positions of a $d \times d$ matrix.

(a) For every $\varepsilon > 0$ there is a set $T \subseteq (\mathbb{Z}/d)^2$ of at most $2w^2/\varepsilon$ pairs such that, for all but at most εd^2 of the ordered pairs (p, p') of mutual helpers, $p' = (r-u, c+v)$ for $p = (c, r)$ and some $(u, v) \in T$.

(b) Let S be a set of matrices over q symbols in which every entry is a function of the entries at the mutual helpers of its position. Then, for every $\varepsilon > 0$, $|S| \le q^{d^2/2+\varepsilon d^2+6d^{2-\varepsilon/(2w^2)}}$. So $|S| \le q^{(1/2+o(1))d^2}$ as $d \to \infty$, uniformly in the relation.

(c) At most wd positions are their own helpers. For every $\varepsilon > 0$ there is a set of at most $\varepsilon d^2 + 2w^6 d/\varepsilon$ positions that meets every transitive triangle*: three distinct positions x, y, z such that x is a helper of y and of z, and y is a helper of z.*

Proof (not formalized). (a) Call $(r - i_m(r),\ j_m(r)) \in (\mathbb{Z}/d)^2$ the *type* of the pair (r, m), and let $n(u, v)$ be the number of pairs (r, m) of type (u, v); these numbers sum to wd. Let $p = (c, r)$ and

$p' = (c', r')$, let p' be the helper of p that belongs to m, and p the helper of p' that belongs to m': $c' = i_m(r)$, $r' = c + j_m(r)$, $c = i_{m'}(r')$, $r = c' + j_{m'}(r')$. Then (r, m) has the type $(r - c', r' - c)$ and (r', m') has the type $(r' - c, r - c')$, its transpose, and $p' = (r - u, c + v)$ for the type (u, v) of (r, m). The pairs (r, m) and (r', m') determine p and p'. So the number of quadruples (p, p', m, m') of this kind is at most $\sum_{(u,v)} n(u, v)\, n(v, u)$. Let $\theta = \varepsilon/(2w)$, and let T be the set of the types (u, v) with $n(u, v) \geq \theta d$ and $n(v, u) \geq \theta d$; it has at most w/θ elements. The terms with $n(v, u) < \theta d$ sum to less than $\theta d \cdot wd$, and so do the terms with $n(u, v) < \theta d$. So fewer than εd^2 quadruples have a type outside T.

(b) For $\varepsilon \geq \frac{1}{2}$ the bound is trivial, so let $\varepsilon < \frac{1}{2}$; then $2w^2/\varepsilon > 2$. Take T from (a), and let E be the set of the positions that have a mutual helper not of the form given there; $|E| \leq \varepsilon d^2$. The matrices of S with given entries on E form a storage code on the relation in which a position of E has no helper and every other position (c, r) has its mutual helpers outside E as helpers, all of the form $(r - u, c + v)$ with $(u, v) \in T$: entries on E are constants. This relation is contained in the one-class relation with the maps $i(r) = r - u$, $j(r) = v$ for $(u, v) \in T$, whose row differences take at most $|T|$ values. By Proposition 6.22(d), with π the identity, such a set has at most $q^{d^2/2 + 6d^{2-1/K}}$ elements, where $K = \max(|T|, 2) \leq 2w^2/\varepsilon$. There are at most $q^{\varepsilon d^2}$ choices of the entries on E.

(c) The position (c, r) is its own helper through m exactly if $c = i_m(r)$ and $j_m(r) = r - c$, which holds for at most one c for each pair (r, m). For the second statement let $z = (c, r)$, let y be the helper of z that belongs to m, and x the one that belongs to m', and let x be the helper of y that belongs to m''. With $r' = c + j_m(r)$, the column of y, this says $i_{m''}(r') = i_{m'}(r)$ and $i_m(r) + j_{m''}(r') = c + j_{m'}(r) = r' - j_m(r) + j_{m'}(r)$, that is, $\alpha(r) = \beta(r')$ for the maps

$$\alpha(r) = \big(i_{m'}(r),\ j_{m'}(r) - j_m(r) - i_m(r)\big), \qquad \beta(r') = \big(i_{m''}(r'),\ j_{m''}(r') - r'\big)$$

from $\mathbb{Z}/d$ to $(\mathbb{Z}/d)^2$. The pair (r, r') determines z, and x lies in the row that is the first coordinate of the common value. So, for fixed (m, m', m''), the number of such configurations is $\sum_\xi |\alpha^{-1}(\xi)|\, |\beta^{-1}(\xi)|$. Let $\theta = \varepsilon/(2w^3)$. At most $1/\theta$ values ξ have $|\alpha^{-1}(\xi)| \geq \theta d$ and $|\beta^{-1}(\xi)| \geq \theta d$, and the other values contribute fewer than $2\theta d^2$ configurations. Over the w^3 triples (m, m', m''): in all but fewer than εd^2 configurations x lies in one of at most $w^3/\theta = 2w^6/\varepsilon$ rows. Take these rows, and the position z of every other configuration. □

So a code that uses only helpers in both directions, as the storage codes of an undirected graph do, cannot exceed the pairing rate on a one-class relation of bounded window; the explicit bound of (b) says something only for very large d, and the statement is about the rate. The relation of the code of rate 7/13 on a graph of degree four [11, Example 7.8] is of this kind, and so is every other undirected construction; the relations of Section 3.5 other than its undirected graphs are directed, and Proposition 6.23 does not decide whether relations like them can be one-class relations. In an affine relation, with $i_m(r) = a_m r + b_m$, $j_m(r) = \alpha_m r + \beta_m$ and every a_m invertible modulo d, part (c) can be read off, and for prime d part (a) as well: the helper maps are the invertible affine maps whose linear part sends the first basis vector to the second, the composition of two of them is never one of them, and one of them has an inverse of this kind only if its linear part is the swap. The parity code on a triangle and the Hamming code of length seven on a tournament, two local codes of rate above one half, both contain transitive triangles; so by (c) copies of them can cover only $o(d^2)$ positions of a one-class relation of bounded window: every copy contains a position of the set of (c), and all positions of a copy are reached from any one of them along helpers, of which a position has at most w.

Part (c) also connects one-class relations with the seventh paper, whose theorems are about relations in which no position is its own helper and there is no transitive triangle. When the entries at the positions of (c) are fixed, the other entries form a storage code on a relation of that kind.

So, for instance, a storage code on a one-class relation of bounded window in which, outside the positions of (c), the positions with three or more helpers form an acyclic set has at most $q^{(1/2+o(1))d^2}$ elements [11, Theorem 5.3]; for $w \le 2$ this is weaker than Proposition 6.22(b), which has an error term linear in d and needs no positions fixed. And the conjecture of that paper for relations with at most three helpers at each position [11, Conjecture 9.1] would give window three in the form $(\frac{1}{2} + o(1))d^2$. With four helpers no statement of this kind holds for relations in general, by the example just cited; that example has mutual helpers only, and these are excluded here by (b).

What the order argument does not reach. Proposition 6.22 answers Problem 9.4(b) for window two, and for every window when the rows $i_m(r)$ have a chain of small deficiency or the offsets are made short by a multiplier. We say that the rows and the offsets of a relation are both *spread* when neither is the case. It cannot be expected to answer the question for such relations in general. For relations with random maps we searched for large acyclic sets, with a solver under a time limit (Appendix B). The table below gives the sizes of the sets found for four families of one-class relations and for a digraph without any structure, in which every position has w helpers at random. In the first family, with zero offsets, an acyclic set of $d(d-1)/2$ positions exists by Proposition 6.21(a) with $g(j) = j + 1$, and the search finds larger ones; on a second control, the pairing together with eleven permutations at zero offsets, where the largest acyclic set has exactly $d(d-1)/2$ positions, it returns that value at $d = 17$, 23 and 29. In the three families whose rows and offsets are both spread the sets found have about $0.4\,d^2$ positions at window twelve for every d in the table, as in the digraph without structure, and the distance to $d(d-1)/2$ grows with d; at window twenty-four they have between $0.25\,d^2$ and $0.32\,d^2$. These are results of a search and not bounds: they indicate, and do not prove, that for such relations and large windows every set that meets all cycles has more than $(\frac{1}{2} + c)d^2$ positions, for some $c > 0$. For one instance the computation gives a proof: in the relation of the family of permutations and offsets at $d = 13$ and $w = 12$, every set of positions that meets all cycles has at least 99 elements, where $d(d+1)/2 = 91$; there the window is almost as large as d.

d	w	zero offsets (control)	permutations and offsets	affine	random	no structure	$d(d-1)/2$
13	12	99	69	64	70	72	78
17	12	176	118	111	118	121	136
23	12	327	211	202	223	219	253
29	12	515	339	323	353	349	406
17	24	160	92	74	73	77	136

Numerical: sizes of the largest acyclic sets found by a search under a time limit (60 seconds for $d \le 17$, 120 seconds otherwise; one run each, and a rerun differs by a few positions) in relations on the d^2 positions. Zero offsets: random permutations as row maps. Permutations and offsets: alternately a random permutation with zero offset and the identity with random offsets. Affine: maps $(ar + b,\ c + \alpha r + \beta)$ with random coefficients. Random: random maps i_m, j_m.

With the pairing and two, three or five further helpers, whose row maps are random permutations and whose offsets are random, the least number of positions that meets every cycle is exactly $d(d+1)/2$ in all eleven instances that we computed, for $9 \le d \le 17$; there the lower and the upper bound of the computation coincide. So for small windows the acyclic-subgraph bound may still settle the question, and for large windows, where a relation with random maps behaves like a random digraph, it very likely cannot; for relations in which whole rows may be read this is proved for degree four (Proposition 6.26). There a bound would need an argument of another kind. A code above the

pairing rate would have to use helpers in one direction, by Proposition 6.23(b), on a relation whose rows and offsets are both spread, by Proposition 6.22, and it could not consist of copies of a code on boundedly many positions that contains a transitive triangle and in which every position is reached from every other along helpers, by Proposition 6.23(c).

Whole rows, and stages. The order of the rows in Proposition 6.22(a) uses only the rows that a position reads, not the entries. Stated for whole rows it bounds what an operator can transpose when its output rows are arbitrary functions of few input rows, and it adds up over the stages of a pipeline. A *row relation* ρ assigns to every $r \in \mathbb{Z}/d$ a set $\rho(r) \subseteq \mathbb{Z}/d$. A set S of $d \times d$ matrices over an alphabet is a *ρ-code* if, for every r, the column r of a matrix of S is a function of its rows in $\rho(r)$; this is a storage code on the relation in which the helpers of (c, r) are all positions of the rows in $\rho(r)$. A storage code on a one-class relation with the row maps i_m is a ρ-code for $\rho(r) = \{i_m(r) : m \le w\}$, whatever its offsets. And if K is a map of the $d \times d$ matrices over the alphabet to themselves whose row r depends only on the rows $\rho(r)$ of its argument, then the set $\{X : K(X) = X^{\mathsf{T}}\}$ of the matrices that K transposes is a ρ-code, and every ρ-code is contained in such a set. Put

$$\sigma(\rho) = \min_{\varphi} \sum_{r} \max \varphi(\rho(r)),$$

the minimum over the bijections $\varphi\colon \mathbb{Z}/d \to \{1, \dots, d\}$, with $\max \emptyset = 0$.

Proposition 6.24 (The order bound for whole rows)**.** Not formalized; a complete proof is given below. *Let ρ be a row relation and S a ρ-code over an alphabet of q symbols.*

(a) $|S| \le q^{\sigma(\rho)}$. *More generally, if every column r is, on S, a function of the rows in $\rho(r)$ and of a quantity $\alpha(X)$ that is the same for all r and takes at most N values, then $|S| \le N\, q^{\sigma(\rho)}$.*

(b) Exactness. *The largest acyclic set of the relation in which the helpers of (c, r) are all positions of the rows in $\rho(r)$ has exactly $d^2 - \sigma(\rho)$ positions. So (a) is the acyclic-subgraph bound of that relation, and no order of the positions gives more than an order of the rows.*

(c) Blocks. *If the rows are partitioned into blocks of at most s rows and every $\rho(r)$ lies inside one block, then $\sigma(\rho) \le (d^2 + sd)/2$. If the blocks have $s_1, \dots, s_b$ rows, π is a bijection of $\mathbb{Z}/d$ and $\rho(r)$ is the whole block of $\pi(r)$, then $\sigma(\rho) = (d^2 + \sum_\kappa s_\kappa^2)/2$, and there is a ρ-code with exactly $q^{\sigma(\rho)}$ elements.*

(d) Degree. *If $|\rho(r)| \le w$ for all r, then $\sigma(\rho) \le w\, d(d+1)/(w+1)$.*

Proof (not formalized). (a) Fix φ and put $m_r = \max \varphi(\rho(r))$. Take $H(c, r) = \varphi(c)$ in the form of the acyclic-subgraph bound stated before Proposition 6.22: the positions (c, r) with $\varphi(c) > m_r$ form an acyclic set, and the others number $\sum_r m_r$. For the second statement apply the first to each of the sets on which α is constant.

(b) Let A be an acyclic set, and number its positions so that helpers come first. Let $\tau(i)$ be the largest number of a position of A in row i, and $\tau(i) = -\infty$ if there is none, and let φ order the rows by τ. If $(c, r) \in A$ and $i \in \rho(r)$, then $i \ne c$, since a position that is its own helper is a cycle, and every position of A in row i is a helper of (c, r); so $\tau(i)$ is less than the number of (c, r), which is at most $\tau(c)$. Hence $\varphi(c) > m_r$ for every position of A, and $|A| \le \sum_r (d - m_r)$.

(c) Order the blocks at random and the rows inside a block arbitrarily. If $\rho(r)$ lies in a block of s' rows, then m_r is at most the place of the last row of that block, whose mean is $s' + (d - s')/2 \le (d + s)/2$. For full blocks and any order φ, let ℓ_κ be the place of the last row of the block B_κ, and number the blocks so that $\ell_1 < \ell_2 < \cdots$. Then $m_r = \ell_\kappa$ for the s_κ indices r with $\pi(r) \in B_\kappa$, and $\ell_\kappa \ge s_1 + \cdots + s_\kappa$,

so $\sum_r m_r \ge \sum_\kappa s_\kappa(s_1+\cdots+s_\kappa) = (d^2+\sum_\kappa s_\kappa^2)/2$, with equality when the blocks follow one another. For the code let B_κ be the blocks and $\Gamma_\kappa = \pi^{-1}(B_\kappa)$. Leave the entries at (c,r) with $c \in B_\kappa$, $r \in \Gamma_\lambda$ free for $\kappa \le \lambda$, and put $X_{c,r} = X_{\pi(r),\pi^{-1}(c)}$ for $c \in B_\lambda$, $r \in \Gamma_\kappa$, $\kappa < \lambda$. For $r \in \Gamma_\kappa$ and any c, either $c \in B_\kappa = \rho(r)$, or $X_{c,r} = X_{\pi(r),\pi^{-1}(c)}$ holds on the set and $\pi(r) \in \rho(r)$. The free entries number $\sum_{\kappa\le\lambda} s_\kappa s_\lambda$.

(d) For a bijection φ chosen uniformly at random the largest of k distinct places has mean $k(d+1)/(k+1)$, which increases with k. □

Part (d) is the bound $1-1/(\Delta+1)$ of the fourth paper [6, Proposition 4.14(b)] with rows as units: an operator whose rows each read at most w rows transposes at most $q^{wd(d+1)/(w+1)}$ matrices, although each of its entries may depend on wd entries. For $w=1$ this is the half bound. For $w=2$ and $d \ge 4$ it is not sharp: the ranking lemma in the proof of Proposition 6.22(b) gives $\sigma(\rho) \le d(d+1)/2+d-1$. For $\rho(r) = \mathbb{Z}/d \setminus \{r\}$ it is sharp: $\sigma(\rho) = d^2-1$, and over a group the matrices whose diagonal sums to zero are a ρ-code.

Proposition 6.25 (Stages)**.** Not formalized; a complete proof is given below.

Let $d = 2^n$ and identify the row indices with $\mathbb{F}_2^n$. Let $K = K_T \circ \cdots \circ K_1$, where K_s maps arrays of rows indexed by a set I_{s-1} to arrays of rows indexed by I_s, the rows of the arrays between the stages being elements of arbitrary sets, and the arrays at the two ends being the $d \times d$ matrices over an alphabet of q symbols, with $I_0 = I_T = \mathbb{F}_2^n$.

(a) Cosets. *Let $I_s = \mathbb{F}_2^n$ for all s, and let the row r of the output of K_s depend only on the input rows in $\pi_s(r) + V_s$, where $\pi_s(x) = P_s x + a_s$ is an affine map of $\mathbb{F}_2^n$, not necessarily invertible, and V_s is a subspace. Put $t = \sum_s \dim V_s$. Then K transposes at most $q^{(d^2+2^t d)/2}$ matrices. For every $t \le n$ this is attained, with $\dim V_s = 1$ for t stages: the partial transpose that exchanges t digits of the row address with the same digits of the column address transposes exactly the $q^{(d^2+2^t d)/2}$ matrices that are symmetric in the other $n-t$ digits.*

(b) Scans. *In (a), let u of the stages be* scans *instead: a scan has an affine bijection λ of $\mathbb{F}_2^n$, the* place *of the row r being the integer whose binary digits are those of $\lambda(r)$, a set of at most q^m buffer states with a fixed first state β_0, and maps f_j, g_j; its output row at the j-th place is $f_j(\beta_j, x_j)$, where x_j is the input row at that place, and $\beta_{j+1} = g_j(\beta_j, x_j)$. Let t be the sum of $\dim V_s$ over the other stages. Then for every integer $0 \le k \le n$ the number of matrices that K transposes is at most $q^{(d^2+2^{t+uk}d)/2+umd/2^k}$. If $m \le d$ and K transposes at least $q^{(1/2+\delta)d^2}$ matrices, then $t + (u+1)\log_2\bigl((1+4u)/(2\delta)\bigr) \ge \log_2 d$.*

(c) Heights. *Let $h_s\colon I_s \to \mathbb{Z}$ be functions with $h_0 = h_T = h$, and let $\omega_s \ge 0$ be such that the row r of the output of K_s depends only on the input rows i with $h_{s-1}(i) \le h_s(r) + \omega_s$. Put $\Omega = \sum_s \omega_s$, and let W be the largest number of elements of $\mathbb{F}_2^n$ on $\Omega+1$ consecutive levels of h. Then K transposes at most $q^{d(d+1)/2+(W-1)d}$ matrices. If h is the number of ones of the address, then $W \le (\Omega+1)\sqrt{2/(\pi n)}\, d$.*

Proof (not formalized). (a) The row r of $K_2(K_1(X))$ depends on the rows of $K_1(X)$ in $\pi_2(r)+V_2$, hence on the rows of X in $\pi_1(\pi_2(r)) + P_1V_2 + V_1$. By induction the row r of $K(X)$ depends on the rows of X in a coset of a subspace of dimension at most t; these cosets are blocks of at most 2^t rows, and Proposition 6.24(a) and (c) apply. For the partial transpose write a row index as (a,v) with v the t digits. It sends X to the matrix with the entry $X_{(a,v'),(b,v)}$ at $((a,v),(b,v'))$, one digit at a time; the exchange of one digit reads, for the output row r, the two rows that agree with r outside that digit. The matrices that it transposes are those with $X_{(b,v'),(a,v)} = X_{(a,v'),(b,v)}$, and there are $q^{2^{2t}d'(d'+1)/2}$ of them, $d' = d/2^t$.

(b) Let the advice $\alpha(X)$ consist, for every scan, of its buffer states at the places j that are multiples of 2^k, $0 < j < d$, when the pipeline runs on X; it takes at most $q^{umd/2^k}$ values. For a fixed value of the advice, the output row of a scan at the place j is a function of the input rows at the places from $2^k\lfloor j/2^k \rfloor$ to j. The rows at these places lie in the coset $r + L^{-1}E_k$, where r is the row at the place j, L is the linear part of λ and E_k is the span of the k lowest digits. So for a fixed value of the advice the pipeline is of the kind of (a) with $t + uk$ in place of t, and Proposition 6.24(a), in its second form, gives the bound. For the last statement let $u \geq 1$ and $k = \lfloor (n-t)/(u+1) \rfloor$, there being nothing to prove if $t \geq n$. Then $2^{t+uk}d \leq d^2(2^t/d)^{1/(u+1)}$ and $umd/2^k < 2ud^2(2^t/d)^{1/(u+1)}$, so $\delta < (\frac{1}{2} + 2u)(2^t/d)^{1/(u+1)}$. For $u = 0$ it is (a).

(c) By induction over the stages, every row i of X that the row r of $K(X)$ depends on has $h(i) \leq h(r) + \Omega$. Let φ order $\mathbb{F}_2^n$ by h. Then $\varphi(i)$ is at most the number of x with $h(x) \leq h(r) + \Omega$, and $\varphi(r)$ is more than the number of x with $h(x) < h(r)$, so the largest place of a row read for r is at most $\varphi(r) + W - 1$; sum over r and apply Proposition 6.24(a). The central binomial coefficient is at most $2^n\sqrt{2/(\pi n)}$. □

What the parts cover. A map that acts on each row separately, by any code, has $\dim V = 0$ in (a), and so has every permutation of the rows that is affine on addresses: the permutations of the address digits, which are the perfect shuffles and unshuffles of the rows at every scale, and the complements of digits. A tape shuffle that moves one digit between the row address and the column address has $\dim V = 1$ (the new row agrees with two old rows outside one digit), and so has a stage in which the row r is any function of the rows r and $r + v$. So by (a) a pipeline that transforms one array through stages of these kinds, with arbitrary codes between them, transposes a set of rate $\frac{1}{2} + \delta$ only if it moves at least $\log_2 d - \log_2(1/(2\delta))$ digits, as the recursion by digit exchanges does. Corollary 6.13 gives the same number for one digit and for linear codes and decoders, and Remark 6.14 explains why commutator rank cannot add the digits up; here the dimensions add up, for all codes. A scan with a buffer of one row, forward or backward, reads all earlier rows, and (b) charges it through advice: scans alone, in any address-affine orders, need about $\log d / \log\log d$ stages. (For scans in one order far more is true: scans in one direction never exceed $d(d+1)/2$, by Proposition 6.24(a) with the order of the scan.)

Part (a) requires that a stage reads rows of one array inside one coset. A stage that combines an array with a differently arranged copy of it does so only with a large coset: for the shuffle parents $\rho(r) = \{\lfloor r/2 \rfloor, \lfloor r/2 \rfloor + d/2\}$ the T-fold composition reads cosets of dimension T, with $\sigma = 144, 160, 192, 256$ at $d = 16$ for $T = 1, \dots, 4$, and with the row r itself added to $\rho(r)$ the only coset is the whole space already for $T = 1$, and $\sigma = 161, 192, 221$ for $T = 1, 2, 3$. Such stages are covered by (c), more weakly. With the number of ones of the address as the height, on every copy, a permutation of the address digits of any copy costs nothing, nor does a stage that combines, row by row, any number of copies arranged by different permutations of the digits; a stage that moves one digit, flips one, or adds one digit to another costs one; and a scan cut every 2^k rows, in an order by permuted digits, costs k, with the advice of (b). So for pipelines without scans that carry copies and mix them the total cost satisfies $\Omega \geq (\delta + 1/(2d))\sqrt{\pi \log_2 d/2} - 1$ at rate $\frac{1}{2} + \delta$: at least that many stages can raise the number of ones, if each raises it by at most one. A height that is invariant under all permutations of the digits has levels that are unions of those of the number of ones, so its W is at least the central binomial coefficient and the bound of (c) is never below $d^2/2 + d^2/\sqrt{2\log_2 d} - d$; whether pipelines of this kind need $\log d$ such stages is open. The least dimension of a coset that contains the rows read is not the quantity that Problem 9.5 asks for: it is subadditive over stages by (a), but a scan, and a stage that reads a row together with its shuffle parents, each raise it to n in one linear-time operation (for the second, the differences $r + \lfloor r/2 \rfloor$ in $\mathbb{F}_2^n$ span the whole space).

Proposition 6.26 (Expanding row relations)**.** Cited: part (b) applies [38, Theorems 1 and 2], the second of which rests on [3]; we use the statements and have not checked their proofs. Part (a) is proved below and is not formalized.

(a) Let ρ be a row relation such that $\pi(r) \in \rho(r)$ for a bijection π and $|\bigcup_{r\in R} \rho(r)| \geq (1+\varepsilon)|R|$ for every set R of at most $d/2$ indices. Then

$$\sigma(\rho) \;\geq\; \frac{d(d+1)}{2} + \sum_{j=0}^{\lfloor d/2 \rfloor} \Big(j - \Big\lfloor \frac{j}{1+\varepsilon} \Big\rfloor \Big) \;\geq\; \frac{d(d+1)}{2} + \frac{\varepsilon}{1+\varepsilon} \cdot \frac{\lfloor d/2 \rfloor (\lfloor d/2 \rfloor + 1)}{2}.$$

(b) There are $\varepsilon > 0$ and, for infinitely many d, three partial maps f_1, f_2, f_3 of $\{0, \ldots, d-1\}$ to itself, each increasing and injective where it is defined, such that $\rho(r) = \{r, f_1(r), f_2(r), f_3(r)\}$ satisfies the hypothesis of (a). For these, $\sigma(\rho) \geq (\frac{1}{2} + \frac{\varepsilon}{8(1+\varepsilon)} - o(1))d^2$.

Proof (not formalized). (a) For a set A of rows let $n(A)$ be the number of r with $\rho(r) \subseteq A$. With $A_j = \varphi^{-1}\{1, \ldots, j\}$ one has $\max \varphi(\rho(r)) > j$ exactly if $\rho(r) \not\subseteq A_j$, so $\sum_r \max \varphi(\rho(r)) = \sum_{j=0}^{d-1} (d - n(A_j))$. Since π is injective, $n(A) \leq |A|$. If $|A| = j \leq d/2$, the set R of the r with $\rho(r) \subseteq A$ has at most $d/2$ elements and its rows lie in A, so $n(A) \leq j/(1+\varepsilon)$.

(b) By the theorem of Bourgain and Yehudayoff on monotone expanders [3], in the form with three monotone matchings of Dujmović, Sidiropoulos and Wood [38, Theorems 1 and 2], there is an infinite family of bipartite graphs on two ordered classes of equal size d, each the union of three matchings without crossing edges, in which every set R of at most $d/2$ vertices of the first class has at least $(1+\varepsilon)|R|$ neighbours. A matching without crossing edges is an increasing partial injection. Adding the identity keeps the expansion and gives the bijection. □

A map K whose row r is a function of the rows $r, f_1(r), f_2(r), f_3(r)$ takes three *gather passes* and a pass that acts on each row together with the three rows gathered next to it: in a gather pass the head on the array and the head on the work tape both move forward only, and the row $f_m(r)$ is copied next to the place of the row r, because f_m is increasing. The passes need, for each m, the two strings of d bits that mark where f_m is defined and which values it takes; we have not checked that a machine can produce these strings for the maps of (b) in linear time. With that reservation, Proposition 6.26 shows that the acyclic-subgraph bound, in any form by Proposition 6.24(b), cannot show that operators made of boundedly many linear-time passes transpose at most $q^{(1/2+o(1))d^2}$ matrices, even with whole rows as units. That such expanders exist answered a question of Galil, Kannan and Szemerédi on the modelling of multitape machines [39, 38]; what is added here is their use as row relations. The shuffle parents belong to the same family of unions of monotone maps and do not expand.

Proposition 6.27 (Products of bounded pieces)**.** Not formalized; a complete proof is given below.

Let ρ be a row relation such that every $\rho(r)$ has at most w elements and every row lies in $\rho(r)$ for at most w indices r. Let the positions be partitioned into pieces of at most m positions, and let S be a product of sets S_Q, one for each piece Q, of assignments of symbols to the positions of Q, such that in S_Q every position of Q is a function of those positions of Q that lie in the rows of ρ of its column. Then $|S| \leq q^{(d^2 + m w^{2m-1} d)/2}$.

Proof (not formalized). Fix a piece Q and let B be the bipartite graph on the column indices and the row indices that occur in Q, as two disjoint classes, in which r is adjacent to i when $i \in \rho(r)$. For $p = (c, r) \in Q$ let the tail of p be the component of the column index r and its head the component of the row index c. If $p' = (i, c') \in Q$ is a helper of p, then $i \in \rho(r)$, so the head of p' is the tail of p.

Number the components, and call p forward if its tail has the smaller number, backward if its head has, and a loop if they are equal. Given the values at the loops and at the backward positions, the forward positions are determined in the order of their tails, since the helpers of a forward position with tail β have head β and are loops, backward positions, or forward positions with a smaller tail. The same holds with forward and backward exchanged, so $\log_q |S_Q| \le (|Q| + \ell_Q)/2$ with ℓ_Q the number of loops. A loop is a position (c, r) whose row index and column index are joined in B, hence by a path of odd length at most $2m - 1$ in the bipartite graph of ρ, and there are at most dw^ℓ paths of length ℓ that start at a column index; so at most $mw^{2m-1}d$ positions of the matrix are loops of their pieces. □

So no tiling by codes on boundedly many positions (a triangle with a parity, a Hamming code on a tournament) exceeds the pairing rate on such a relation; if only $|\rho(r)| \le w$ is assumed, the same argument gives $q^{(1/2+o(1))d^2}$, since at most wd/D rows lie in more than D of the sets $\rho(r)$ and the pieces that meet them cover at most mwd^2/D positions. The statement is about products. It says nothing about intersections of local codes whose checks depend on each other across pieces, which is how the codes of the thirteenth paper exceed one half on relations that are locally bipartite [13] (Section 3.5). What these four propositions leave is Problem 9.4(c): whether operators whose rows read boundedly many rows transpose more than the pairing rate. By Proposition 6.22(b) they do not for two rows, by Proposition 6.24(c) and Proposition 6.25 not when the rows read lie in small blocks or change the height by little, and by Proposition 6.26 the order argument does not decide it for three increasing maps and the identity.

7 The construction side

This section says what the two theorems exclude, what they leave, and what would have to be added to make them statements about machines. Readings of the one-hop model as statements about tapes are labelled as readings.

7.1 Mechanisms that fail

Table 2 lists upper bounds on the code dimension for decoders with structure on one index, in the linear one-hop model, whatever is done on the other index. Each row is a theorem of Sections 5 and 6.

Two design rules follow, for the one-hop model.

- *Synchronous access is the pairing.* Reading any number of source rows in step with the output, at any offsets and through any convolution, gives the symmetric matrices and no more. With offsets of at most B the same holds, up to a term of order Bd, for sets of matrices and for functions that are not linear (Proposition 6.21(a)); and with offsets of any size that take boundedly many values it holds up to a term $o(d^2)$, whatever rows are read (Proposition 6.22(c)).

- *Uniform passes are the pairing.* A stretch on which translated outputs share their decoder gains nothing, whatever its window. The bound rises only with changes of regime, by $d/2$ symbols for each. On a cycle, a change that does not affect a whole row is charged by the entry: half a symbol for each output entry whose decoder changes from one row to the next (Proposition 6.15), which is not a bound for structure on one index and is not in Table 2.

Structure on one index	Upper bound	By
At most two Kronecker terms	$d(d+1)/2$	Corollary 5.5
Commuting maps: any convolutions of any number of rows read in step	$d(d+1)/2$	Corollary 5.7
One p-group in characteristic p: one dyadic hierarchy over $\mathbb{F}_2$	$d(d+1)/2$	Corollary 5.7
All affine index maps, $d = p^m$, characteristic p	$d(d+1)/2$	Corollary 5.8
Expansion factor c along some flag	$(1-\frac{1}{2c})d^2 + O(d)$	Corollary 5.9
Uniform decoders with M interval pieces, cycle or line	$(d^2+Md)/2$	Theorems 6.4 and 6.9
Copy rules with M decoder classes of any shape	$(d^2+Md)/2$	Proposition 6.8
s modulation frequencies	$\frac{s}{s+1}(d^2+d)$	Corollary 6.12
One address digit of size q treated uniformly	$\frac{1}{2}(d^2+d^2/q)$	Corollary 6.13
Sets: every column a function of at most two rows, by any map	$d(d+1)/2+d-1$	Proposition 6.22(b)
Sets: every entry a function of entries at $s+1$ offsets from its row, in any rows	$d^2/2 + 6d^{2-1/(s+1)}$	Proposition 6.22(c)
Sets: every column a function of whole rows inside one block of s rows; after t moved digits (Proposition 6.25), $s = 2^t$	$(d^2+sd)/2$	Proposition 6.24(c)
Sets: every column a function of at most w whole rows	$\frac{w}{w+1}(d^2+d)$	Proposition 6.24(d)

Table 2: Upper bounds on the code dimension in the linear one-hop model. The other index is arbitrary. The last four rows bound the logarithm, to the base of the alphabet size, of the number of matrices of a set, for arbitrary functions.

7.2 A reading for tapes

The following is a reading, not a theorem about machines. A pass in which the position read and the position written advance at the same speed, by a linear finite-state rule, computes a convolution with an unbounded window. On a line this is a two-sided rule in the sense of Theorem 6.9, with one class per stretch of unchanged regime, under two provisos: the regime changes at the same positions for every output row (change points that depend on the row need their common refinement), and the rule of each regime is applied to the whole stored word, with no state carried across a change. On a cycle the common refinement is not needed: Proposition 6.15 lets every output row have its own change points and charges half a symbol for each. We have not proved a form of Proposition 6.15 on a line. A state of q symbols carried across a change alters K by a rank of at most q per output row, so by Lemma 6.1 it costs a further term of order qd per change. Under the provisos, Theorem 6.9 bounds the gain over the pairing value by $d/2$ symbols per regime change, so rate $\frac{1}{2}+\delta$ needs at least $2\delta d$ regime changes, and for equal pieces the bound is attained by the block-symmetric matrices, that is, by the coarse step of the folklore recursion. Other speed ratios are decimations. On a cycle a single ratio is one twist class of Theorem 6.11; mixtures of ratios fall under Corollary 5.9 or Corollary 5.8.

This reading matches what the series proved about motion (Section 3.2): locally affine motion is excluded on the geometric side by the second and fourth papers, and here uniform decoding is excluded on the coding side, in a model that has no geometry at all. The seventh paper has a statement of the same kind for storage codes on relations: Cayley relations $\mathrm{Cay}(G, \{s, s^{-1}, t\})$ without transitive triangles give no gain over the pairing rate, for every finite group [11, Corollary 5.12]. Theorem 6.4 allows any degree and any window, but only one hop and only codes that are subspaces;

the decoders may be arbitrary maps (Section 6.7).

7.3 What a linear-time machine would need

Combining Section 4.5 with Table 2: in addition to what Section 4.5 lists (for a linear code, positive characteristic or coefficients that cannot be rescaled to be bounded; curved motion; expanding cone relations; growing paired distance), a linear-time transposer, read in the one-hop model, needs on *both* indices a family of maps that

- has no invariant flag and no pair of flags with bounded expansion (Theorem 5.1);
- commutes with no non-derogatory translation up to $o(d)$ boundary columns, and intertwines none with boundedly many pairwise commuting others (Theorems 6.4 and 6.11);
- treats no address digit uniformly (Corollary 6.13).

The digit masks at every scale, which the folklore recursion uses, have these properties: their number grows with d, and no shift respects them. A bounded number of classes is not enough for rate near one, whatever their shape, because M classes leave a flag with $g(i) \le Mi$ and hence rate at most $1 - 1/(2M) + O(1/d)$. A later paper of the series sharpens this, for decoders that need not be linear: two classes of any shape have rate at most $\frac{41}{64} + O(d^{-1/2})$, and $M \le d/2$ classes at most $1 - 1/(4\lceil\sqrt{2M}\,\rceil)$, so that rate $1 - \varepsilon$, for $\varepsilon \le \frac{1}{8}$, needs more than $d/2$ classes or more than $1/(128\varepsilon^2)$ [19, Theorems 3.4 and 3.5]. What the theorems leave open is how large a code a given number of classes can carry when the classes have no almost-period. It can exceed the value $(d^2 + Md)/2$ of every copy rule and of every interval rule at $d = 6$ (Proposition 8.2); at $d = 5$ over $\mathbb{F}_2$ it does not, on the word of a solver for one partition (Remark 8.1). A gain of one dimension at $d = 6$ exists over every field in which $3 \neq 0$, and it persists in proportion when the number of classes grows with d: for $d = 6k$ and $2k$ classes there are codes of dimension $25k^2$, against $24k^2$ [14, Theorem 3.2 and Corollary 4.4]. For every fixed number $M \ge 2$ of classes the gain grows with d as well: for two classes it is at least $d/4 - 1$ at every $d = 4k \ge 8$, and for M classes there are partitions with a gain of order $Md/8$, over every field (Proposition 6.17 and Corollary 6.19). None of these gains exceeds the order of dM, which is the order of the term $Md/2$ that intervals give. Whether a gain survives in rate for $o(d)$ classes is Problem 9.1(a).

7.4 From the one-hop model to machines

Three steps separate Theorem 6.4 from a statement about oblivious machines.

1. *A hypothesis on cone systems.* The coding bounds of the series use properties of the cone relation, and Section 3.5 shows that the relation-level properties proved so far cannot carry the conclusion. "Translated receivers share their decoder on every stretch of periodic head motion" is a property of the rule. It has to be stated for the cone systems of real schedules, where the receivers are output bits and the decoders are compositions of the transition function. For such decoders, which are not linear, and for the inputs that share a value of advice, which do not form a subspace, the sharing of a decoder is not enough when the decoder sees the whole stored word (Section 6.7). Whether it is enough for decoders that read boundedly many bits, as the decoders of bounded cones do, is open already in the one-hop model. So is the stronger statement in which the decoders may differ and only the positions read by translated receivers are translates; that is a property of the relation after all, though not one of those that Section 3.5 shows to be insufficient (Problem 9.4(b)). The stronger statement is proved when

every output entry reads at most two stored entries, and for every window when the offsets of the entries read take boundedly many values (Proposition 6.22).

2. *Depth reduction.* The genie of Section 3.3 reveals values at chosen steps. The revealed word is advice that depends on position, and Theorem 6.4 has no place for it. Revision `b2c389a` of this paper said that a version of Theorem 6.4 with advice would have to charge each revealed bit as a regime change, at half a row per change. Both halves of that sentence need correction, already in the one-hop model.

   - *Advice need not change the regime.* The subsequent memo [14] bounds the cost of advice in the one-hop model, in theorems proved in Lean. If every row may add to its class-uniform term a linear function of its own of $\alpha(X)$, for a linear map α of rank q, the bound of Theorem 6.4 rises by at most q dimensions [14, Theorem 5.2]. And every code can gain the full rank: a code together with the image of a projection p that vanishes on it is a code with advice p, of dimension larger by the rank of p [14, Proposition 5.3]. Over a finite field the advice may be any map to a set of $|F|^q$ values, and its value may select the colouring and the decoders and add an offset to every row: if every colouring that it selects has at most $s \geq 1$ class changes, the code has dimension at most $(d^2 + ds)/2 + q$ [14, Theorem 5.5]. So advice that enters the rows additively induces no change of class.
   - *Half a row is the price of a change of class.* A change of class changes the decoders of all d entries of a row. Each place where the decoder of a single output entry changes from one row to the next raises the bound by at most half a symbol (Proposition 6.15); altering the decoder of one entry at one row is two such changes. Charged at half a row each, εd^2 revealed symbols would cost $\varepsilon d^3/2$, more than the whole space once $d > 2/\varepsilon$.

   Corollary 6.16 combines the two corrections, over a finite field. If the advice has q symbols and every rule that it selects changes, from one row to the next, at no more than E entries, then $2 \dim \mathcal{X} \leq d^2 + d + E + 2q$, and the matrices of the code that share a value of the advice number at most $|F|^{(d^2+d+E)/2}$. The rate of such a set is the analogue, in this model, of the function $r(\varepsilon)$ of Section 4.1: in it the advice costs nothing, and only the changes count. So what remains to be bounded in this model is E, the number of changes of decoders in the rules that the advice selects, whether the advice induces them or not. For machines the step is open: there the decoders are compositions of the transition function and need not be linear, and for a decoder that is not linear an offset that depends on the row is itself a break of uniformity [14, Remark 5.6]. A later paper makes this exact: with decoders that are arbitrary maps, one decoder for all rows is correct on a set of at least $q^{d^2}/(2d)$ matrices over an alphabet of q symbols, and on the affine space of the matrices with a given unit vector as row 0 [19, Theorem A.1, Corollary A.2 and Proposition 4.5]. So Corollary 6.16 has no counterpart for such decoders when they see the whole matrix, and a statement for machines has to use that the cones a genie leaves are of bounded size (Section 6.7 and Problem 9.4).

3. *Many hops.* Remark 6.14 bounds t uniform hops by a loss that grows linearly in t. A machine that runs for cL steps has unboundedly many hops, and Remark 6.14 shows that one cheap operation can break every digit symmetry at once. So an argument for machines has to count something that an unshuffle does not supply for free. The first bound for two hops that is not of this additive kind is in a later paper of the series. Let K_1 and K_2 be linear maps of $\mathrm{M}_d(F)$ such that column c of $K_i X$ is a function of $\sigma^c X$ that depends only on the class of c, each map with its own partition, into M_1 and M_2 classes, and with decoders of any window. If $K_2 K_1$ is the transposition on all of $\mathrm{M}_d(F)$, then M_1 and M_2 cannot both stay bounded as

d grows; with two classes each, $d \le 16$; and for bounded M_1, M_2 the rank of the difference between the transposition and K_2K_1 is not $O(d)$, when the characteristic does not divide d [19, Theorem 6.3, Corollary 6.4 and Theorem 6.6]. Both maps act on the d^2 entries with no further storage, and the argument is for two of them. The number of classes is again not something that a machine pays for. Let d be even and let π be the perfect unshuffle, $\pi(c) = \lfloor c/2 \rfloor$ for even c and $\pi(c) = \lfloor c/2 \rfloor + d/2$ for odd c. The map that puts column $\pi(c)$ of X in column c, a perfect shuffle of the columns and a linear-time operation on tapes, is of this kind. Two columns c and e can share a decoder only if $\pi(c) - c = \pi(e) - e$ in $\mathbb{Z}/d$, and these differences take $d - 1$ values.

We do not know how to take any of these steps for machines.

8 Small sizes

This section is numerical, except for Proposition 8.2, which is a formal statement about one explicit code, and for the results of Section 6.6 and of the subsequent memo [14] that are cited at the end of Remark 8.1 and at the end of the section. It reports exact computations over $\mathbb{F}_2$ for class-uniform rules (Definition 6.3) on a cycle. No proof in this paper uses them.

8.1 Method

For a partition of $\mathbb{Z}/d$ and a target dimension D, the existence of a class-uniform rule with a code of dimension at least D is a rank condition: the $d^2 \times d^2$ matrix of $\tau - K$, whose row (c, r) is determined by the functional of the class of c for the coordinate r, must have rank at most $d^2 - D$. We decide it in two ways.

- *Exhaustively*, for $d \le 4$, by enumerating subspaces and testing the criterion that a subspace $\mathcal{X}$ is a code if and only if, for every class J, $\sum_{c \in J} \sigma^c X_c = 0$ with $X_c \in \mathcal{X}$ implies $\sum_{c \in J}(\text{row } c \text{ of } X_c) = 0$ (`tools/exhaustive_pigeonhole_checks.py`).
- *By a SAT solver*, for $d = 5, 6$: the rank condition is written as a factorization of the matrix through $\mathbb{F}_2^{d^2 - D}$, with the second factor in reduced echelon form, and given to CryptoMiniSat (`tools/class_uniform_sat.py`). A satisfying assignment is a rule together with its code; the script rebuilds the matrix from the rule, recomputes its kernel, and `tools/verify_class_uniform_code.py` checks the saved code against the definition, sharing no code with the search. An answer "unsatisfiable" rests on the solver, and we say so wherever we report one; as controls, the solver confirms the bound of Theorem 6.4 for $d = 4$ and for $\{0, 1\}, \{2, 3\}, \{4\}$ at $d = 5$. A time-out says nothing: the solver also times out on instances that are known to be satisfiable. Some searches were restricted to codes that contain the space $\mathcal{H}^*$ of matrices with $x_{c'} = \sigma^{c' - c} x_c$ for all c, c' in one class, which is itself a code of dimension Md and, for interval classes, is the space $\mathcal{H}$ of Section 6. This makes the search faster and can only lower the result. The code of Proposition 8.2 was found without the restriction and contains $\mathcal{H}^*$.

8.2 Results

Table 3 lists the largest dimension of a code for partitions of $\mathbb{Z}/d$, up to the affine maps $c \mapsto uc + t$ of the index. For $d \le 5$ every partition with $1 < M < d$ classes is listed; for $d = 6$, the partitions into two classes and the nine partitions into three classes for which the bound exceeds $(d^2 + Md)/2$. "Bound" is the best value of Theorem 6.4 over shift sets of at most four elements, rounded down. Every code counted in the last column for $d \ge 5$ is saved in `tools/data/`. The script `tools/small_sizes.py`

prints the table from these files, with the bounds recomputed and every saved code checked again, and lists the partitions; `tools/README.md` records the runs of the solver, including those that timed out.

d	classes	pairing	$\lfloor (d^2+Md)/2 \rfloor$	bound	largest code
3	$\{0\},\{1,2\}$	6	7	7	$= 7$
4	$\{0\},\{1,2,3\}$	10	12	12	$= 12$
	$\{0,1\},\{2,3\}$	10	12	12	$= 12$
	$\{0,2\},\{1,3\}$	10	12	12	$= 12$
	$\{0\},\{1\},\{2,3\}$	10	14	14	$= 14$
	$\{0\},\{2\},\{1,3\}$	10	14	14	$= 14$
5	$\{0\},\{1,2,3,4\}$	15	17	17	$= 17$
	$\{0,1\},\{2,3,4\}$	15	17	17	$= 17$
	$\{0\},\{1,2\},\{3,4\}$	15	20	20	$= 20$
	$\{0\},\{1\},\{2,3,4\}$	15	20	20	19^*
	$\{0\},\{1,4\},\{2,3\}$	15	20	21	20^*
	$\{0\},\{1\},\{2\},\{3,4\}$	15	22	22	$= 22$
6	$\{0\},\{1,\ldots,5\}$	21	24	24	$= 24$
	$\{0,1\},\{2,\ldots,5\}$	21	24	24	$= 24$
	$\{0,1,2\},\{3,4,5\}$	21	24	24	$= 24$
	$\{0,2,4\},\{1,3,5\}$	21	24	24	$= 24$
	$\{0,2\},\{1,3,4,5\}$	21	24	27	≥ 24
	$\{0,3\},\{1,2,4,5\}$	21	24	27	≥ 23
	$\{0,1,3\},\{2,4,5\}$	21	24	28	≥ 25
	$\{0,1\},\{2,4\},\{3,5\}$	21	27	30	≥ 28
	$\{0,1\},\{2,5\},\{3,4\}$	21	27	30	≥ 28
	$\{0,2\},\{1,4\},\{3,5\}$	21	27	30	≥ 28
	$\{0\},\{1,2,4\},\{3,5\}$	21	27	30	≥ 28
	$\{0\},\{1,5\},\{2,3,4\}$	21	27	30	≥ 27
	$\{0\},\{1,2,5\},\{3,4\}$	21	27	30	≥ 27
	$\{0\},\{1,3,4\},\{2,5\}$	21	27	30	≥ 27
	$\{0\},\{4\},\{1,2,3,5\}$	21	27	30	≥ 27
	$\{0\},\{3\},\{1,2,4,5\}$	21	27	30	≥ 26

Table 3: Largest codes of class-uniform rules over $\mathbb{F}_2$. "=" marks an exact value: a code of that dimension was found, and it equals the bound of Theorem 6.4. "$\geq$" marks the largest code found. A star marks a value that was found and whose optimality rests on the solver (Remark 8.1).

Remark 8.1 (Numerical: what the table shows)**.** (a) *Intervals.* For interval classes the bound $\lfloor (d^2+Md)/2 \rfloor$ of Theorem 6.4 is attained in every case computed but one. For unequal pieces the codes that attain it are not block-symmetric, and at $d = 4$ with pieces $\{0\},\{1,2,3\}$ no copy rule attains it (Remark 6.7). The exception is $\{0\},\{1\},\{2,3,4\}$ at $d = 5$: a code of dimension 19 exists; by Lemma 6.5 a code of dimension 20 would contain $\mathcal{H}$; and the solver reports that no code of dimension 20 contains $\mathcal{H}$. So the largest code has dimension 19, on the word of the solver, and the bound of Theorem 6.4 is not attained for every partition into intervals. (b) *No almost-period, $d = 5$.* For $\{0\},\{1,4\},\{2,3\}$ every single shift leaves four components, and the bound of Theorem 6.4 is 21, from a set of three shifts. A code of dimension $20 = (d^2+Md)/2$ exists, and the solver reports that none of dimension 21 does. (c) *No almost-period, $d = 6$.* For $\{0,1,3\},\{2,4,5\}$ every nonzero shift leaves four components, and the best bound of Theorem 6.4 is 28, from a set of three shifts.

A code of dimension 25 exists, one more than $(d^2 + Md)/2 = 24$ (Proposition 8.2); searches for dimension 26 timed out. Four partitions into three classes have codes of dimension 28, one more than $(d^2 + Md)/2 = 27$. (d) The gain over $(d^2 + Md)/2$ in (c) is one dimension in each case. The data, at $d \leq 6$, do not distinguish a gain of bounded size from one that grows with d. For every fixed number $M \geq 2$ of classes it can grow, and this is proved, not numerical, though for other partitions than those of the table. For two classes the gain is at least $d/4 - 1$ at every $d = 4k \geq 8$, over every field (Proposition 6.17, which extends [14, Theorem 4.8] to characteristic 2); the codes over $\mathbb{F}_2$ at $d = 8$, 12 and 16 are saved with those of Table 3 (Remark 6.20). For every $M \geq 2$ there are partitions into M classes with a gain of order $Md/8$ (Corollary 6.19). And when the number of classes grows in proportion to d, the blow-ups of the code of Proposition 8.2 are codes of dimension $25k^2$ for $2k$ classes on $\mathbb{Z}/6k$, a gain of k^2 [14, Theorem 4.2 and Corollary 4.4].

Proposition 8.2 (Two classes with no almost-period beat every interval rule)**.** `Lean`: `exists_classUniformCode_gt`, `isClassUniformCode_aperiodicCode`, `finrank_aperiodicCode`, `aperiodicCode`, `aperiodicCls_eq_true_iff` [`AperiodicExample.lean`]

For $d = 6$ over $\mathbb{F}_2$ and the classes $\{0, 1, 3\}$ and $\{2, 4, 5\}$ of $\mathbb{Z}/6$ there is a code of a class-uniform rule of dimension 25. In particular the bound $(d^2 + Md)/2$ of Theorem 6.4 for M interval classes, and of Proposition 6.8 for copy rules with M classes of any shape, does not hold for coded rules on arbitrary classes.

Outline of the formal proof. The code is the span of 25 explicit matrices, and the two decoders are explicit (`AperiodicExampleData.lean`, generated from a certificate by `tools/gen_aperiodic_example.py`). That every generator satisfies the rule at every row, 900 equations over $\mathbb{F}_2$, is checked by the kernel (`aperiodicGen_row`); by linearity the span is a code (`isClassUniformCode_aperiodicCode`). Each generator has an entry equal to 1 where all the others vanish (`aperiodicGen_pivot`), so the generators are independent and the dimension is 25 (`finrank_aperiodicCode`). Then $2 \cdot 25 = 50 > 48 = 6^2 + 2 \cdot 6$ (`exists_classUniformCode_gt`). The certificate is used only as data. □

The code of Proposition 8.2 was found by the solver. It contains the 12-dimensional space $\mathcal{H}^*$, it meets its translate by one column in exactly $2 \cdot 25 - 36 = 14$ dimensions, the least that Lemma 6.1 allows, and it meets its translates by two and three columns in 18 and 17 dimensions (`tools/analyse_code.py`). We do not have a description of the code itself, or a way to enlarge it at a fixed number of classes, and one example at $d = 6$ says nothing about rates: the term $Md/2$ is a quarter of $(d^2 + Md)/2$ at this size. What the example does show is that the union bound in Theorem 6.4 is not an artefact, and that masks without an almost-period carry a coding gain that no copy rule and no interval rule has. Problem 9.1 asks how large it can be.

The subsequent memo [14] proves two further facts about this example, in Lean, and it finds larger gains of a different kind.

- *The gain does not come from characteristic* 2. The same two classes carry a code of dimension 25 over every field in which $3 \neq 0$, in particular over $\mathbb{Q}$ [14, Theorem 3.2]. Whether they do in characteristic 3 is open.

- *The example can be blown up.* If a colouring of $\mathbb{Z}/d$ with M classes carries a code of dimension D, then a colouring of $\mathbb{Z}/dk$ with kM classes carries a code of dimension k^2D [14, Theorem 4.2]. Blowing up multiplies the gain over $(d^2 + Md)/2$ and the product dM by the same factor k^2, so blowing up this one example gives growing gains only together with a growing number of classes. Blowing up a family by a fixed factor is another matter (Corollary 6.19).

- *Two classes can gain more than one dimension.* For $d = 4k \geq 8$ the classes $\{0, 2, \ldots, d-4\}$ and the rest carry an explicit rule, each of whose decoders reads one anti-diagonal, with a code of dimension $(d^2 + 2d)/2 + d/4 - 1$. The memo proves this over fields in which $2 \neq 0$, in its text, and proves in Lean at $d = 8$, 12 and 16 that the classes carry a code of this dimension [14, Theorems 4.8 and 4.9]; Proposition 6.17 proves it over every field, and Corollary 6.19 derives from it an unbounded gain for every fixed number of classes. Unlike the code of Proposition 8.2, these codes have a description, and it is one that can be analysed.

9 Open problems

The question (T) is open. For oblivious machines it is the question about the function $r(\varepsilon)$ of Section 4.1. The problems below are the precise points at which this paper stops; the first three are finite or algebraic and can be attacked directly.

Problem 9.1 (Classes with no almost-period)**.** *For a partition $\mathcal{P}$ of $\mathbb{Z}/d$ into M classes let $D(\mathcal{P})$ be the largest dimension of a code of a class-uniform rule (Definition 6.3), over a given field. Theorem 6.4 gives $D(\mathcal{P}) \leq (d^2 + Md)/2$ when the classes are intervals. For general classes this fails (Proposition 8.2), and for every fixed number $M \geq 2$ of classes it fails by an amount that grows with d (Proposition 6.17 and Corollary 6.19).*

(a) For two classes, is $D(\mathcal{P}) \leq (\frac{1}{2} + o(1))d^2$? More generally, is this so for $o(d)$ classes? For two classes of any shape the best upper bound known is $\frac{41}{64}d^2 + \frac{5}{3}d^{3/2}$ [19, Theorem 3.4]; before that paper it was $\frac{3}{4}d^2 + O(d)$, from Theorem 5.1, together with what Theorem 6.4 gives for the best shift set. For $M \leq d/2$ classes that paper proves the bound $(1 - 1/(4\lceil\sqrt{2M}\,\rceil))d^2$ [19, Theorem 3.5], which is below the value $(1 - \frac{1}{2M})d^2 + O(d)$ of Theorem 5.1 when $M \geq 11$. The largest gains known for a fixed number of classes are of order d.

(b) Determine D for the classes $\{0, 1, 3\}, \{2, 4, 5\}$ of $\mathbb{Z}/6$. By Theorem 6.4 it is at most 28 over every field. It is at least 25 over every field in which $3 \neq 0$: over $\mathbb{F}_2$ by Proposition 8.2, and in general by [14, Theorem 3.2]. In characteristic 3 no code of dimension 25 is known.

(c) Fix the number $M \geq 2$ of classes. Is there a constant C_M such that $D(\mathcal{P}) \leq (d^2 + Md)/2 + C_M d$ for every d and every partition $\mathcal{P}$ of $\mathbb{Z}/d$ into M classes? Or is there a family of partitions into M classes, with codes that have a description that can be analysed, in which $D(\mathcal{P}) - (d^2 + Md)/2$ grows faster than d?

A positive answer to (a) would be the half bound for all piecewise-uniform decoders in the one-hop model; it needs a count sharper than the union bound of Theorem 6.4, which loses a factor $s/(s+1)$ for $s+1$ translates. A negative answer would be a coding gain of positive rate over the pairing value from $o(d)$ classes, native to masks and rotations, the first of its kind, and its cost on a tape would become the question. Section 8 records what exact computation says for small d.

The two bounds quoted in (a) come from a later paper of the series [19], which proves them in Lean, over every field, for codes that are subspaces and decoders that are arbitrary maps (Section 6.7). Its count is sharper than the union bound in the way asked for above. The code is intersected with one translate, the intersection lies in the space of matrices that are covariant on the components of that shift, this space is invariant under rotation, and the next intersection is taken inside it; so with shift sets $E_0 = \{0\}$ and $E_j = E_{j-1} \cup (E_{j-1} + t_j)$ for $1 \leq j \leq k$, a code has dimension at most $\sum_{j<k} n_j/2^{j+1} + n_k/2^k$, where n_j is d times the number of components for E_j [19, Theorem 3.2]. When E_2 has four elements, Theorem 6.4 for E_2 gives $\frac{3}{4}n_0 + \frac{1}{4}n_2$ where this gives

$\frac{1}{2}n_0 + \frac{1}{4}n_1 + \frac{1}{4}n_2$, and $n_0 = d^2$. The value $\frac{41}{64}$ comes from the shifts $s, 3s, 9s$ and an average over s. For almost every partition into two classes the same paper has a slightly smaller constant, and it proves that the certificates of this method cannot go below $(\frac{5}{8} - o(1))d^2$, and those of a variant not below $(\frac{9}{16} - o(1))d^2$, for almost every such partition [19, Theorems C.5, C.6 and C.7]. So (a) is open, and going below these values needs more than the certificates of that method.

Part (c) has a history. Revision `b2c389a` of this paper asked only for a family of partitions and codes, with a description that can be analysed, in which $D(\mathcal{P}) - (d^2 + Md)/2$ grows, without fixing the number of classes. The memo [14] answered that question as it was worded: the blow-ups of the code of Proposition 8.2 are codes of dimension $25k^2$ for $2k$ classes on $\mathbb{Z}/6k$, so that $D(\mathcal{P}) - (d^2 + Md)/2 \geq k^2$ there [14, Theorem 4.2 and Corollary 4.4]. The family is described given its base code, which a solver found. A blow-up multiplies the excess of a code over $(d^2 + Md)/2$ and the product dM by the same factor [14, Corollary 4.3], so the blow-ups of one finite example gain only together with a growing number of classes. The memo and revision `14b157a` of this paper therefore asked whether $D(\mathcal{P}) - (d^2 + Md)/2$ is bounded for a fixed number of classes [14, Problem 6.2]. It is not. An update of the memo showed this for $M = 2$ over every field in which $2 \neq 0$, with an explicit rule whose code has a gain of $d/4 - 1$ [14, Theorem 4.8 and Corollary 4.10], and left open characteristic 2 and every $M \geq 3$. Proposition 6.17 is that theorem with its two coefficients 2 and $\frac{1}{2}$ replaced by λ and λ^{-1}, which makes it hold over every field. Corollary 6.19 blows the two-class codes up by a fixed factor, with a code of its own at each position, and gives an unbounded gain for every $M \geq 2$ over every field. What remains is the rate of growth, which is (c) as now stated; for $M = 2$ it is, in substance, [14, Problem 6.3].

By Corollary 6.19 a constant C_M as in (c) is at least $\lfloor M/2 \rfloor/4$, over every field; for two classes it is at least $\frac{1}{4}$. For the two classes of Proposition 6.17, Theorem 6.4 bounds the gain by $d/2$. In general Theorem 6.4 does not answer (c): with two shifts it bounds the gain of a two-class partition by $(\chi - 2)d/2$, where the number χ of components can be of order d, and with three or more shifts its bound on $D(\mathcal{P})$ is at least $2d^2/3$ [14, Section 6]. A positive answer to (c) for $M = 2$ would prove (a) for two classes. A family of two-class partitions with a gain of order d^2 would be a negative answer to (a).

For partitions with a bounded period the answer to (c) is positive, and for some of them the constant is known. Let a partition p of $\mathbb{Z}/s$ be lifted to the partition $p^{\uparrow}$ of $\mathbb{Z}/d$, $d = ms$, in which c has the class of c mod s. Theorem 6.4 with the shifts 0 and s bounds its largest code by $(d^2 + sd)/2$. If m is prime to s and not divisible by the characteristic, and the field contains the m-th roots of unity, the exact value is

$$D(p^{\uparrow}) = \frac{d^2 - ms^2}{2} + m\, D(p) = \frac{d^2}{2} + \frac{d}{s}\Big(D(p) - \frac{s^2}{2}\Big);$$

over a field without these roots of unity the right side is a lower bound, and it is an upper bound when $D(p)$ is taken over the algebraic closure [19, Theorems 5.1 and 5.3]. So the gain of a partition with period s is at most $(s - M)d/2$, for these lifts it is a fixed multiple of d, which is zero when $D(p)$ is the value $(s^2 + Ms)/2$ of intervals, and a gain that grows faster than d needs partitions without a bounded period; the classes of Proposition 6.17 have no period smaller than d. From the classes of Proposition 8.2 the lift gives two classes on $\mathbb{Z}/6m$, for m prime to 6, with a binary code of dimension $d^2/2 + 7d/6$, and this is the whole code of the lifted rule [19, Corollary 5.2 and Theorem B.2]. Its gain over $(d^2 + 2d)/2$ is $d/6$; the rule of Proposition 6.17 gains $d/4 - 1$, at the sizes divisible by four.

Problem 9.2 (Unequal pieces)**.** *For which partitions of $\mathbb{Z}/d$ into M intervals, and over which fields, is the bound $\lfloor (d^2 + Md)/2 \rfloor$ of Theorem 6.4 attained? It is attained for equal pieces (Proposition 6.6).*

Over $\mathbb{F}_2$ it is attained in every case computed in Section 8 except one, the pieces $\{0\},\{1\},\{2,3,4\}$ at $d=5$, where the largest code has dimension $19<20$ on the word of a solver. Some of the codes that attain it are not codes of copy rules.

The question is about linear decoders. If the decoders may be arbitrary maps and the code is still a subspace, the bound is attained over $\mathbb{F}_2$ for every partition into M intervals with $d-M$ even, and the same construction gives a code of dimension $(d^2+Md)/2$ for every partition into M classes of any shape with $d-M$ even [19, Theorem 4.3]. For the pieces $\{0\},\{1\},\{2,3,4\}$ at $d=5$ this is a code of dimension 20 with decoders that are not linear, where linear decoders reach 19 on the word of a solver.

Problem 9.3 (Both sides, all scales)**.** *Let both the A_a and the B_a lie in the span of masks of dyadic intervals composed with rotations, with at most M_j intervals of length $d/2^j$ in use at scale j. Bound $\dim\ker(\tau-K)$ in terms of the profile $(M_j)_j$. The folklore recursion shows that the full profile suffices for dimension d^2; Theorem 6.4 and Corollary 6.13 treat one side and one scale at a time.*

Problem 9.4 (Shared decoders on cone systems)**.** *(a) State and prove a version of Theorem 6.4 for the cone systems of oblivious linear-time schedules: on a stretch of periodic head motion the outputs share their decoder, and a revealed bit or an irregular step changes the decoders of some outputs. Does a bound of the form $r(\varepsilon)\le\frac12+C(\varepsilon+\eta)$ hold for schedules whose motion is periodic outside a set of steps of density η? (Section 7.4.)*

(b) Fix a window $w\ge3$ and an alphabet of q symbols. Does every storage code on a one-class relation of window w (Section 6.7) have at most $q^{(1/2+o(1))d^2}$ elements as d grows, or even at most $q^{d^2/2+Cd}$ with a constant C that depends on w and q? And is this so at least when one function serves all rows, that is, when the function for the position (c,r) does not depend on c? (For $w\le2$ the answer to both is yes, with $C=\frac32$: Proposition 6.22(b). For $w=2$ and a function for each position the least constant lies between 1 and $\frac32$, by the examples after that proposition; for $w=1$ it is $\frac12$.)

(c) Fix $w\ge3$ and an alphabet of q symbols, and let ρ assign to every column index at most w rows. Does every ρ-code (Section 6.7), that is, every set of matrices in which each column is a function of its w rows, whole, have at most $q^{(1/2+o(1))d^2}$ elements as d grows? (For $w\le2$ it has at most $q^{d(d+1)/2+d-1}$, by the proof of Proposition 6.22(b); in general at most $q^{wd(d+1)/(w+1)}$, by Proposition 6.24(d).)

Part (a) is the problem as the earlier revisions stated it; part (b) was added in revision `3911d87`, for every window, and is restricted here to the windows that remain open. The form of the bound in (a) comes from the one-hop model with affine rules. By Corollary 6.16, over a finite field, if every rule that the advice selects changes, from one row to the next, at no more than E entries, then the matrices of a code that share a value of the advice number at most $|F|^{(d^2+d+E)/2}$, whatever the advice is. The rate of such a set, the analogue of $r(\varepsilon)$, is therefore at most $\frac12+(d+E)/(2d^2)$, and $E=O((\varepsilon+\eta)d^2)$ gives a bound of the form above. So the question is whether the revealed bits and the irregular steps of a schedule amount to that many changes of decoders at single outputs, on the cone systems of real schedules and for decoders that are compositions of a transition function. Revision `b2c389a` of this paper stated the count as "a genie that reveals εL bits changes the regime at most εL times", and its Section 7.4 priced a change of regime at half a row. Half a row is the price of a change that affects all d entries of a row, and at that price the count gives nothing.

For decoders that are not affine the count of Corollary 6.16 is not available at all, and this is the reason for part (b). With a view of the whole matrix, one decoder for all rows, with no change

of any kind, is correct on a set of at least $q^{d^2}/(2d)$ matrices [19, Theorem A.1 and Corollary A.2] (Section 6.7). The inputs that share a value of advice form a set and not a subspace. So for them, and for decoders that are compositions of a transition function, the sharing of a decoder bounds nothing when the decoder sees the whole stored word. What that example uses, and the decoders of a machine do not have, is the size of the view. The cones that a genie leaves have bounded size, so a decoder reads boundedly many bits, and with them it can tell apart only boundedly many positions (Section 6.7). Whether the sharing of a decoder bounds a set of matrices when the decoders read boundedly many symbols of a fixed alphabet is not known, and part (b) asks it in the one-hop model, in two forms. In the weaker form one function serves all rows and reads translated positions, which is what a stretch of periodic motion gives. Since a set of matrices can contain marker rows, one function can then imitate boundedly many classes of rows, where the marker rows are free: with markers the value $(d^2+d)/2$ of subspaces is exceeded by a multiple of d that grows with the window, over every finite field (Section 6.7), so the constant C of (b) has to grow with w. In the stronger form only the positions are translates, and every position may have its own function. Over a fixed alphabet there are boundedly many functions of w symbols, so the stronger form is the question for boundedly many classes of any shape, drawn entry by entry. It contains what is left of the weaker form among the inputs that share a revealed word: there the outputs of a periodic stretch whose cones meet the same pattern of revealed bits, with the same values, compute the same function of the input bits that they read, so that the revealed word divides the outputs into boundedly many classes of a shape that the advice chooses. The hypothesis of the stronger form is a property of the relation alone, and so of the head motion. What is proved is in Propositions 6.21 to 6.23: windows one and two; every window when the offsets, or the differences of the rows from a permutation, are bounded or take boundedly many values; codes that use only helpers in both directions; and, for the weaker form, subspaces. What these results leave is a window of at least three on a relation whose rows and offsets are both spread. For small windows the acyclic-subgraph bound, which gives all of them but the last, may still be enough; for large windows the computations reported in Section 6.7, in the paragraph on what the order argument does not reach, indicate that it is not. A code above the pairing rate would have to use helpers in one direction, on a relation of that kind.

The conclusion asked for in (b) is that of the all-code question of the fifth paper [22, Question 9.1], with other hypotheses. That question is about relations of bounded incidence degree that are separated, and for those the answer is no (Section 3.5). A one-class relation has the symmetry of its positions instead; it may have loops and cycles of length two, and one position may help d others. We have not determined whether relations like those of the counterexamples of Section 3.5 can be one-class relations; the undirected ones cannot, in the sense of Proposition 6.23(b). In every one-class relation of bounded window few positions are their own helpers, and $o(d^2)$ positions meet all transitive triangles (Proposition 6.23(c)). A rule with M classes of any shape whose decoders have window w gives a relation that is contained in a one-class relation of window Mw. So a positive answer to the stronger form of (b) would give the half bound of Problem 9.1(a) for every fixed number of classes and decoders of bounded window over a fixed finite field, for sets as well as subspaces. For subspaces the weaker form is enough. A code of a rule with M classes whose decoders have window w can be replaced by a code of at least the same dimension in which one row of each class is free, with decoders of window at most $2w+1$ [19, Theorem 4.1]; fixing $k = \lceil \log_q M \rceil$ of these rows as markers (Section 6.7) gives a storage code with one function for all rows, on a one-class relation of window $M(2w+1)+k$, with q^{-kd} times as many elements. So a positive answer to the weaker form, for every window, gives Problem 9.1(a) for rules with a fixed number of classes and decoders of bounded window over a fixed finite field. A negative answer to either form would be a code above the pairing rate, with bounded windows, on positions that move with the output row.

Part (c) was added with Propositions 6.24 to 6.27. A storage code on a one-class relation of

window w is a ρ-code with at most w rows for each column, whatever its offsets, so a positive answer to (c) for every w gives the bound $q^{(1/2+o(1))d^2}$ of (b) for every window; it does not give the form $d^2/2 + Cd$. In the other direction (c) allows more than (b): an entry may depend on all entries of its rows. What is proved is the case in which the rows of each column lie in a small block, and with it pipelines of stages that read rows inside address cosets, where t moved digits give exactly $(d^2 + 2^t d)/2$; pipelines with buffered scans and with differently arranged copies, more weakly (Proposition 6.25); and products of codes on boundedly many positions (Proposition 6.27). What is left are the relations on which the acyclic-subgraph bound stays a constant above one half whatever order is used, among them those that expand and contain a bijection. There are such relations that are unions of the identity and three increasing maps, so that an operator with these rows takes boundedly many forward passes, given the strings that mark the maps (Proposition 6.26 and the paragraph after it). A positive answer to (c) for $w \geq 4$ therefore needs an argument that is not an order argument, and a negative answer on increasing maps, with functions computable in linear time, would be an operator of boundedly many passes that transposes more than the pairing rate. By Proposition 6.27 a code above the pairing rate cannot be a product of local codes. One family in which to look for either answer: for a group Γ of order d and a bounded set $\Sigma \subseteq \Gamma$, index the rows and columns by Γ and let the helpers of (c, r) be the positions (rs, ct) for boundedly many pairs (s, t) with $s \in \Sigma$; these relations are contained in the relation of $\rho(r) = r\Sigma$, and for $\Gamma = \mathbb{Z}/d$ they are the one-class relations with constant offsets and constant row differences, which Proposition 6.22(d) settles.

For adaptive machines a later paper of the series puts the function $r(\varepsilon)$ on the machine's own runs. At the canonical entry width, for all but an exponentially small share of the inputs that a linear-time transposer handles, and for every set B of steps of the run on such an input, the relation between entries that is left when B is cut out carries storage codes whose deficit is at most a small multiple of the input length plus a constant times $|B|$ [25, Theorem H.5 and Corollary H.6]. So a bound on storage codes that is stated for relations, as part (b) is, can be applied to a single run of an adaptive machine wherever the relation of that run satisfies its hypothesis.

Problem 9.5 (What an unshuffle does not supply)**.** *Corollary 6.13 requires a transposer to break the symmetry of every address digit, and Remark 6.14 shows that commutator rank cannot turn this into a factor* $\log d$*, because one unshuffle breaks all digits at once. Find a quantity that is subadditive over the steps of a machine, is large for the swap at every scale separately, and is not raised at all scales by a single linear-time operation. This is the operator form of the multi-scale problem of the sixteenth paper [15, Problem 11.1].*

For pipelines whose stages read rows inside cosets of the address space, the dimension of the coset adds up over the stages and bounds the set that is transposed, for all codes: one for each digit moved, and $\log_2 d$ for the swap (Proposition 6.25(a)). This is not the quantity asked for. It does not look at scales separately, and a single linear-time operation can raise it to its maximum: a scan with a buffer of one row, or a stage that reads a row r together with its shuffle parents $\lfloor r/2 \rfloor$ and $\lfloor r/2 \rfloor + d/2$ (the remarks after Proposition 6.25). For the first the proposition has a price in advice, and for the second only the weaker count by the number of ones of the address.

Problem 9.6 (Relative rigidity on ball patterns)**.** *The eighth paper asks whether there is $\rho > 0$ such that, for every linear-time schedule, every fixed D and all large d, the transposition matrix is not the sum of a matrix of rank below ρL and a matrix supported on the depth-D ball pattern of the schedule [4, Problem 10.6]; a positive answer alone would exclude oblivious linear-time transposers with linear data paths [5, Corollary 9.3]. The classes of this paper are defined by structure on one tensor side. Which ball patterns of schedules are contained in such a class, after a change of bases on one side? For those, Theorems 5.1 and 6.4 give $\rho = \frac{1}{2} - o(1)$.*

A Index of Lean declarations

Every declaration of this paper's library cited in the text, grouped by the Lean module that contains it, relative to the namespace `SwapRigidity`. All of them depend only on the axioms `propext`, `Classical.choice` and `Quot.sound` (Appendix B).

`SwapRigidity.AperiodicExample` `aperiodicCls_eq_true_iff`, `aperiodicCode`, `aperiodicGen_pivot`, `aperiodicGen_row`, `exists_classUniformCode_gt`, `finrank_aperiodicCode`, `isClassUniformCode_aperiodicCode`.

`SwapRigidity.BlockSymmetric` `blockSymmetric`, `blockSymmetric_attains_bound`, `isClassUniformCode_blockSymmetric`, `two_mul_finrank_blockSymmetric`.

`SwapRigidity.ClassUniform` `IsClassUniformCode`, `IsClassUniformCode.finrank_le`, `IsClassUniformCode.row_eq_of_rot_mem`, `IsClassUniformCode.two_mul_finrank_le`, `IsClassUniformCode.two_mul_finrank_le_of_const`, `Link`, `rot`.

`SwapRigidity.Entry` `IsClassUniformCode.exists_isEntryRuleCode`, `IsEntryRuleCode`, `IsEntryRuleCode.apply_succ_eq`, `IsEntryRuleCode.two_mul_finrank_le`, `IsEntryRuleCode.two_mul_finrank_le_of_changes`, `IsEntryRuleCode.two_mul_finrank_le_of_rows`.

`SwapRigidity.EntryAdvice` `IsEntryRuleCodeWithAdvice`, `IsEntryRuleCodeWithAdvice.natCard_fiber_le`, `IsEntryRuleCodeWithAdvice.natCard_pow_two_mul_finrank_le`, `entryRuleSpace`.

`SwapRigidity.Flag` `card_staircase`, `eq_zero_of_mem_swapCode_of_staircase`, `exists_staircase_finrank_eq`, `finrank_swapCode_conj`, `finrank_swapCode_le_of_flag`, `finrank_swapCode_le_of_flag_left`, `finrank_swapCode_le_of_staircase`, `finrank_swapCode_staircase`, `finrank_swapCode_swap`, `mem_swapCode`, `mem_swapCode_staircase_iff`, `staircaseRestrict_injective`, `swapCode`, `swapCodeEquiv`, `two_mul_finrank_swapCode_le_of_triangular`, `two_mul_finrank_swapCode_le_of_triangular_left`, `two_mul_sum_fin_succ`.

`SwapRigidity.Line` `IsLineUniformCode`, `IsLineUniformCode.row_succ_eq`, `IsLineUniformCode.two_mul_finrank_le`, `isLineUniformCode_of_coeff`, `isLineUniformCode_symmetric`, `lineView`, `truncShift`.

`SwapRigidity.Pigeonhole` `card_mul_finrank_add_le_iInf_comap`, `finrank_ker_commutator_sub_le`, `finrank_range_commutator_comp_le`, `finrank_range_commutator_le`, `mul_finrank_le_add_finrank_iInf_comap`, `two_mul_finrank_ker_sub_le`, `two_mul_finrank_ker_sub_le_ker`, `two_mul_finrank_ker_sub_le_of_commute`.

`SwapRigidity.Swap` `card_le_finrank_ker_swap_commutator`, `two_mul_finrank_ker_swap_sub_one`, `two_mul_finrank_range_swap_sub_one`.

`SwapRigidity.Twist` `finrank_le_sum_finrank_twistSpace`, `succ_mul_finrank_swapCode_le`, `succ_mul_finrank_swapCode_le_of_twist_classes`, `twistSpace`, `two_mul_finrank_swapCode_le_of_twist`.

B Building and verifying the formalization

The Lean project is the directory `lean/` next to this paper. It uses Lean and Mathlib, both at version v4.34.1 (`lean-toolchain`, `lakefile.toml`, `lake-manifest.json`), and requires the shared `lean-machine/` library of the repository through a relative path, which supplies Mathlib at the pinned revision. It uses no library of an earlier paper. Keep the repository's directory layout so that `lean-machine/` stays in place.

Build. With `elan` installed, run in `lean/`:

```
lake exe cache get && lake build
```

The first command downloads the compiled Mathlib cache; the second compiles `SwapRigidity`.

Axioms and index. The file `lean/PaperAxioms.lean`, generated from the LaTeX sources of this paper by `tools/lean_index.py`, runs `#print axioms` on every declaration cited in this paper (Appendix A):

```
lake env lean PaperAxioms.lean
```

Every declaration depends at most on `propext`, `Classical.choice` and `Quot.sound`. No Lean file of this library uses `sorry`, `admit` or `native_decide`, declares an `axiom`, bypasses the kernel via `unsafe`, `implemented_by` or `extern`, or raises `maxHeartbeats`. Running `python3 tools/lean_index.py` from the paper's root directory regenerates both `paper/sections/leanindex.tex` and `lean/PaperAxioms.lean` (`-index-only` writes only the index; `-check` verifies that every cited name resolves and that both generated files are current).

Where things are. The library `lean/SwapRigidity/` has the following modules, all imported by the root module `lean/SwapRigidity.lean`.

- `Flag.lean` (Section 5): the swap code, the elimination on staircase patterns, invariance under changes of bases, the coordinate form of Theorem FL, the corollary for a pair of flags, the exchange of sides, and attainment.
- `Pigeonhole.lean` (Section 6): the pigeonhole lemma in its three forms, the rank bound for commutators and its subadditivity under composition, and the dimension count for an intersection of preimages.
- `Swap.lean`: the dimensions of the kernel and image of $\tau - 1$ and the bound $\dim \ker[\tau, T] \geq d$.
- `ClassUniform.lean`: class-uniform codes and Theorem TP on a cycle.
- `Entry.lean` (Proposition 6.15): rules given entry by entry, and the bound for decoders that change at single entries.
- `EntryAdvice.lean` (Corollary 6.16): the same rules with advice of any kind, over a finite field.
- `BlockSymmetric.lean`: the block-symmetric code, its dimension, and attainment of the bound.
- `Line.lean`: two-sided rules on a line and the line form of Theorem TP.
- `Twist.lean`: the twist-class theorem.

- `AperiodicExample.lean` and the generated `AperiodicExampleData.lean` (Proposition 8.2): an explicit class-uniform code of dimension 25 for $d = 6$ over $\mathbb{F}_2$. The data file contains definitions only. The finite facts about the data are proved by `decide`, which the kernel checks; nothing is taken on trust from the certificate `tools/data/aperiodic_d6_dim25.json` or from the generator `tools/gen_aperiodic_example.py` (run it with `-check` to confirm that the data file is current).

Computations. Every script that was used for this paper is in `tools/`, and the saved codes are in `tools/data/`. The file `tools/README.md` lists the scripts with their commands and the versions used, and records the solver runs behind Table 3, including those that timed out. Each script documents its usage at its top.

- `tools/exhaustive_flag_checks.py` and `tools/exhaustive_pigeonhole_checks.py` (Python with NumPy): the exhaustive and randomized checks behind the numerical remarks of Sections 5 and 6 and the entries for $d \leq 4$ of Table 3.
- `tools/three_term_example.py`: the examples of Remark 5.6.
- `tools/digit_and_composition_checks.py`: small instances of Corollary 6.13 and of Remark 6.14, both of which are proved in the text.
- `tools/entry_form_check.py`: small instances of Proposition 6.15, which is proved in Lean apart from a count.
- `tools/two_class_family.py`: the rule of Proposition 6.17, which is proved in the text, over five small fields, with every rank stated in the proof; the saved codes over $\mathbb{F}_2$; and small instances of Corollary 6.19 (Remark 6.20).
- `tools/set_bounds_check.py`: small instances of Proposition 6.21, which is proved in the text. The largest code of every recovery relation at $d = 2$ over two symbols, against the acyclic-subgraph bound and both parts of the proposition; the step of the two proofs over all pairs of matrices, for 40 random relations at $d = 3$ over two symbols and 60 at $d = 2$ over three; for the cyclic instance at $d \leq 12$, its hypothesis, its exponent and the absence of cycles on random relations; the count of cycles in part (b) on random maps; the remark on marker rows, on every matrix of the set at $d = 4$ and on a sample at $d = 16$; three controls that must fail; and the count for the perfect shuffle in Section 7.4.
- `tools/order_bounds_check.py`: Propositions 6.22 and 6.23, which are proved in the text. The ranking lemma on every multigraph with at most five vertices and as many edges, and its sharpness for at most four; the sets of positions of Proposition 6.22 on random relations, with their sizes; for window two, the exact least number of positions that meets every cycle, for random relations at $d = 3$ and $d = 4$, against the bound; the count of Proposition 6.22(c) and the multiplier of its second statement on random instances; and the two counts in the proof of Proposition 6.23 against enumeration; the two local codes named after Proposition 6.23; and the examples after Proposition 6.22: the cycle for the ranking lemma up to seven vertices, the relation at $d = 3$ with its code, the codes with $d^2/2 + d$ free entries at $d = 4$ and 6, and the relation of whole rows at $d = 4$. With two controls.
- `tools/row_spread_check.py` (needs NumPy): Propositions 6.24 and 6.25 and Proposition 6.26(a), which are proved in the text. The quantity σ by dynamic programming

over sets of rows against all orders; the largest acyclic set against $d^2 - \sigma$, exhaustively over sets of positions for $d \leq 4$; the exact value for blocks, the bound for rows inside blocks, and the two codes named in and after Proposition 6.24; pipelines of affine stages, also with singular linear parts, against the coset of Proposition 6.25(a); for scans, that inputs with the same advice that agree on the coset give the same output row, with a control; for heights, the bound of part (c) on pipelines over two copies; the inequality of Proposition 6.26(a) with the exact expansion constant; the numbers quoted for the shuffle parents; and a simulation, with rows as units, of the gather passes from the marking strings. Not checked by the script: the bound of Proposition 6.25(b) itself and its last inequality (at sizes where all inputs can be enumerated the bound is weaker than Proposition 6.24(a)), and Proposition 6.26(b), which is cited. Proposition 6.27 and the count in Proposition 6.25(a) for partial transposes were checked by the referees' own programs, which are not shipped.

- `tools/acyclic_set_search.py`: the searches for acyclic sets that are quoted in Section 6.7, in the paragraph on what the order argument does not reach. It needs a solver (OR-tools). The searches for large sets run under a time limit, so their results are sizes of sets that were found, a rerun can give slightly different ones, and `tools/README.md` records the output that is quoted.
- `tools/class_uniform_sat.py` (needs the package `pycryptosat`): one run of the solver, for a partition and a target dimension.
- `tools/small_sizes.py`: Table 3 with its bounds, recomputed, and with every saved code checked again; the list of partitions up to affine maps; the search that raises the target dimension, which produced the saved codes of the table; and controls that repeat the answers of the solver quoted in Remark 8.1.
- `tools/verify_class_uniform_code.py`: checks every saved code in `tools/data/` against the definition.
- `tools/analyse_code.py`: the intersections of the code of Proposition 8.2 with its translates, quoted after that proposition.

Apart from the certificate of Proposition 8.2, which Lean checks, these computations are not part of the formal development, and no proof depends on them.

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